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Unconditional energy stable and mass-conserving Gauge–Uzawa MSAV methods for the Cahn–Hilliard–Navier–Stokes model

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ABSTRACT

We develop linear, fully decoupled first- and second-order time-discretization schemes for the Cahn–Hilliard–Navier–Stokes system, by integrating the multiple scalar auxiliary variables (MSAV) approach with the Gauge–Uzawa (GU) method. The schemes are proved to ensure both mass conservation and unconditional energy stability. Moreover, by reformulating the system into matrix form, we present two linearly decoupled numerical solution algorithms through Gaussian elimination: a commonly adopted decoupled method with constant-coefficient matrices and an alternative approach involving solution-dependent coefficient matrices. Both schemes involve only solving a series of Poisson-type equations alongside a straightforward linear algebraic system at each time step. Notably, the proposed methods eliminate the need for artificial pressure boundary conditions. Besides, a rigorous error analysis of the first-order scheme is conducted, in which the idea of high-order consistency analysis is employed for the intermediate velocity by introducing an auxiliary momentum equation. It is shown that the scheme achieves optimal temporal convergence rates for the phase function and velocity across various norms. Finally, numerical experiments are performed to validate the theoretical findings and demonstrate the performance of the proposed schemes.

1. Introduction

The well-known Cahn–Hilliard–Navier–Stokes (CHNS) system serves as the fundamental system for phase field models of two-phase flows [1,2]. This system can be expressed as follows:

$$\phi_t + (\boldsymbol{u}\cdot\nabla)\phi = M\Delta\mu, \quad \text{in } \Omega\times J, \tag{1.1a}$$
$$\mu = -\lambda\Delta\phi + \lambda F'(\phi), \quad \text{in } \Omega\times J, \tag{1.1b}$$
$$\boldsymbol{u}_t + \boldsymbol{u}\cdot\nabla\boldsymbol{u} - \nu\Delta\boldsymbol{u} + \nabla p = \mu\nabla\phi, \quad \text{in } \Omega\times J, \tag{1.1c}$$
$$\nabla\cdot\boldsymbol{u} = 0, \quad \text{in } \Omega\times J, \tag{1.1d}$$
$$\frac{\partial\phi}{\partial\boldsymbol{n}} = \frac{\partial\mu}{\partial\boldsymbol{n}} = 0,\ \boldsymbol{u} = \boldsymbol{0}, \quad \text{on } \partial\Omega\times J, \tag{1.1e}$$

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with initial conditions $\phi(\boldsymbol{x},0)=\phi^0, \boldsymbol{u}(\boldsymbol{x},0)=\boldsymbol{u}^0$, where $\boldsymbol{n}$ is the outward unit normal vector. Here, ϕ represents the phase field, μ is the chemical potential, $\boldsymbol{u}$ is the fluid velocity, and p is the pressure. The mobility coefficient M is a positive constant and $\nu>0$ denotes the viscosity of the fluid. The double-well potential is defined as $F(\phi)=\frac{1}{4\epsilon^2}\left(1-\phi^2\right)^2$, where ϵ characterizes the interfacial width. For simplicity, we set the mixing coefficient $\lambda=1$. Let $\Omega\subset\mathbb{R}^2$ be an open, bounded and connected domain with a smooth boundary, and let $J=(0,T]$. Throughout this paper, we consider the classical incompressible CHNS model (1.1) with matched densities, constant viscosity, and constant mobility. We refer the readers to Anderson et al. [3], Khanwale et al. [4] for the derivation of the model and its extensions (e.g., to variable-density, variable-viscosity and even variable mobility cases), and to Feng [5], Giorgini et al. [6] for the existence and uniqueness results. The above system satisfies the following energy dissipation law:

$$\frac{dE(\phi,\boldsymbol{u})}{dt}=-M\|\nabla\mu\|^2-\nu\|\nabla\boldsymbol{u}\|^2 \text{ with } E(\phi,\boldsymbol{u})=\int_\Omega\left\{\frac{1}{2}|\boldsymbol{u}|^2+\frac{1}{2}|\nabla\phi|^2+F(\phi)\right\}d\boldsymbol{x}. \tag{1.2}$$

The CHNS system, a cornerstone of hydrodynamic phase field models, has received significant attention in recent research [7–11]. Despite remarkable progress in the development of the CHNS system, the construction of efficient, fully decoupled, linear, and unconditionally energy stable numerical schemes remains a significant challenge. One of the key difficulties lies in handling nonlinear terms while ensuring unconditional energy dissipation at the discrete level. To date, numerous time discretization methods have been proposed to design energy stable schemes for phase field models, such as the convex splitting approach [12], the exponential time differencing method [13], the invariant energy quadratization (IEQ) approach [14], and the scalar auxiliary variable (SAV) approach [15]. Similarly, for the Navier–Stokes equations, it is essential to handle the nonlinear term effectively while ensuring that the scheme remains unconditionally energy stable. Recently, SAV schemes have attracted significant attention in the Navier–Stokes model [16–18].

The second numerical challenge in the CHNS system arises from the strong coupling between fluid dynamics and phase field variables. In the monolithic approach, the fluid velocity, pressure, and phase variables have to be solved together, which incurs a significant computational cost. Consequently, it is highly desirable to decouple the CHNS system into smaller subsystems and solve them sequentially. Building upon the concepts of zero-energy contribution (ZEC) and IEQ methods, Yang et al. [19,20] developed a decoupling strategy by employing auxiliary variables, yielding fully decoupled linear schemes for coupled phase field systems. Notably, Li and Shen [21] introduced a class of linear, fully decoupled, and unconditionally energy stable time discretization schemes for the CHNS model. By integrating the multiple scalar auxiliary variables (MSAV) approach with (rotational) pressure-correction techniques [22,23], their framework explicitly addresses nonlinear coupling terms while maintaining computational efficiency. Their work further established rigorous error estimates for the first-order scheme, representing the first such analysis for linear, fully decoupled schemes that explicitly handle all nonlinear terms in the CHNS system.

The third challenge originates from the coupling between velocity and pressure, enforced by the incompressibility constraint $\nabla\cdot\boldsymbol{u}=0$ in (1.1d). This coupling leads to a saddle-point structure, which requires compatibility between the discrete velocity and pressure spaces, as defined by the inf-sup condition [24,25]. This compatibility introduces additional numerical complexities [26] that need to be addressed in the analysis and implementation of the numerical scheme. To resolve these challenges, decoupled schemes are widely favored for their computational efficiency and simplicity. Pressure-correction methods [22,27] decouple the system by either treating the pressure term explicitly or ignoring it in the first sub-step and correcting it at the second sub-step to enforce incompressibility. In contrast, velocity-correction methods reverse the roles of velocity and pressure treatment. However, projection-type methods impose artificial boundary conditions on pressure, potentially generating boundary layers and compromising accuracy [27]. For a comprehensive overview of related methodologies, we refer to Guermond et al. [23].

The Gauge method is founded on the Hodge decomposition (also known as the Helmholtz–Hodge decomposition) [26], which states that any sufficiently smooth, rapidly decaying vector field $\boldsymbol{a}$ can be decomposed into a divergence-free term $\boldsymbol{u}$ (the solenoidal part) and the gradient of a scalar potential ψ (the irrotational part), expressed as:

$$\boldsymbol{a}=\boldsymbol{u}-\nabla\psi, \tag{1.3}$$

where $\nabla\cdot\boldsymbol{u}=0$ and the two components $\boldsymbol{u}$ and $\nabla\psi$ are orthogonal. A key advantage of this formulation lies in the non-physical nature of ψ, which provides flexibility in prescribing boundary conditions. However, the Gauge method suffers from the non-variational boundary term and the requirement of high-order polynomial space for the auxiliary variable. To address these issues, Nochetto and Pyo [28] proposed the fully-discrete Gauge–Uzawa finite element method (GU-FEM) for the Navier–Stokes equations, which introduces an artificial velocity function that vanishes on the boundary. They proved the unconditional stability and error estimates for both velocity and pressure via a variational approach under some time stepsize condition. Subsequently, for the first-order time-discretization GU method, Pyo [29] obtained optimal accuracy for velocity in the $\ell^\infty(0,T;L^2(\Omega))\cap\ell^2(0,T;H^1(\Omega))$-norm via energy estimate. Compared with standard projection methods, the GU method does not require the imposition of artificial boundary conditions on the pressure, which can otherwise compromise accuracy. Moreover, it avoids the need to specify an appropriate initial pressure value. An inappropriate choice of initial pressure may lead to incompatibility between the ellipticity of the Lagrange multiplier p and $\nabla\cdot\boldsymbol{u}=0$. In comparison with the Gauge method, there are no complex boundary conditions to deal with, which makes the calculation much simpler. Although the classical second-order GU method [30] achieves superior numerical accuracy, it imposes stringent stability constraints. To address this issue, Pyo et al. [31] developed the second-order stabilized Gauge–Uzawa (SGU) method, which is unconditionally stable. They proved an optimal convergence rate for velocity in the $\ell^2(0,T;L^2(\Omega))$-norm. To date, comprehensive convergence analyses of GU methods remain scarce and continues to present significant challenges, even for the Navier–Stokes equation.

This paper aims to develop a class of linear, unconditionally energy stable schemes for the CHNS system by integrating the MSAV approach with the GU method. Most importantly, we shall establish rigorous error estimates for the first-order scheme. Our principal contributions are threefold:

- Two novel fully decoupled and linear schemes: We construct fully decoupled first- and second-order linear schemes which only require solving a sequence of Poisson-type equations with constant coefficients along with a simple linear algebraic system at each time step, ensuring computational efficiency and simplifying implementation.
- Unconditional energy stability: The two proposed schemes are unconditionally energy stable with respect to the corresponding modified energy. To enhance the consistency between the modified energy and the original discrete energy, we incorporate a relaxation technique [32,33]. To the best of our knowledge, the proposed GU-MSAV schemes are the first attempt to achieve full temporal accuracy while remaining unconditionally energy stable for the CHNS system.
- Rigorous error analysis: We perform a rigorous error analysis for the semi-discrete scheme using energy estimates. Unlike the projection approach [21], which updates the velocity via a time evolution equation to provide an additional time step on the gradient of the pressure, the GU method updates the velocity via (1.3) without involving the time derivative. This introduces analytical complexities, as discussed in Remark 3.3. To achieve an optimal convergence rate, we construct an auxiliary "momentum" equation based on high-order consistency analysis.

The remainder of the paper is organized as follows. In Section 2, we construct first- and second-order linearly decoupled schemes, prove their unconditional energy stability, and outline the procedures for efficient implementation. In Section 3, rigorous error estimates are developed for the first-order temporal scheme, with particular attention given to the intermediate equation derived from the high-order consistency analysis due to its critical role in the estimate of velocity error. Section 4 presents numerical experiments to verify the theoretical results. Concluding remarks are given in the last section.

2. Linear and energy stable numerical schemes

In this section, we begin by introducing some key notation and some useful lemmas. Following a brief review of the GU method, we reformulate the CHNS model into an equivalent system by incorporating the MSAV method alongside the GU approach. We then construct first- and second-order linear, decoupled schemes, and give a detailed implementation procedure. Furthermore, we prove that both schemes are mass-conserving and unconditionally energy stable.

2.1. Notation and preliminaries

First, we denote $W^{m,p}(\Omega)$, with $m \geq 0$ and $1 \leq p \leq +\infty$, as the usual Sobolev space, which is equipped with the norm $\|\cdot\|_{W^{m,p}(\Omega)}$ and semi-norm $|\cdot|_{W^{m,p}(\Omega)}$. In particular, it reduces to the usual notation $H^m(\Omega)$ when $p = 2$ with the corresponding norm $\|\cdot\|_m$, and when $m = 0$, it corresponds to the classical Lebesgue space $L^p(\Omega)$ with the norm $\|\cdot\|_{L^p}$. We also denote by $(\cdot,\cdot)$ the $L^2(\Omega)$ inner product (for scalar- or vector-valued functions) with the induced L^2 norm $|\boldsymbol{v}| = (\boldsymbol{v},\boldsymbol{v})^{\frac{1}{2}}$. Let $H_0^1(\Omega)$ be the completion of $C_0^\infty(\Omega)$ with respect to the norm $\|\cdot\|_1$, and denote $H^{-1}(\Omega)$ as the dual space of $H_0^1(\Omega)$ with the norm $\|\cdot\|_{-1}$. Finally, let H be a function space defined by

$$H = \left\{\boldsymbol{v} \in L^2(\Omega): \quad \nabla\cdot\boldsymbol{v} = 0 \text{ in } \Omega, \quad \boldsymbol{v}\cdot\boldsymbol{n} = 0 \text{ on } \partial\Omega\right\}.$$

For details, we refer readers to the monographs [24,25,34].

Next, we introduce some useful lemmas.

Lemma 1 ([21,35]). *If $\boldsymbol{v} \in H_0^1(\Omega)$, then*

(i) there exists a positive constant c_1, depending only on Ω, such that

$$\|\boldsymbol{v}\| \leq c_1\|\nabla\boldsymbol{v}\|. \quad \textit{(Poincaré inequality)}$$

(ii) there holds

$$\|\nabla\cdot\boldsymbol{v}\| \leq \|\nabla\boldsymbol{v}\|. \quad \textit{(Divergence-gradient relation)}$$

Lemma 2 (Properties of trilinear form Li et al. [17]). *Define the trilinear form $b(\boldsymbol{u},\boldsymbol{v},\boldsymbol{w}) = \int_\Omega(\boldsymbol{u}\cdot\nabla)\boldsymbol{v}\cdot\boldsymbol{w}d\boldsymbol{x}$. Then,*

(i) the trilinear form is skew-symmetric with respect to its last two arguments

$$b(\boldsymbol{u},\boldsymbol{v},\boldsymbol{w}) = -b(\boldsymbol{u},\boldsymbol{w},\boldsymbol{v}), \quad \forall\boldsymbol{u}\in H, \quad \boldsymbol{v},\boldsymbol{w}\in H_0^1(\Omega),$$

thus

$$b(\boldsymbol{u},\boldsymbol{v},\boldsymbol{v}) = 0, \quad \forall\boldsymbol{u}\in H, \quad \boldsymbol{v}\in H_0^1(\Omega).$$

(ii) there exists a positive constant c_2, depending only on Ω, such that $\forall\boldsymbol{u},\boldsymbol{v}\in H\cap H_0^1(\Omega)$, and $\boldsymbol{w}\in H_0^1(\Omega)$, then

$$b(\boldsymbol{u},\boldsymbol{v},\boldsymbol{w}) \leq \begin{cases} c_2\|\boldsymbol{u}\|_1\|\boldsymbol{v}\|_1\|\boldsymbol{w}\|_1, \\ c_2\|\boldsymbol{u}\|_1^{1/2}\|\boldsymbol{u}\|^{1/2}\|\boldsymbol{v}\|_1^{1/2}\|\boldsymbol{v}\|^{1/2}\|\boldsymbol{w}\|_1. \end{cases}$$

If $\boldsymbol{u}$ or $\boldsymbol{v}$ has higher regularity, we further have

$$b(\boldsymbol{u},\boldsymbol{v},\boldsymbol{w}) \le \begin{cases} c_2\|\boldsymbol{u}\|_2\|\boldsymbol{v}\|\|\boldsymbol{w}\|_1, & \forall \boldsymbol{u}\in H^2(\Omega)\cap H, \quad \boldsymbol{v}\in H,\ \boldsymbol{w}\in H_0^1(\Omega), \\ c_2\|\boldsymbol{u}\|_2\|\boldsymbol{v}\|_1\|\boldsymbol{w}\|, & \forall \boldsymbol{u}\in H^2(\Omega)\cap H, \quad \boldsymbol{w},\boldsymbol{v}\in H_0^1(\Omega), \\ c_2\|\boldsymbol{u}\|_1\|\boldsymbol{v}\|_2\|\boldsymbol{w}\|, & \forall \boldsymbol{v}\in H^2(\Omega)\cap H, \quad \boldsymbol{u},\boldsymbol{w}\in H_0^1(\Omega), \\ c_2\|\boldsymbol{u}\|\|\boldsymbol{v}\|_2\|\boldsymbol{w}\|_1, & \forall \boldsymbol{v}\in H^2(\Omega)\cap H, \quad \boldsymbol{u}\in H,\ \boldsymbol{w}\in H_0^1(\Omega). \end{cases}$$

Lemma 3 (Discrete Grönwall inequality [36]). *Let $\{a_n, b_n, c_n, d_n, \gamma_n, \Delta t_n\}$ be nonnegative real sequences such that*

$$a_{n+1} - a_n + b_{n+1}\Delta t_{n+1} + c_{n+1}\Delta t_{n+1} - c_n\Delta t_n \le a_n d_n \Delta t_n + \gamma_{n+1}\Delta t_{n+1}$$

for all $0 \le n \le m$. Then

$$a_{m+1} + \sum_{n=0}^{m+1} b_n\Delta t_n \le \exp\Big(\sum_{n=0}^{m} d_n\Delta t_n\Big)\Big\{a_0 + (b_0+c_0)\Delta t_0 + \sum_{n=1}^{m+1}\gamma_n\Delta t_n\Big\}.$$

Lemma 4 (Interpolation inequalities [24]). *There exist positive constants c_3 and c_4 such that*

$$\|v\|_{L^\infty} \le c_3\|v\|_1^{1/2}\|v\|_2^{1/2}, \quad \|v\|_{L^4} \le c_4\|v\|^{1/2}\|v\|_1^{1/2},$$

Lemma 5 (Properties of the projection operator [37]). *Let P_H be the projection operator from $L^2(\Omega)$ into H. Then there exists a positive constant c_5, depending only on Ω, such that*

$$\|P_H\boldsymbol{u}\|_1 \le c_5\|\boldsymbol{u}\|_1, \quad \forall \boldsymbol{u}\in H^1(\Omega).$$

Throughout the paper, we use C, with or without subscript, to denote a generic positive constant, independent of Δt, which could have different values at different places.

2.2. The first-order scheme

In this subsection, we turn to constructing a first-order, linearly decoupled numerical scheme based on the GU method and MSAV-ZEC approach.

2.2.1. Construction of the GU-MSAV-BDF1 scheme

Define $E_1(\phi) = \int_\Omega F(\phi)d\boldsymbol{x}$ and assume that there exists a positive constant δ_0 such that $E_1(\phi)+\delta_0 > 0$. Then, we introduce the following two scalar auxiliary variables

$$R(t) = \sqrt{E_1(\phi)+\delta_0} \quad \text{and} \quad Q(t) = \exp(-t/T).$$

Based on the Helmholtz–Hodge decomposition (1.3) and the aforementioned scalar auxiliary variables, the GU method described in Nochetto and Pyo [28], Pyo and Shen [30], Nochetto and Pyo [35] imposes ψ as a solution to the parabolic equation

$$\psi_t - \nu\Delta\psi = -p, \tag{2.1}$$

and thus, the original model problem (1.1) can be reformulated as follows:

$$\phi_t + \frac{R}{\sqrt{E_1(\phi)+\delta_0}}(\boldsymbol{u}\cdot\nabla)\phi = M\Delta\mu, \qquad \text{in } \Omega\times J, \tag{2.2a}$$

$$\mu = -\Delta\phi + \frac{R}{\sqrt{E_1(\phi)+\delta_0}}F'(\phi), \qquad \text{in } \Omega\times J, \tag{2.2b}$$

$$R_t = \frac{1}{2\sqrt{E_1(\phi)+\delta_0}}\Big(\int_\Omega F'(\phi)\phi_t d\boldsymbol{x} + (\mu, \boldsymbol{u}\cdot\nabla\phi) - (\boldsymbol{u}, \mu\nabla\phi)\Big), \qquad \text{in } \Omega\times J, \tag{2.2c}$$

$$\boldsymbol{a}_t + \exp\big(\frac{t}{T}\big)Q(\boldsymbol{u}\cdot\nabla)\boldsymbol{u} = \nu\Delta\boldsymbol{a} + \frac{R}{\sqrt{E_1(\phi)+\delta_0}}\mu\nabla\phi, \qquad \text{in } \Omega\times J, \tag{2.2d}$$

$$\Delta\psi = -\nabla\cdot\boldsymbol{a}, \qquad \text{in } \Omega\times J, \tag{2.2e}$$

$$\boldsymbol{u} = \boldsymbol{a} + \nabla\psi, \qquad \text{in } \Omega\times J, \tag{2.2f}$$

$$Q_t = -\frac{1}{T}Q + \exp\big(\frac{t}{T}\big)(\boldsymbol{u}\cdot\nabla\boldsymbol{u}, \boldsymbol{u}), \qquad \text{in } \Omega\times J. \tag{2.2g}$$

The auxiliary variables ψ and $\boldsymbol{a}$ satisfy the following boundary conditions:

$$\partial_{\boldsymbol{n}}\psi = 0, \quad \boldsymbol{a}\cdot\boldsymbol{n} = 0, \quad \boldsymbol{a}\cdot\boldsymbol{\tau} = -\partial_{\boldsymbol{\tau}}\psi, \tag{2.3}$$

where $\boldsymbol{\tau}$ is the unit tangent vector. This set of boundary conditions has been shown to be effective in Weinan and Liu [38], Wang and Liu [39]. A primary advantage of the Gauge formulation is that it treats ψ as a non-physical variable.

To balance the nonlinear terms appearing in the CHNS system (1.1) at the discrete level, based on the ZEC idea [40], two zero terms $(\mu, \boldsymbol{u}\cdot\nabla\phi)-(\boldsymbol{u},\mu\nabla\phi)$ and $\exp\left(\frac{t}{T}\right)(\boldsymbol{u}\cdot\nabla\boldsymbol{u},\boldsymbol{u})$ are respectively added to (2.2c) and (2.2g). Furthermore, the modified energy dissipation law can be expressed as follows:

$$\frac{dE_{\text{sav}}(\phi,\boldsymbol{u},R,Q)}{dt}=-M\|\nabla\mu\|^2-\nu\|\nabla\boldsymbol{u}\|^2-\frac{1}{T}Q^2,$$

where the energy functional is defined as

$$E_{\text{sav}}(\phi,\boldsymbol{u},R,Q)=\int_\Omega\frac{1}{2}\left(|\boldsymbol{u}|^2+|\nabla\phi|^2\right)d\boldsymbol{x}+\frac{1}{2}Q^2+R^2. \tag{2.4}$$

Thereafter, we shall construct efficient numerical schemes for (2.2) that are unconditionally energy stable with respect to (2.4).

For a given positive integer N_t, we set $t_n=n\Delta t$ for $n=0,1,\dots,N_t$, and denote $J_n=[t_n,t_{n+1}]$. Let $(\cdot)^n$ represent the value of the variable at $t=t_n$, and define $\delta_t v^{n+1}=v^{n+1}-v^n$. Upon discretizing (2.2d)–(2.2f) in time via the first-order backward differentiation formula (BDF1) and implicit-explicit approach, we obtain

$$\frac{\boldsymbol{a}^{n+1}-\boldsymbol{a}^n}{\Delta t}+\exp\left(\frac{t_{n+1}}{T}\right)Q^{n+1}(\boldsymbol{u}^n\cdot\nabla)\boldsymbol{u}^n=\nu\Delta\boldsymbol{a}^{n+1}+\frac{R^{n+1}}{\sqrt{E_1(\phi^n)+\delta_0}}\mu^n\nabla\phi^n, \tag{2.5a}$$

$$\Delta\psi^{n+1}=-\nabla\cdot\boldsymbol{a}^{n+1}, \tag{2.5b}$$

$$\boldsymbol{u}^{n+1}=\boldsymbol{a}^{n+1}+\nabla\psi^{n+1}, \tag{2.5c}$$

with

$$\partial_{\boldsymbol{n}}\psi^{n+1}=0,\quad \boldsymbol{a}^{n+1}\cdot\boldsymbol{n}=0,\quad \boldsymbol{a}^{n+1}\cdot\boldsymbol{\tau}=-\partial_{\boldsymbol{\tau}}\psi^n.$$

We point out that the momentum equation in (2.5) is linear with respect to $\boldsymbol{a}^{n+1}$, and that the explicit boundary condition $\boldsymbol{a}^{n+1}\cdot\boldsymbol{\tau}=-\partial_{\boldsymbol{\tau}}\psi^n$ plays a crucial role in decoupling the equations for $\boldsymbol{a}^{n+1}$ and ψ^{n+1} [39]. However, the calculation of $\partial_{\boldsymbol{\tau}}\psi^n$, which involves numerical differentiation, leads to a loss of accuracy, particularly at the corners of $\partial\Omega$ where $\boldsymbol{\tau}$ is not well defined. It also makes the use of finite element methods problematic for spatial discretization.

To overcome the aforementioned shortcomings, Nochetto et al. [28] proposed the GU method for the Navier–Stokes equations, which combines the advantages of the Gauge method and the Uzawa iteration while avoiding boundary derivatives. In the SAV framework, we also introduce a new vector variable $\widehat{\boldsymbol{u}}^{n+1}$ with zero boundary value, defined by

$$\widehat{\boldsymbol{u}}^{n+1}=\boldsymbol{a}^{n+1}+\nabla\psi^n. \tag{2.6}$$

Substituting (2.6) into (2.5), we have

$$\frac{\widehat{\boldsymbol{u}}^{n+1}-\boldsymbol{u}^n}{\Delta t}+\exp\left(\frac{t_{n+1}}{T}\right)Q^{n+1}(\boldsymbol{u}^n\cdot\nabla)\boldsymbol{u}^n=\nu\Delta\widehat{\boldsymbol{u}}^{n+1}-\nu\nabla\Delta\psi^n+\frac{R^{n+1}}{\sqrt{E_1(\phi^n)+\delta_0}}\mu^n\nabla\phi^n.$$

To address the challenge posed by $\nabla\Delta\psi^n$, Nochetto et al. [28] introduced the auxiliary variables $s^{n+1}=\Delta\psi^{n+1}$ and $\rho^{n+1}=\delta_t\psi^{n+1}$, so that it holds that:

$$s^{n+1}=s^n-\nabla\cdot\widehat{\boldsymbol{u}}^{n+1},\quad -\Delta\rho^{n+1}=\nabla\cdot\widehat{\boldsymbol{u}}^{n+1}.$$

This formulation not only simplifies the treatment of high-order derivatives but also preserves the structure of the Uzawa iteration, making the scheme more efficient and consistent for numerical implementation.

Remark 2.1. The projection methods either require a suitable initial value or artificial boundary conditions on pressure, which can introduce complications [23]. However, with the Gauge formula (2.1), we impose initial and boundary conditions on the non-physical variable ψ^{n+1} instead of pressure p^{n+1}, ensuring that the conditions imposed on ψ^{n+1} do not affect the value of p^{n+1}. Regarding the initial value s^0, there is no specific restriction on s, so we simply take $s^0=0$.

Remark 2.2. Note that in (2.5), the convection term $(u\cdot\nabla)u$, the coupling term $\mu\nabla\phi$, and the energy-related term $E_1(\phi)$ are treated explicitly, while the viscous diffusion term Δa and the auxiliary variables R, Q are handled implicitly. This implicit-explicit combination constitutes a fully linear scheme, which can ensure both stability and computational efficiency, particularly for long-term energy stability with a relatively larger time stepsize. Notably, equation (2.5) only serves as an intermediate formulation for the derivation of the Gauge–Uzawa method, and the proposed scheme in (2.7d) also retains the same implicit-explicit treatments.

Finally, we obtain the GU-MSAV-BDF1 scheme for the CHNS system: given $(\phi,\mu,\widehat{\boldsymbol{u}},\rho,s,\boldsymbol{u},R,Q)^n$, solve $(\phi,\mu,\widehat{\boldsymbol{u}},\rho,s,\boldsymbol{u},R,Q)^{n+1}$ via

$$\frac{\phi^{n+1}-\phi^n}{\Delta t}+\frac{R^{n+1}}{\sqrt{E_1(\phi^n)+\delta_0}}(\boldsymbol{u}^n\cdot\nabla)\phi^n=M\Delta\mu^{n+1},\quad \left.\frac{\partial\phi^{n+1}}{\partial\boldsymbol{n}}\right|_{\partial\Omega}=0, \tag{2.7a}$$

$$\mu^{n+1}=-\Delta\phi^{n+1}+\frac{R^{n+1}}{\sqrt{E_1(\phi^n)+\delta_0}}F'(\phi^n),\quad \left.\frac{\partial\mu^{n+1}}{\partial\boldsymbol{n}}\right|_{\partial\Omega}=0, \tag{2.7b}$$

$$\frac{R^{n+1}-R^n}{\Delta t}=\frac{1}{2\sqrt{E_1(\phi^n)+\delta_0}}\left(\left(F'(\phi^n),\frac{\phi^{n+1}-\phi^n}{\Delta t}\right)+\left(\mu^{n+1},\boldsymbol{u}^n\cdot\nabla\phi^n\right)\right)-\frac{1}{2\sqrt{E_1(\phi^n)+\delta_0}}\left(\widehat{\boldsymbol{u}}^{n+1},\mu^n\nabla\phi^n\right), \tag{2.7c}$$

$$\frac{\widehat{\boldsymbol{u}}^{n+1}-\boldsymbol{u}^n}{\Delta t}+\exp\left(\frac{t_{n+1}}{T}\right)Q^{n+1}\boldsymbol{u}^n\cdot\nabla\boldsymbol{u}^n-\nu\Delta\widehat{\boldsymbol{u}}^{n+1}+\nu\nabla s^n=\frac{R^{n+1}}{\sqrt{E_1(\phi^n)+\delta_0}}\mu^n\nabla\phi^n,\quad \widehat{\boldsymbol{u}}^{n+1}\,|_{\partial\Omega}=0, \tag{2.7d}$$

$$-\Delta\rho^{n+1}=\nabla\cdot\widehat{\boldsymbol{u}}^{n+1},\quad \left.\frac{\partial\rho^{n+1}}{\partial\boldsymbol{n}}\right|_{\partial\Omega}=0, \tag{2.7e}$$

$$s^{n+1}=s^n-\nabla\cdot\widehat{\boldsymbol{u}}^{n+1}, \tag{2.7f}$$

$$\boldsymbol{u}^{n+1}=\widehat{\boldsymbol{u}}^{n+1}+\nabla\rho^{n+1}, \tag{2.7g}$$

$$\frac{Q^{n+1}-Q^n}{\Delta t}=-\frac{Q^{n+1}}{T}+\exp\left(\frac{t_{n+1}}{T}\right)(\boldsymbol{u}^n\cdot\nabla\boldsymbol{u}^n,\widehat{\boldsymbol{u}}^{n+1}). \tag{2.7h}$$

Note that the added term $\left(\mu^{n+1},\boldsymbol{u}^n\cdot\nabla\phi^n\right)-\left(\widehat{\boldsymbol{u}}^{n+1},\mu^n\nabla\phi^n\right)$ in (2.7c) serves as a first-order approximation to the corresponding continuous identity $(\mu,\boldsymbol{u}\cdot\nabla\phi)-(\boldsymbol{u},\mu\nabla\phi)=0$. However, it plays an important role in the energy stability analysis.

Remark 2.3. Although the relations $R(t)=\sqrt{E_1(\phi)+\delta_0}$ and $Q(t)=\exp(-\frac{t}{T})$ hold in the continuous case, the time-discrete variables may not satisfy these relations exactly, owing to truncation error in time. This drawback may compromise the consistency between the numerical scheme and the original model. For simplicity, we only present the relaxation step for R [32,33]; the correction for Q can be handled in an analogous manner. Specifically, we rename the auxiliary variable computed in (2.7c) as $\widehat{R}^{n+1}$, and introduce an additional correction step to update R^{n+1} as follows:

Update the scalar auxiliary variable R^{n+1} via a relaxation step as

$$R^{n+1}=\kappa_0\widehat{R}^{n+1}+\left(1-\kappa_0\right)\sqrt{E_1(\phi^{n+1})+\delta_0}. \tag{2.8}$$

Here κ_0 is determined by the following optimization problem

$$\kappa_0=\min_{\kappa\in[0,1]}\kappa,\quad \text{s.t. } |R^{n+1}|^2-|\widehat{R}^{n+1}|^2\le\Delta t\eta\|\nabla\mu^{n+1}\|^2, \tag{2.9}$$

where $\eta\in(0,M]$ is an artificial parameter that can be manually assigned. In fact, the above optimization problem can be simplified as

$$\kappa_0=\min_{\kappa\in[0,1]}\kappa,\quad \text{s.t. } f(\kappa)=a\kappa^2+b\kappa+c\le 0,$$

where the coefficients are

$$\begin{cases} a=\left(\widehat{R}^{n+1}-\sqrt{E_1(\phi^{n+1})+\delta_0}\right)^2,\\ b=2\sqrt{E_1(\phi^{n+1})+\delta_0}\left(\widehat{R}^{n+1}-\sqrt{E_1(\phi^{n+1})+\delta_0}\right),\\ c=E_1(\phi^{n+1})+\delta_0-|\widehat{R}^{n+1}|^2-\Delta t\eta\|\nabla\mu^{n+1}\|^2. \end{cases}$$

If $a=0$, we set $\kappa_0=0$. Otherwise, noting that $f(1)=a+b+c=-\Delta t\eta\|\nabla\mu^{n+1}\|^2\le 0$, the quadratic function $f(\kappa)$ has at least one real root. Then the quadratic optimization problem can be solved as

$$\kappa_0=\max\left\{0,(-b-\sqrt{b^2-4ac})/(2a)\right\}.$$

Remark 2.4. One may compute the pressure whenever necessary as

$$p^{n+1}=\nu s^{n+1}-\frac{\rho^{n+1}}{\Delta t}.$$

Furthermore, applying the divergence operator to (2.7g) and noticing (2.7e), we obtain that

$$\nabla\cdot\boldsymbol{u}^{n+1}=0, \tag{2.10}$$

which is consistent with the original model and is also critical in our subsequent analysis.

2.2.2. *Efficient implementations*

We observe that the above scheme is linear, but still a coupled system via the auxiliary variables. To compute it efficiently, we propose the following implementation process, leveraging the scalar auxiliary variables to eliminate all nonlinear terms and achieve the desired fully decoupled structure.

Denote

$$\xi_1^{n+1}=\frac{R^{n+1}}{\sqrt{E_1(\phi^n)+\delta_0}},\quad \xi_2^{n+1}=\exp\left(\frac{t_{n+1}}{T}\right)Q^{n+1}. \tag{2.11}$$

Then, we can transform the system (2.7a)–(2.7d) and (2.7h) into the following system:

$$\begin{pmatrix} I & -M\Delta t\Delta & 0 & \Delta t(\boldsymbol{u}^n\cdot\nabla)\phi^n & 0\\ \Delta & I & 0 & -F'(\phi^n) & 0\\ 0 & 0 & I-\nu\Delta t\Delta & -\Delta t\mu^n\nabla\phi^n & \Delta t\boldsymbol{u}^n\cdot\nabla\boldsymbol{u}^n\\ \mathcal{L}_1 & \mathcal{L}_2 & \mathcal{L}_3 & \sqrt{E_1(\phi^n)+\delta_0} & 0\\ 0 & 0 & \mathcal{L}_4 & 0 & \exp(-\frac{t_{n+1}}{T})(1+\frac{\Delta t}{T}) \end{pmatrix}\begin{pmatrix} \phi^{n+1}\\ \mu^{n+1}\\ \widehat{\boldsymbol{u}}^{n+1}\\ \xi_1^{n+1}\\ \xi_2^{n+1} \end{pmatrix}=\begin{pmatrix} \phi^n\\ 0\\ \boldsymbol{u}^n-\Delta t\nu\nabla s^n\\ R^n-\frac{(F'(\phi^n),\phi^n)}{2\sqrt{E_1(\phi^n)+\delta_0}}\\ Q^n \end{pmatrix}, \tag{2.12}$$

where the operators $\mathcal{L}_i$ $(i = 1, \dots, 4)$ are defined as

$$\mathcal{L}_1 \boldsymbol{v} = -\frac{1}{2\sqrt{E_1(\phi^n)+\delta_0}}\big(F'(\phi^n), \boldsymbol{v}\big), \quad \mathcal{L}_2 \boldsymbol{v} = -\frac{\Delta t}{2\sqrt{E_1(\phi^n)+\delta_0}}\big(\boldsymbol{u}^n \cdot \nabla \phi^n, \boldsymbol{v}\big),$$
$$\mathcal{L}_3 \boldsymbol{v} = \frac{\Delta t}{2\sqrt{E_1(\phi^n)+\delta_0}}\big(\mu^n \nabla \phi^n, \boldsymbol{v}\big), \quad \mathcal{L}_4 \boldsymbol{v} = -\Delta t \exp\Big(\frac{t_{n+1}}{T}\Big)(\boldsymbol{u}^n \cdot \nabla \boldsymbol{u}^n, \boldsymbol{v}).$$

Based on the Gaussian elimination method, we propose several strategies to efficiently solve the system 2.12 and subsequently solve the scheme (2.7).

Method I. To clarify this process, we eliminate ξ_1^{n+1} and ξ_2^{n+1} from the system given in 2.12, resulting in a reduced system solely with the variables of ϕ^{n+1}, μ^{n+1} and $\widehat{\boldsymbol{u}}^{n+1}$:

$$\begin{pmatrix} I & -M\Delta t\Delta & 0 \\ \Delta & I & 0 \\ 0 & 0 & I - \nu\Delta t\Delta \end{pmatrix}\begin{pmatrix} \phi^{n+1} \\ \mu^{n+1} \\ \widehat{\boldsymbol{u}}^{n+1} \end{pmatrix} = \begin{pmatrix} \phi^n - \xi_1^{n+1}\Delta t(\boldsymbol{u}^n \cdot \nabla)\phi^n \\ -\xi_1^{n+1} F'(\phi^n) \\ \boldsymbol{u}^n - \Delta t \nu \nabla s^n + \xi_1^{n+1}\Delta t \mu^n \nabla \phi^n - \xi_2^{n+1}\Delta t \boldsymbol{u}^n \cdot \nabla \boldsymbol{u}^n \end{pmatrix}. \tag{2.13}$$

From (2.13), it is evident that ϕ^{n+1} and μ^{n+1} are independent of ξ_2^{n+1}. To efficiently solve (2.13), based on (2.11), we split ϕ^{n+1}, μ^{n+1} and $\widehat{\boldsymbol{u}}^{n+1}$ into a set of linear combinations with respect to ξ_1^{n+1} and ξ_2^{n+1} that reads as

$$\phi^{n+1} = \phi_0^{n+1} + \xi_1^{n+1}\phi_1^{n+1}, \tag{2.14a}$$
$$\mu^{n+1} = \mu_0^{n+1} + \xi_1^{n+1}\mu_1^{n+1}, \tag{2.14b}$$
$$\widehat{\boldsymbol{u}}^{n+1} = \widehat{\boldsymbol{u}}_0^{n+1} + \xi_1^{n+1}\widehat{\boldsymbol{u}}_1^{n+1} + \xi_2^{n+1}\widehat{\boldsymbol{u}}_2^{n+1}. \tag{2.14c}$$

Then, inserting (2.14a)–(2.14c) into (2.13), and comparing the coefficients of ξ_1^{n+1} and ξ_2^{n+1}, we can obtain $\{\phi_i^{n+1}, \mu_i^{n+1}\}_{i=0}^1$, $\{\widehat{\boldsymbol{u}}_i^{n+1}\}_{i=0}^2$ and $\rho^{n+1}, s^{n+1}, \boldsymbol{u}^{n+1}$ as follows:

Step 1. Compute $\{\phi_i^{n+1}, \mu_i^{n+1}\}_{i=0}^1$ via

$$\phi_0^{n+1} - \phi^n = M\Delta t\Delta\mu_0^{n+1}, \mu_0^{n+1} = -\Delta\phi_0^{n+1}, \quad \frac{\partial \phi_0^{n+1}}{\partial \boldsymbol{n}}\bigg|_{\partial\Omega} = \frac{\partial \mu_0^{n+1}}{\partial \boldsymbol{n}}\bigg|_{\partial\Omega} = 0, \tag{2.15a}$$

$$\phi_1^{n+1} + \Delta t(\boldsymbol{u}^n \cdot \nabla)\phi^n = M\Delta t\Delta\mu_1^{n+1}, \quad \mu_1^{n+1} = -\Delta\phi_1^{n+1} + F'(\phi^n), \quad \frac{\partial \phi_1^{n+1}}{\partial \boldsymbol{n}}\bigg|_{\partial\Omega} = \frac{\partial \mu_1^{n+1}}{\partial \boldsymbol{n}}\bigg|_{\partial\Omega} = 0. \tag{2.15b}$$

Step 2. Solve $\{\widehat{\boldsymbol{u}}_i^{n+1}\}_{i=0}^2$ using

$$\widehat{\boldsymbol{u}}_0^{n+1} - \boldsymbol{u}^n - \nu\Delta t\Delta\widehat{\boldsymbol{u}}_0^{n+1} + \nu\Delta t\nabla s^n = 0, \quad \widehat{\boldsymbol{u}}_0^{n+1}\big|_{\partial\Omega} = 0, \tag{2.16a}$$

$$\widehat{\boldsymbol{u}}_1^{n+1} - \nu\Delta t\Delta\widehat{\boldsymbol{u}}_1^{n+1} = \Delta t\mu^n\nabla\phi^n, \quad \widehat{\boldsymbol{u}}_1^{n+1}\big|_{\partial\Omega} = 0, \tag{2.16b}$$

$$\widehat{\boldsymbol{u}}_2^{n+1} + \Delta t\boldsymbol{u}^n \cdot \nabla\boldsymbol{u}^n - \nu\Delta t\Delta\widehat{\boldsymbol{u}}_2^{n+1} = 0, \quad \widehat{\boldsymbol{u}}_2^{n+1}\big|_{\partial\Omega} = 0. \tag{2.16c}$$

Step 3. Solve the auxiliary variables $\{\xi_i^{n+1}\}_{i=1}^2$. Using the split form for the variables ϕ^{n+1}, μ^{n+1} and $\widehat{\boldsymbol{u}}^{n+1}$, one can rewrite (2.7c) and (2.7h) as the following 2×2 linear algebraic system:

$$\begin{cases} A_1\xi_1 + A_2\xi_2 = A_0, \\ B_1\xi_1 + B_2\xi_2 = B_0, \end{cases} \tag{2.17}$$

where

$$\begin{cases} A_0 = \frac{R^n}{\Delta t} + \frac{1}{2\sqrt{E_1(\phi^n)+\delta_0}}\Big(\big(F'(\phi^n), \frac{\phi_0^{n+1}-\phi^n}{\Delta t}\big) + \big(\mu_0^{n+1}, \boldsymbol{u}^n \cdot \nabla\phi^n\big) - \big(\widehat{\boldsymbol{u}}_0^{n+1}, \mu^n\nabla\phi^n\big)\Big), \\ A_1 = \frac{\sqrt{E_1(\phi^n)+\delta_0}}{\Delta t} - \frac{1}{2\sqrt{E_1(\phi^n)+\delta_0}}\Big(\big(F'(\phi^n), \frac{\phi_1^{n+1}}{\Delta t}\big) + \big(\mu_1^{n+1}, \boldsymbol{u}^n \cdot \nabla\phi^n\big) - \big(\widehat{\boldsymbol{u}}_1^{n+1}, \mu^n\nabla\phi^n\big)\Big), \\ A_2 = \frac{1}{2\sqrt{E_1(\phi^n)+\delta_0}}\big(\widehat{\boldsymbol{u}}_2^{n+1}, \mu^n\nabla\phi^n\big), \\ B_0 = \frac{Q^n}{\Delta t} + \exp\Big(\frac{t_{n+1}}{T}\Big)\big(\boldsymbol{u}^n \cdot \nabla\boldsymbol{u}^n, \widehat{\boldsymbol{u}}_0^{n+1}\big), \\ B_1 = -\exp\Big(\frac{t_{n+1}}{T}\Big)\big(\boldsymbol{u}^n \cdot \nabla\boldsymbol{u}^n, \widehat{\boldsymbol{u}}_1^{n+1}\big), \\ B_2 = \frac{\exp\left(-\frac{t_{n+1}}{T}\right)}{\Delta t} + \frac{\exp\left(-\frac{t_{n+1}}{T}\right)}{T} - \exp\Big(\frac{t_{n+1}}{T}\Big)\big(\boldsymbol{u}^n \cdot \nabla\boldsymbol{u}^n, \widehat{\boldsymbol{u}}_2^{n+1}\big). \end{cases}$$

Step 4. Once the values of $\{\phi_i^{n+1}, \mu_i^{n+1}\}_{i=0}^1$, $\{\widehat{\boldsymbol{u}}_i^{n+1}\}_{i=0}^2$ and $\{\xi_i^{n+1}\}_{i=1}^2$ are known, update ϕ^{n+1}, μ^{n+1} and $\widehat{\boldsymbol{u}}^{n+1}$ from (2.14a)–(2.14c). In addition, calculate ρ^{n+1}, s^{n+1} via (2.7e) and (2.7f), respectively.

Step 5. Calculate $\boldsymbol{u}^{n+1}$ using (2.7g) and compute R^{n+1} and Q^{n+1} by (2.11).

Remark 2.5. Using arguments analogous to those in Li and Shen [21], we can demonstrate that the algebraic system (2.17) admits a unique solution. At each time step, the computational cost involves solving a Cahn–Hilliard solver with the same constant coefficients

in (2.15a) and (2.15b), three linear Poisson-type equations with the same constant coefficients in (2.16a)–(2.16c) and determine $\{\xi_i^{n+1}\}_{i=1}^2$ by solving a 2×2 linear algebraic system (2.17) with negligible computational cost. In Step 4, we only need to solve one elliptic equation. In summary, the decoupling of all equations, combined with the constant-coefficient nature of the system, leads to highly efficient calculations in practice.

We summarize Method I in Algorithm 1 for the scheme (2.7), which is the method utilized in our numerical experiments.

Algorithm 1 Efficient Implementation of the GU-MSAV-BDF1 Scheme (2.7) Based on Method I.

Input: Initial values $\phi^0, \boldsymbol{u}^0, R^0 = \sqrt{E_1(\phi^0)+\delta_0}, Q^0 = 1$ and $s^0 := 0$.
Output: Numerical solutions $\left(\phi, \mu, \widehat{\boldsymbol{u}}, \rho, s, \boldsymbol{u}, R, Q\right)^{n+1}$.
for $n = 0$ to $N_t - 1$ **do**
  Step 1: Compute $\{\phi_i^{n+1}, \mu_i^{n+1}\}_{i=0}^1$ via (2.15a)–(2.15b).
  Step 2: Solve $\{\widehat{\boldsymbol{u}}_i^{n+1}\}_{i=0}^2$ using (2.16a)–(2.16c).
  Step 3: Determine $\{\xi_i^{n+1}\}_{i=1}^2$ by solving (2.17).
  Step 4: Update ϕ^{n+1}, μ^{n+1} and $\widehat{\boldsymbol{u}}^{n+1}$ via (2.14a)–(2.14c), then compute ρ^{n+1}, s^{n+1} through (2.7e)–(2.7f).
  Step 5: Calculate $\boldsymbol{u}^{n+1}$ from (2.7g) and update R^{n+1}, Q^{n+1} using (2.11).
end for

Method II. Alternatively, we can directly solve ϕ^{n+1} and μ^{n+1} without decomposing them, and decompose only $\widehat{\boldsymbol{u}}^{n+1}$ in terms of ξ_1. By employing Gaussian elimination method, we can simplify (2.13) into the following form:

$$\begin{pmatrix} \mathcal{M}_1 & \mathcal{M}_2 & 0 & 0 & 0 \\ \mathcal{M}_3 & \mathcal{M}_4 & 0 & 0 & 0 \\ 0 & 0 & \mathcal{M}_5 & -\Delta t\mu^n\nabla\phi^n & 0 \\ \mathcal{L}_1 & \mathcal{L}_2 & 0 & Z^n & 0 \\ 0 & 0 & \mathcal{L}_4 & 0 & \exp(-\frac{t_{n+1}}{T})(1+\frac{\Delta t}{T}) \end{pmatrix} \begin{pmatrix} \phi^{n+1} \\ \mu^{n+1} \\ \widehat{\boldsymbol{u}}^{n+1} \\ \xi_1^{n+1} \\ \xi_2^{n+1} \end{pmatrix} = \begin{pmatrix} \phi^n - \frac{\Delta t}{Z^n} S^n (\boldsymbol{u}^n\cdot\nabla)\phi^n \\ \frac{F'(\phi^n)}{Z^n} S^n \\ \boldsymbol{U}^n \\ S^n \\ Q^n \end{pmatrix}, \tag{2.18}$$

where the operator $\mathcal{M}_5$ is defined as $\mathcal{M}_5\boldsymbol{v} = \boldsymbol{v} - \nu\Delta t\Delta\boldsymbol{v} - \Delta t\frac{(\boldsymbol{u}^n\cdot\nabla)\boldsymbol{u}^n\mathcal{L}_4\boldsymbol{v}}{\exp(-\frac{t_{n+1}}{T})(1+\frac{\Delta t}{T})}$, and $Z^n = \sqrt{E_1(\phi^n)+\delta_0} + \Delta t\mathcal{L}_3\mathcal{M}_5^{-1}\mu^n\nabla\phi^n$. By the definition of $\mathcal{L}_4$, $\mathcal{M}_5$ is positive definite. Furthermore, one can show that $\Delta t\mathcal{L}_3\mathcal{M}_5^{-1}\mu^n\nabla\phi^n \geq 0$ and $Z^n > 0$. Based on $\mathcal{M}_5$ and Z^n, define the other operators $\mathcal{M}_i$ $(i = 1, \dots, 4)$ and quantities as follows:

$$\mathcal{M}_1 : \boldsymbol{v} \to \boldsymbol{v} - \frac{\Delta t}{Z^n}(\boldsymbol{u}^n\cdot\nabla)\phi^n\mathcal{L}_1\boldsymbol{v}, \quad \mathcal{M}_2 : \boldsymbol{v} \to -M\Delta t\Delta\boldsymbol{v} - \frac{\Delta t}{Z^n}(\boldsymbol{u}^n\cdot\nabla)\phi^n\mathcal{L}_2\boldsymbol{v},$$
$$\mathcal{M}_3 : \boldsymbol{v} \to \Delta\boldsymbol{v} + \frac{1}{Z^n}F'(\phi^n)\mathcal{L}_1\boldsymbol{v}, \quad \mathcal{M}_4 : \boldsymbol{v} \to \boldsymbol{v} + \frac{1}{Z^n}F'(\phi^n)\mathcal{L}_2\boldsymbol{v},$$

and

$$\boldsymbol{U}^n = \boldsymbol{u}^n - \Delta t\nu\big(s^n - \Delta t\frac{Q^n((\boldsymbol{u}^n\cdot\nabla)\boldsymbol{u}^n)}{\exp(-\frac{t_{n+1}}{T})(1+\frac{\Delta t}{T})},$$
$$S^n = R^n - \frac{(F'(\phi^n),\phi^n)}{2\sqrt{E_1(\phi^n)+\delta_0}} - \mathcal{L}_3\mathcal{M}_5^{-1}\boldsymbol{U}^n.$$

From the system (2.18), the following observations are immediate:

- The modified phase field subsystem is decoupled from ξ_1^{n+1} and ξ_2^{n+1}.
- The velocity $\widehat{\boldsymbol{u}}^{n+1}$ only depends on ξ_1^{n+1}.
- The velocity and the phase field system are decoupled.

Based on the above observations, we propose the following solution strategy:

Step 1. Solve $\{\widehat{\boldsymbol{u}}_i^{n+1}\}_{i=0}^1$ using:

$$\mathcal{M}_5\widehat{\boldsymbol{u}}_0^{n+1} = \boldsymbol{U}^n, \quad \mathcal{M}_5\widehat{\boldsymbol{u}}_1^{n+1} = \Delta t\mu^n\nabla\phi^n. \tag{2.19}$$

Compute Z^n and S^n as:

$$Z^n = \sqrt{E_1(\phi^n)+\delta_0} + \mathcal{L}_3\widehat{\boldsymbol{u}}_1^{n+1}, \quad S^n = R^n - \frac{(F'(\phi^n),\phi^n)}{2\sqrt{E_1(\phi^n)+\delta_0}} - \mathcal{L}_3\widehat{\boldsymbol{u}}_0^{n+1}. \tag{2.20}$$

Step 2. Solve the decoupled Cahn–Hilliard-type system for ϕ^{n+1}, μ^{n+1}:

$$\begin{pmatrix} \mathcal{M}_1 & \mathcal{M}_2 \\ \mathcal{M}_3 & \mathcal{M}_4 \end{pmatrix} \begin{pmatrix} \phi^{n+1} \\ \mu^{n+1} \end{pmatrix} = \begin{pmatrix} \phi^n - \frac{\Delta t}{Z^n} S^n (\boldsymbol{u}^n\cdot\nabla)\phi^n \\ \frac{F'(\phi^n)}{Z^n} S^n \end{pmatrix}. \tag{2.21}$$

Step 3. Compute ξ_1^{n+1} via

$$\xi_1^{n+1} = \frac{1}{Z^n}\left(-\mathcal{L}_1\phi^{n+1} - \mathcal{L}_2\mu^{n+1} + S^n\right). \tag{2.22}$$

Using the computed $\{\widehat{\boldsymbol{u}}_i^{n+1}\}_{i=1}^2$ and ξ_1^{n+1}, update $\widehat{\boldsymbol{u}}^{n+1}$ by

$$\widehat{\boldsymbol{u}}^{n+1} = \widehat{\boldsymbol{u}}_0^{n+1} + \xi_1^{n+1}\widehat{\boldsymbol{u}}_1^{n+1}. \tag{2.23}$$

Step 4. Solve for ξ_2^{n+1} through

$$\xi_2^{n+1} = \exp\left(\frac{t_{n+1}}{T}\right)\frac{T}{T+\Delta t}\left(-\mathcal{L}_4\widehat{\boldsymbol{u}}^{n+1} + Q^n\right). \tag{2.24}$$

Once ξ_1^{n+1} and ξ_2^{n+1} have been computed, update R^{n+1} and Q^{n+1} using (2.11).

Step 5. Solve ρ^{n+1}, s^{n+1} through (2.7e) and (2.7f), respectively. Finally, compute $\boldsymbol{u}^{n+1}$ from (2.7g).

Remark 2.6. For the velocity step, Method I solves three Poisson problems that share the same constant-coefficient matrix for (2.16a)–(2.16c). Therefore, matrix assembly and factorization need to be implemented only once for all time levels. Conversely, Method II involves solving only two Poisson-type systems; however, this approach necessitates the use of time-dependent coefficient matrices (2.19). The time dependence arises only through low-rank, symmetric positive semidefinite outer-product updates induced by the discrete convection and SAV couplings. In particular,

$$\mathcal{M}_5 = \mathcal{A} + \frac{(\Delta t)^2 e^{2t_{n+1}/T}}{1+\Delta t/T}\,\boldsymbol{w}\,\boldsymbol{w}^\top, \quad \mathcal{A} = I - \nu\Delta t\Delta, \quad \boldsymbol{w} = (\boldsymbol{u}^n\cdot\nabla)\boldsymbol{u}^n,$$

i.e., $\mathcal{M}_5$ can be viewed as a rank-one, symmetric positive semidefinite perturbation of the constant-coefficient operator $\mathcal{A}$. Accordingly, it admits an efficient evaluation of $\mathcal{M}_5^{-1}$ via the Sherman-Morrison-Woodbury identity [41]. An analogous low-rank structure also holds for the Cahn–Hilliard substep in (2.21). In our experiments, Method I performs better than Method II with respect to the computational efficiency; therefore, Method I is adopted for all subsequent simulations.

Remark 2.7. While Method II is less computationally efficient than Method I, it nevertheless reveals the stabilization mechanism inherent in the SAV formulation. Compared to the traditional fully explicit time discretization method, the SAV couplings contribute several symmetric positive semidefinite terms at the discrete level, including for the Navier–Stokes and Cahn–Hilliard components. For example, eliminating the SAV variable ξ_2 adds $\frac{\Delta t^2 e^{2t_{n+1}/T}}{1+\Delta t/T}(\boldsymbol{u}^n\cdot\nabla)\boldsymbol{u}^n(\boldsymbol{u}^n\cdot\nabla\boldsymbol{u}^n,\cdot)$ to $\mathcal{M}_5$. More importantly, Method II motivates multiplying these terms by a suitable stabilization factor θ to enhance stabilization; see also the analogous treatment for the Navier–Stokes equations in Lan et al. [42].

Method II is summarized in Algorithm 2.

Algorithm 2 Efficient Implementation of the GU-MSAV-BDF1 Scheme (2.7) Based on Method II.

Input: Initial values $\phi^0, \boldsymbol{u}^0, R^0 = \sqrt{E_1(\phi^0)+\delta_0}, Q^0 = 1$ and $s^0 := 0$.
Output: Numerical solutions $\left(\phi, \mu, \widehat{\boldsymbol{u}}, \rho, s, \boldsymbol{u}, p, R, Q\right)^{n+1}$.
for $n = 0$ to $N_t - 1$ **do**
  Step 1: Calculate $\{\widehat{\boldsymbol{u}}_i^{n+1}\}_{i=0}^1$ from (2.19), and Z^n, S^n from (2.20).
  Step 2: Solve ϕ^{n+1}, μ^{n+1} via the Cahn–Hilliard-type system (2.21).
  Step 3: Determine the auxiliary variable ξ_1^{n+1} using (2.22), then update $\widehat{\boldsymbol{u}}^{n+1}$ via (2.23).
  Step 4: Calculate ξ_2^{n+1} through (2.24) and then compute R^{n+1} and Q^{n+1} using (2.11).
  Step 5: Solve ρ^{n+1}, s^{n+1} through (2.7e) and (2.7f), respectively. Finally, solve $\boldsymbol{u}^{n+1}$ from (2.7g).
end for

2.2.3. Mass conservation and energy stability

Now, we turn to prove that the GU-MSAV-BDF1 scheme (2.7) is mass-conserving and unconditionally energy stable.

Theorem 2.8. *The time-discrete GU-MSAV-BDF1 scheme (2.7) satisfies the discrete mass conservation law:*

$$(\phi^{n+1}, 1) = (\phi^n, 1), \quad \forall\ 0 \le n \le N_t - 1.$$

Moreover, the discrete modified energy dissipation law holds in the sense that

$$E_{\mathrm{num},1}^{n+1}(\phi, \boldsymbol{u}, R, Q, s) - E_{\mathrm{num},1}^{n}(\phi, \boldsymbol{u}, R, Q, s) \le -M\Delta t\|\nabla\mu^{n+1}\|^2 - \frac{\nu}{2}\Delta t\|\nabla\widehat{\boldsymbol{u}}^{n+1}\|^2 - \frac{\Delta t}{T}|Q^{n+1}|^2,$$

where

$$E_{\mathrm{num},1}^{n+1}(\phi, \boldsymbol{u}, R, Q, s) = \frac{1}{2}\|\nabla\phi^{n+1}\|^2 + |R^{n+1}|^2 + \frac{1}{2}\|\boldsymbol{u}^{n+1}\|^2 + \frac{1}{2}\nu\Delta t\|s^{n+1}\|^2 + \frac{1}{2}|Q^{n+1}|^2. \tag{2.25}$$

Proof. First, multiplying (2.7a) by Δt, and then taking the inner product with 1 and applying the zero boundary conditions and $\nabla \cdot \boldsymbol{u}^{n+1} = 0$, we have

$$(\phi^{n+1} - \phi^n, 1) = 0.$$

Therefore, the discrete mass conservation law is proved.

Next, we proceed to show that the scheme also satisfies the discrete energy dissipation law. Taking the inner products of (2.7a) with μ^{n+1}, (2.7b) with $\frac{1}{\Delta t}\delta_t\phi^{n+1}$ respectively and multiplying (2.7c) with $2R^{n+1}$, we can obtain

$$\frac{\|\nabla\phi^{n+1}\|^2 - \|\nabla\phi^n\|^2}{2\Delta t} + \frac{|R^{n+1}|^2 - |R^n|^2}{\Delta t} \le -\frac{R^{n+1}}{\sqrt{E_1(\phi^n)+\delta_0}}\big(\widehat{\boldsymbol{u}}^{n+1}, \mu^n\nabla\phi^n\big) - M\|\nabla\mu^{n+1}\|^2. \tag{2.26}$$

Next, taking the inner product of (2.7d) with $\widehat{\boldsymbol{u}}^{n+1}$, we arrive at

$$\begin{aligned}&\frac{\|\widehat{\boldsymbol{u}}^{n+1}\|^2 - \|\boldsymbol{u}^n\|^2 + \|\widehat{\boldsymbol{u}}^{n+1} - \boldsymbol{u}^n\|^2}{2\Delta t} + \exp\Big(\frac{t_{n+1}}{T}\Big)Q^{n+1}\big(\boldsymbol{u}^n\cdot\nabla\boldsymbol{u}^n, \widehat{\boldsymbol{u}}^{n+1}\big)\\ &+ \frac{\nu}{2}\|\nabla\widehat{\boldsymbol{u}}^{n+1}\|^2 + \frac{\nu}{2}(\|s^{n+1}\|^2 - \|s^n\|^2) \le \frac{R^{n+1}}{\sqrt{E_1(\phi^n)+\delta_0}}\big(\mu^n\nabla\phi^n, \widehat{\boldsymbol{u}}^{n+1}\big),\end{aligned} \tag{2.27}$$

where (2.7f) and Lemma 1 (ii) are applied. Specifically, the following estimate has been applied:

$$\begin{aligned}(s^n, \nabla\cdot\widehat{\boldsymbol{u}}^{n+1}) &= (s^n, s^n - s^{n+1}) = \frac{1}{2}(\|s^n\|^2 - \|s^{n+1}\|^2) + \frac{1}{2}\|\nabla\cdot\widehat{\boldsymbol{u}}^{n+1}\|^2\\ &\le \frac{1}{2}(\|s^n\|^2 - \|s^{n+1}\|^2) + \frac{1}{2}\|\nabla\widehat{\boldsymbol{u}}^{n+1}\|^2.\end{aligned}$$

Furthermore, taking the inner product of (2.7g) with $\boldsymbol{u}^{n+1}$, it yields

$$\|\boldsymbol{u}^{n+1}\|^2 - \|\widehat{\boldsymbol{u}}^{n+1}\|^2 + \|\boldsymbol{u}^{n+1} - \widehat{\boldsymbol{u}}^{n+1}\|^2 = 0, \tag{2.28}$$

and multiplying (2.7h) with Q^{n+1}, we have

$$\frac{|Q^{n+1}|^2 - |Q^n|^2 + |\delta_t Q^{n+1}|^2}{2\Delta t} = -\frac{|Q^{n+1}|^2}{T} + \exp\Big(\frac{t_{n+1}}{T}\Big)Q^{n+1}\big(\boldsymbol{u}^n\cdot\nabla\boldsymbol{u}^n, \widehat{\boldsymbol{u}}^{n+1}\big). \tag{2.29}$$

Finally, adding (2.26) and (2.27)–(2.29) together, we complete the proof. □

Remark 2.9. Theorem 2.8 immediately implies that if $\Delta t \le 1$, there exists a uniform positive constant $\widehat{L} = \|\nabla\phi^0\|^2 + 2|R^0|^2 + \|\boldsymbol{u}^0\|^2 + \nu\|s^0\|^2 + |Q^0|^2$ satisfying

$$\|\boldsymbol{u}^{n+1}\|^2 + \nu\Delta t\|s^{n+1}\|^2 + \nu\Delta t\sum_{k=0}^{n}\|\nabla\widehat{\boldsymbol{u}}^{n+1}\|^2 + \Delta t\sum_{k=0}^{n}\|\nabla\mu^{k+1}\|^2 + \|\nabla\phi^{n+1}\|^2 + |R^{n+1}|^2 + |Q^{n+1}|^2 \le \widehat{L}.$$

Due to the Poincaré-Friedrichs inequality and the mass-conservation of ϕ, we see

$$\|\phi^{n+1}\| \le C\|\nabla\phi^{n+1}\| + \Big\|\int_\Omega \phi^0\, d\boldsymbol{x}/|\Omega|\Big\| := \overline{L}.$$

For simplicity, we set $L = \max\{\overline{L}, \widehat{L}\}$. These estimates will be used repeatedly in what follows.

2.3. The second-order scheme

We then extend the methodology to second-order accuracy based on the second-order backward differentiation formula (BDF2) for the CHNS model and demonstrate its unconditional energy stability.

2.3.1. Construction of the GU-MSAV-BDF2 scheme

Building on previous work [30,31,43], we outline the derivation starting from the BDF2-discretized Stokes equation.

$$\frac{3\boldsymbol{u}^{n+1} - 4\boldsymbol{u}^n + \boldsymbol{u}^{n-1}}{2\Delta t} + \nabla p^{n+1} - \nu\Delta\boldsymbol{u}^{n+1} = \boldsymbol{f}(t^{n+1}). \tag{2.30}$$

Noting that $\rho^{n+1} = \delta_t\psi^{n+1}$, we introduces artificial variables $(\widehat{\boldsymbol{u}}^{n+1}, \psi^{n+1})$ satisfying

$$\widehat{\boldsymbol{u}}^{n+1} = \boldsymbol{u}^{n+1} - \nabla(\psi^{n+1} - 2\psi^n + \psi^{n-1}) = \boldsymbol{u}^{n+1} - \nabla(\rho^{n+1} - \rho^n). \tag{2.31}$$

Substituting (2.31) into (2.30) yields

$$\frac{3\widehat{\boldsymbol{u}}^{n+1} - 4\boldsymbol{u}^n + \boldsymbol{u}^{n-1}}{2\Delta t} + \nabla p^{n+1} - \nu\Delta\widehat{\boldsymbol{u}}^{n+1} + \nabla\Big(\frac{3(\rho^{n+1} - \rho^n)}{2\Delta t} - \nu\Delta(\rho^{n+1} - \rho^n)\Big) = \boldsymbol{f}(t^{n+1}). \tag{2.32}$$

Applying BDF2 to the Gauge equation (2.1), we obtain

$$\frac{3\rho^{n+1} - \rho^n}{2\Delta t} - \nu\Delta\psi^{n+1} = -p^{n+1}. \tag{2.33}$$

Combining (2.32) and (2.33), it yields

$$\frac{3\widehat{\boldsymbol{u}}^{n+1}-4\boldsymbol{u}^{n}+\boldsymbol{u}^{n-1}}{2\Delta t}-\nu\Delta\widehat{\boldsymbol{u}}^{n+1}-\nabla\Big(\frac{\rho^{n}}{\Delta t}-\nu\Delta(2\psi^{n}-\psi^{n-1})\Big)=\boldsymbol{f}(t^{n+1}). \tag{2.34}$$

Similarly to the GU-MSAV-BDF1 scheme, we introduce $s^{n+1}=\Delta\psi^{n+1}$ to deal with the third-order term $\nabla\Delta\psi^{n}$, and

$$s^{n+1}=2s^{n}-s^{n-1}-\nabla\cdot\widehat{\boldsymbol{u}}^{n+1}.$$

Denote $(\cdot)^{*}=2(\cdot)^{n}-(\cdot)^{n-1}$ as the second-order linear extrapolation. Finally, given $\big(\phi,\mu,\widehat{\boldsymbol{u}},\rho,s,\boldsymbol{u},R,Q\big)^{n-1}$ and $\big(\phi,\mu,\widehat{\boldsymbol{u}},\rho,s,\boldsymbol{u},R,Q\big)^{n}$, we discretize the CHNS system using the GU-MSAV-BDF2 scheme:

$$\frac{3\phi^{n+1}-4\phi^{n}+\phi^{n-1}}{2\Delta t}+\frac{R^{n+1}}{\sqrt{E_1(\phi^{*})+\delta_0}}\big(\boldsymbol{u}^{*}\cdot\nabla\big)\phi^{*}=M\Delta\mu^{n+1},\quad \frac{\partial\phi^{n+1}}{\partial\boldsymbol{n}}\Big|_{\partial\Omega}=0, \tag{2.35a}$$

$$\mu^{n+1}=-\Delta\phi^{n+1}+\frac{R^{n+1}}{\sqrt{E_1(\phi^{*})+\delta_0}}F'\big(\phi^{*}\big),\quad \frac{\partial\mu^{n+1}}{\partial\boldsymbol{n}}\Big|_{\partial\Omega}=0, \tag{2.35b}$$

$$\frac{3R^{n+1}-4R^{n}+R^{n-1}}{2\Delta t}=\frac{1}{2\sqrt{E_1(\phi^{*})+\delta_0}}\Big(\big(F'\big(\phi^{*}\big),\frac{3\phi^{n+1}-4\phi^{n}+\phi^{n-1}}{2\Delta t}\big)+\big(\mu^{n+1},\boldsymbol{u}^{*}\cdot\nabla\phi^{*}\big)\Big)-\frac{1}{2\sqrt{E_1(\phi^{*})+\delta_0}}\big(\widehat{\boldsymbol{u}}^{n+1},\mu^{*}\nabla\phi^{*}\big), \tag{2.35c}$$

$$\frac{3\widehat{\boldsymbol{u}}^{n+1}-4\boldsymbol{u}^{n}+\boldsymbol{u}^{n-1}}{2\Delta t}+\exp\Big(\frac{t_{n+1}}{T}\Big)Q^{n+1}\Big(\boldsymbol{u}^{*}\cdot\nabla\boldsymbol{u}^{*}-\nabla\big(\frac{1}{\Delta t}\rho^{n}-\nu s^{n}\big)\Big)-\nu\Delta\widehat{\boldsymbol{u}}^{n+1}+\nu\nabla(s^{n}-s^{n-1})$$
$$=\frac{R^{n+1}}{\sqrt{E_1(\phi^{*})+\delta_0}}\mu^{*}\nabla\phi^{*},\quad \widehat{\boldsymbol{u}}^{n+1}\,|_{\partial\Omega}=0, \tag{2.35d}$$

$$\Delta\rho^{n+1}=\Delta\rho^{n}-\nabla\cdot\widehat{\boldsymbol{u}}^{n+1},\quad \frac{\partial\rho^{n+1}}{\partial\boldsymbol{n}}\Big|_{\partial\Omega}=0, \tag{2.35e}$$

$$s^{n+1}=2s^{n}-s^{n-1}-\nabla\cdot\widehat{\boldsymbol{u}}^{n+1}, \tag{2.35f}$$

$$\boldsymbol{u}^{n+1}=\widehat{\boldsymbol{u}}^{n+1}+\nabla(\rho^{n+1}-\rho^{n}), \tag{2.35g}$$

$$\frac{3Q^{n+1}-4Q^{n}+Q^{n-1}}{2\Delta t}=-\frac{Q^{n+1}}{T}+\exp\Big(\frac{t_{n+1}}{T}\Big)\Big((\boldsymbol{u}^{*}\cdot\nabla\boldsymbol{u}^{*},\widehat{\boldsymbol{u}}^{n+1})-(\nabla(\frac{1}{\Delta t}\rho^{n}-\nu s^{n}),\widehat{\boldsymbol{u}}^{n+1})\Big). \tag{2.35h}$$

Remark 2.10. We use the GU-MSAV-BDF1 scheme to compute the quantities $\big(\phi,\mu,\widehat{\boldsymbol{u}},\rho,s,\boldsymbol{u},R,Q\big)^{1}$. The GU-MSAV-BDF2 scheme (2.35) can be efficiently implemented in a manner similar to the first-order scheme (2.7) by solving a sequence of linear systems with constant coefficients. Following an analogous procedure to Li and Shen [21], we demonstrate that the GU-MSAV-BDF2 scheme (2.35a)–(2.35h) admits a unique solution. Detailed proofs are omitted for brevity.

Remark 2.11. The pressure p^{n+1} is computed as:

$$p^{n+1}=-\frac{3\rho^{n+1}-\rho^{n}}{2\Delta t}+\nu s^{n+1}.$$

Compared to (2.7h), the term $(\nabla\big(\frac{1}{\Delta t}\rho^{n}-\nu s^{n}\big),\widehat{\boldsymbol{u}}^{n+1})$ introduced in (2.35h) serves as a second-order approximation to zero, which holds due to the divergence-free condition $\nabla\cdot\boldsymbol{u}=0$. This term does not contribute to energy and the introduction of this term greatly facilitates the subsequent stability analysis. Furthermore, combining (2.35e) and (2.35g), we derive the divergence-free condition:

$$\nabla\cdot\boldsymbol{u}^{n+1}=0. \tag{2.36}$$

Remark 2.12. Similar to the first-order case in Remark 2.3, let $\widehat{R}^{n+1}$ denote the auxiliary variable computed in (2.35c). The scalar auxiliary variable R^{n+1} is then updated via the relaxation step:

$$R^{n+1}=\chi_0\widehat{R}^{n+1}+\big(1-\chi_0\big)\sqrt{E_1(\phi^{n+1})+\delta_0}. \tag{2.37}$$

Here χ_0 is determined by the constrained optimization problem

$$\chi_0=\min_{\chi\in[0,1]}\chi,\quad \text{s.t. }\frac{1}{2}(|R^{n+1}|^{2}+|2R^{n+1}-R^{n}|^{2})-\frac{1}{2}\Big(|\widehat{R}^{n+1}|^{2}+|2\widehat{R}^{n+1}-R^{n}|^{2}\Big)\le\Delta t\eta\|\nabla\mu^{n+1}\|^{2}. \tag{2.38}$$

We simplify the above optimization problem as

$$\chi_0=\min_{\chi\in[0,1]}\chi,\ \text{s.t. }g(\chi)=d\chi^{2}+e\chi+k\le 0,$$

where the coefficients are

$$\begin{cases} d=\frac{5}{2}\big(\widehat{R}^{n+1}-\sqrt{E_1(\phi^{n+1})+\delta_0}\big)^{2},\\ e=\Big(\widehat{R}^{n+1}-\sqrt{E_1(\phi^{n+1})+\delta_0}\Big)\Big(5\sqrt{E_1(\phi^{n+1})+\delta_0}-2R^{n}\Big),\\ k=\frac{1}{2}\Big(\big|E_1(\phi^{n+1})+\delta_0\big|+|2\sqrt{E_1(\phi^{n+1})+\delta_0}-R^{n}|^{2}-|\widehat{R}^{n+1}|^{2}-|2\widehat{R}^{n+1}-R^{n}|^{2}\Big)-\Delta t\eta\|\nabla\mu^{n+1}\|^{2}. \end{cases}$$

Following Remark 2.3, the optimal χ_0 is computed as:

$$\chi_0 = \max\left\{0, (-e - \sqrt{e^2 - 4\,d\,k})/(2\,d)\right\}.$$

2.3.2. Mass conservation and energy stability

The second-order scheme is also mass-conserving and unconditionally energy stable as we show below.

Theorem 2.13. *The time-discrete GU-MSAV-BDF2 scheme (2.35) satisfies the discrete mass conservation law:*

$$(\phi^{n+1}, 1) = (\phi^n, 1), \quad \forall\ 0 \le n \le N_t - 1. \tag{2.39}$$

Moreover, the discrete modified energy dissipation law holds in the sense that

$$E^{n+1}_{\text{num},2}(\phi, \boldsymbol{u}, R, Q, s) - E^{n}_{\text{num},2}(\phi, \boldsymbol{u}, R, Q, s) \le -\frac{\Delta t}{T}\left|Q^{n+1}\right|^2 - M\Delta t\|\nabla\mu^{n+1}\|^2 - \frac{\nu}{2}\Delta t\|\nabla\widehat{\boldsymbol{u}}^{n+1}\|^2,$$

where

$$\begin{aligned} E^{n+1}_{\text{num},2}(\phi, \boldsymbol{u}, R, Q, s) =& \tfrac{1}{4}\|\boldsymbol{u}^{n+1}\|^2 + \tfrac{1}{4}\|2\boldsymbol{u}^{n+1} - \boldsymbol{u}^n\|^2 + \tfrac{1}{2}\nu\Delta t(\|s^{n+1} - s^n\|^2) + \tfrac{1}{4}\|\nabla\phi^{n+1}\|^2 + \tfrac{1}{4}\|2\nabla\phi^{n+1} - \nabla\phi^n\|^2 \\ &+ \tfrac{1}{2}|R^{n+1}|^2 + |2R^{n+1} - R^n|^2 + \tfrac{1}{4}|Q^{n+1}|^2 + \tfrac{1}{4}|2Q^{n+1} - Q^n|^2. \end{aligned} \tag{2.40}$$

Proof. Thanks to Theorem 2.8, we have $(\phi^1, 1) = (\phi^0, 1)$. Then for $n \ge 1$, multiplying (2.35a) by Δt, and then taking the inner product with 1 and applying the zero boundary conditions and the incompressibility condition, we have

$$(3\phi^{n+1} - 4\phi^n + \phi^{n-1}, 1) = 0.$$

By mathematical induction, we can easily show the mass conservation, i.e., Eq. (2.39).

Next, we proceed to show that the scheme also satisfies the discrete energy dissipation law. Taking the inner products of (2.35a) with $2\Delta t\mu^{n+1}$, (2.35b) with $(3\phi^{n+1} - 4\phi^n + \phi^{n-1})$ respectively and multiplying (2.35c) with $4\Delta t R^{n+1}$, we can obtain

$$\begin{aligned} &\frac{1}{2}\left(\|\nabla\phi^{n+1}\|^2 + \left\|2\nabla\phi^{n+1} - \nabla\phi^n\right\|^2 - \|\nabla\phi^n\|^2 - \|2\nabla\phi^n - \nabla\phi^{n-1}\|^2\right) + |R^{n+1}|^2 + |2R^{n+1} - R^n|^2 - |R^n|^2 - |2R^n - R^{n-1}|^2 \\ &\le -2M\Delta t\left\|\nabla\mu^{n+1}\right\|^2 - \frac{2\Delta t R^{n+1}}{\sqrt{E_1(\phi^*) + \delta_0}}\left(\widehat{\boldsymbol{u}}^{n+1}, \mu^*\nabla\phi^*\right), \end{aligned} \tag{2.41}$$

where we use

$$2(3a - 4b + c, a) = |a|^2 + |2a - b|^2 - |b|^2 - |2b - c|^2 + |a - 2b + c|^2. \tag{2.42}$$

Taking the inner product (2.35d) with $2\Delta t\widehat{\boldsymbol{u}}^{n+1}$ leads to

$$\begin{aligned} &\left(3\widehat{\boldsymbol{u}}^{n+1} - 4\boldsymbol{u}^n + \boldsymbol{u}^{n-1}, \widehat{\boldsymbol{u}}^{n+1}\right) + 2\nu\Delta t\left\|\nabla\widehat{\boldsymbol{u}}^{n+1}\right\|^2 \\ &= 2\exp\left(\frac{t^{n+1}}{T}\right)Q^{n+1}\left(\nabla(\rho^n - \nu\Delta t s^n) - \Delta t\boldsymbol{u}^*\cdot\nabla\boldsymbol{u}^*, \widehat{\boldsymbol{u}}^{n+1}\right) - 2\nu\Delta t\left(\nabla(s^n - s^{n-1}), \widehat{\boldsymbol{u}}^{n+1}\right) + \frac{2\Delta t R^{n+1}}{\sqrt{E_1(\phi^*) + \delta_0}}\left(\mu^*\nabla\phi^*, \widehat{\boldsymbol{u}}^{n+1}\right). \end{aligned} \tag{2.43}$$

The first term on the left hand side of (2.43) can be transformed as follows:

$$\begin{aligned} &\left(3\widehat{\boldsymbol{u}}^{n+1} - 4\boldsymbol{u}^n + \boldsymbol{u}^{n-1}, \widehat{\boldsymbol{u}}^{n+1}\right) \\ &= 3\|\widehat{\boldsymbol{u}}^{n+1} - \boldsymbol{u}^{n+1}\|^2 + \tfrac{1}{2}\|\boldsymbol{u}^{n+1}\|^2 + \tfrac{1}{2}\|2\boldsymbol{u}^{n+1} - \boldsymbol{u}^n\|^2 - \tfrac{1}{2}\|\boldsymbol{u}^n\|^2 - \tfrac{1}{2}\|2\boldsymbol{u}^n - \boldsymbol{u}^{n-1}\|^2 + \tfrac{1}{2}\|\boldsymbol{u}^{n+1} - 2\boldsymbol{u}^n + \boldsymbol{u}^{n-1}\|^2, \end{aligned} \tag{2.44}$$

Here, we have used the identities in (2.42) and

$$\left(\widehat{\boldsymbol{u}}^{n+1} - \boldsymbol{u}^{n+1}, \boldsymbol{u}^{n+1}\right) = 0, \quad \left(3\boldsymbol{u}^{n+1} - 4\boldsymbol{u}^n + \boldsymbol{u}^{n-1}, \widehat{\boldsymbol{u}}^{n+1} - \boldsymbol{u}^{n+1}\right) = 0,$$

which follow from (2.35g) and (2.36).

Combining (2.35f) and Lemma 1 (ii), for the first term on the right hand side of (2.43), we have

$$\begin{aligned} -2\nu\Delta t\left(\nabla(s^n - s^{n-1}), \widehat{\boldsymbol{u}}^{n+1}\right) &= -2\nu\Delta t\left(s^n - s^{n-1}, s^{n+1} - 2s^n + s^{n-1}\right) \\ &\le -\nu\Delta t(\|s^{n+1} - s^n\|^2 - \|s^n - s^{n-1}\|^2) + \nu\Delta t\|\nabla\widehat{\boldsymbol{u}}^{n+1}\|^2. \end{aligned} \tag{2.45}$$

Then combining (2.43) with (2.44), (2.45), we have

$$\begin{aligned} &\frac{1}{2}\|\boldsymbol{u}^{n+1}\|^2 + \frac{1}{2}\|2\boldsymbol{u}^{n+1} - \boldsymbol{u}^n\|^2 + \nu\Delta t\|s^{n+1} - s^n\|^2 + \nu\Delta t\|\nabla\widehat{\boldsymbol{u}}^{n+1}\|^2 \\ &\le \tfrac{1}{2}\|\boldsymbol{u}^n\|^2 + \tfrac{1}{2}\|2\boldsymbol{u}^n - \boldsymbol{u}^{n-1}\|^2 + \nu\Delta t\|s^n - s^{n-1}\|^2 + \frac{2\Delta t R^{n+1}}{\sqrt{E_1(\phi^*)+\delta_0}}\left(\mu^*\nabla\phi^*, \widehat{\boldsymbol{u}}^{n+1}\right) \\ &\quad + 2\exp\left(\frac{t^{n+1}}{T}\right)Q^{n+1}\left(\nabla(\rho^n - \nu\Delta t s^n) - \Delta t\boldsymbol{u}^*\cdot\nabla\boldsymbol{u}^*, \widehat{\boldsymbol{u}}^{n+1}\right), \end{aligned} \tag{2.46}$$

and multiplying (2.35h) by $2\Delta t Q^{n+1}$ and using (2.42), we have

$$\begin{aligned}&\frac{1}{2}\left|Q^{n+1}\right|^2+\frac{1}{2}\left|2Q^{n+1}-Q^n\right|^2-\frac{1}{2}|Q^n|^2-\frac{1}{2}\left|2Q^n-Q^{n-1}\right|^2\\&\leq-\frac{2\Delta t}{T}\left|Q^{n+1}\right|^2+2\exp\left(\frac{t^{n+1}}{T}\right)Q^{n+1}\left(\nabla(\rho^n-\nu\Delta t s^n)-\Delta t\boldsymbol{u}^*\cdot\nabla\boldsymbol{u}^*,\widehat{\boldsymbol{u}}^{n+1}\right).\end{aligned}\tag{2.47}$$

Thus, combining (2.41) with (2.46), (2.47), we get

$$\begin{aligned}&\frac{1}{2}\|\nabla\phi^{n+1}\|^2+\frac{1}{2}\|2\nabla\phi^{n+1}-\nabla\phi^n\|^2+\frac{1}{2}\|\boldsymbol{u}^{n+1}\|^2+\frac{1}{2}\|2\boldsymbol{u}^{n+1}-\boldsymbol{u}^n\|^2+\nu\Delta t\|s^{n+1}-s^n\|^2\\&+|R^{n+1}|^2+|2R^{n+1}-R^n|^2+\frac{1}{2}|Q^{n+1}|^2+\frac{1}{2}|2Q^{n+1}-Q^n|^2\\&\leq\frac{1}{2}\|\nabla\phi^n\|^2+\frac{1}{2}\|2\nabla\phi^n-\nabla\phi^{n-1}\|^2+\frac{1}{2}\|\boldsymbol{u}^n\|^2+\frac{1}{2}\|2\boldsymbol{u}^n-\boldsymbol{u}^{n-1}\|^2+\nu\Delta t\|s^n-s^{n-1}\|^2\\&\quad+|R^n|^2+|2R^n-R^{n-1}|^2+\frac{1}{2}|Q^n|^2+\frac{1}{2}|2Q^n-Q^{n-1}|^2-\frac{2\Delta t}{T}\left|Q^{n+1}\right|^2-2M\Delta t\|\nabla\mu^{n+1}\|^2-\nu\Delta t\|\nabla\widehat{\boldsymbol{u}}^{n+1}\|^2,\end{aligned}$$

which leads to the desired result. □

Remark 2.14. Within the MSAV-ZEC framework, we propose an alternative numerical strategy for a second-order GU scheme for the CHNS system. To the best of our knowledge, this is the first second-order GU method built on MSAV-ZEC method, and the resulting GU-MSAV-BDF2 scheme (2.35) is unconditionally energy stable with second-order temporal accuracy for both velocity and pressure. For the single Navier–Stokes equation, Pyo et al. [44] once developed the classical second-order GU method and proved that it is energy stable under the condition

$$\Delta t\nu^2\|\nabla s^n\|^2\leq K,\tag{2.48}$$

for some constant $K>0$. The stability condition (2.48) depends sensitively on the numerical solution s^n and is therefore not meaningful in some situations. By redefining the auxiliary variable ψ, Pyo et al. subsequently proposed the SGU method [31], which removes this solution-dependent constraint; they established second-order convergence for the velocity and first-order (suboptimal) estimate for the pressure.

3. Temporal error estimates of the GU-MSAV-BDF1 scheme

In this section, we conduct an error analysis for the first-order semi-discrete scheme (2.7). The introduced auxiliary variable ψ in the GU method poses significant challenges for analysis compared with projection-based approaches [21]. In particular, we construct auxiliary equations to facilitate the error analysis for the Navier–Stokes equation, which constitutes the most technically challenging aspect of the proof.

Let $(\phi(t_{n+1}),\mu(t_{n+1}),\boldsymbol{u}(t_{n+1}))$ be the exact solution of (1.1), and $(\phi,\mu,\widehat{\boldsymbol{u}},\boldsymbol{u},R,Q)^{n+1}$ be the solution of the scheme (2.7). We denote the corresponding errors by

$$\begin{aligned}&\widehat{e}_{\boldsymbol{u}}^{n+1}=\boldsymbol{u}(t_{n+1})-\widehat{\boldsymbol{u}}^{n+1},\quad e_{\boldsymbol{u}}^{n+1}=\boldsymbol{u}(t_{n+1})-\boldsymbol{u}^{n+1},\quad e_{\phi}^{n+1}=\phi(t_{n+1})-\phi^{n+1},\\&e_{\mu}^{n+1}=\mu(t_{n+1})-\mu^{n+1},\quad e_{Q}^{n+1}=Q(t_{n+1})-Q^{n+1},\quad e_{R}^{n+1}=R(t_{n+1})-R^{n+1}.\end{aligned}$$

Theorem 3.1. *Suppose* $\phi\in H^2\left(0,T;H^1(\Omega)\right)\cap W^{1,\infty}\left(0,T;H^1(\Omega)\right)$, $\mu\in H^1\left(0,T;L^2(\Omega)\right)\cap L^\infty\left(0,T;H^1(\Omega)\right)$, *and* $\boldsymbol{u}\in H^2\left(0,T;L^2(\Omega)\right)\cap W^{1,\infty}\left(0,T;H^2(\Omega)\right)$, *then for the semi-discrete GU-MSAV-BDF1 scheme, if* Δt *is small enough, s.t.,* $\Delta t\leq\min\{\frac{1}{K_2},\frac{1}{2C_1}\}$, *for* $\forall\ 0\leq m\leq N_t-1$, *there holds*

$$\|e_{\phi}^{m+1}\|_1^2+\Delta t\sum_{n=0}^{m}\|\nabla e_{\mu}^{n+1}\|_1^2+|e_R^{m+1}|^2+|e_Q^{m+1}|^2+\|e_{\boldsymbol{u}}^{n+1}\|^2+\Delta t\sum_{n=0}^{m}\|\nabla\widehat{e}_{\boldsymbol{u}}^{n+1}\|^2\leq C(\Delta t)^2.$$

The detailed proof of this theorem is presented in C, following estimates for the Cahn–Hilliard equation (Lemma 7); and estimates for the Navier–Stokes equation (Lemmas 13 and 14).

Remark 3.2. Noting that

$$R^{n+1}=\sqrt{E_1(\phi(t_{n+1}))+\delta_0}+C\Delta t,\quad\sqrt{E_1(\phi^{n+1})+\delta_0}=\sqrt{E_1(\phi(t_{n+1}))+\delta_0}+C\Delta t.$$

We have

$$R^{n+1}=\kappa_0\widehat{R}^{n+1}+\left(1-\kappa_0\right)\sqrt{E_1(\phi^{n+1})+\delta_0}=R\left(t_{n+1}\right)+C\Delta t,$$

which implies that the relaxation step in (2.8) does not affect the temporal accuracy even if $\kappa_0\neq1$.

3.1. Estimates on phase variables and auxiliary variable R

The purpose of this subsection is to estimate the errors of ϕ and μ. To begin with, based on the stability result in Remark 2.9, we review the H^2 bound for ϕ^{n+1} and L^2 bound for μ^{n+1} without assuming the Lipschitz condition on $F(\phi)$.

Lemma 6 ([21]). *Assume (ϕ^{n+1}, μ^{n+1}) be the solution to the scheme (2.7a) and (2.7b), then for $0 \le n \le N_t - 1$, there exists a positive constant $\tilde{C}$ independent of Δt such that*

$$\|\Delta\phi^{n+1}\|^2 + \|\mu^{n+1}\|^2 \le \tilde{C}.$$

Next, we establish the error estimate of the scheme (2.7a)–(2.7c) and obtain the following lemma. The detailed proof is presented in Appendix A.

Lemma 7. *Under the assumptions of Theorem 3.1, we have*

$$\begin{aligned}
&\frac{1}{2\Delta t}\left(\|e_\phi^{n+1}\|_1^2 - \|e_\phi^n\|_1^2\right) + \frac{5M}{8}\|e_\mu^{n+1}\|_1^2 + \frac{1}{\Delta t}\left(|e_R^{n+1}|^2 - \left|e_R^n\right|^2\right) \\
&\quad \le C\|e_\phi^n\|_1^2 + C\|e_\phi^{n+1}\|_1^2 + C|e_R^{n+1}|^2 + C\|\boldsymbol{u}\|^2_{H^1(J_n;H^1(\Omega))}\Delta t + \frac{M}{8}\|e_\mu^n\|^2 \\
&\qquad - \frac{e_R^{n+1}}{\sqrt{E_1(\phi^n)+\delta_0}}\left(\hat{e}_{\boldsymbol{u}}^{n+1}, \mu^n\nabla\phi^n\right) + C\left\|e_{\boldsymbol{u}}^n\right\|^2 + C\|R\|^2_{H^2(J_n)}\Delta t + C\left(\|\phi\|^2_{H^1(J_n;H^1(\Omega))} + \|\phi\|^2_{H^2(J_n;L^2(\Omega))} + \|\mu\|^2_{H^1(J_n;L^2(\Omega))}\right)\Delta t.
\end{aligned}$$

3.2. Estimates on velocity and auxiliary variable Q

While the introduced Gauge variable offers algorithmic advantages, it complicates our analysis-particularly in constructing the error equation, which distinguishes it from projection methods. Rather than directly comparing (2.7d) with (1.1c), we follow the approach in Pyo [29], Wang and Liu [39] and introduce an important auxiliary Eq. (3.7). This allows us to estimate the intermediate velocity error effectively; otherwise, the convergence rate would be downgraded.

Combining (2.1) and (2.3), we obtain the heat equation of $\psi(\boldsymbol{x}, t)$ as follows:

$$\begin{aligned}
&\psi_t - \nu\Delta\psi = -p, && \text{in } \Omega\times J,\\
&\partial_\nu\psi = 0, && \text{on } \partial\Omega\times J,
\end{aligned} \tag{3.1}$$

with $\psi(\boldsymbol{x}, 0) = 0$. Let $(\boldsymbol{v}, q)$ be the solutions of the time-dependent Stokes equation

$$\begin{aligned}
&\boldsymbol{v}_t + \nabla q - \Delta\boldsymbol{v} = \boldsymbol{0}, && \text{in } \Omega\times J,\\
&\nabla\cdot\boldsymbol{v} = 0, && \text{in } \Omega\times J,\\
&\boldsymbol{v} = \Delta t\nabla\psi_t, && \text{on } \partial\Omega\times J,
\end{aligned} \tag{3.2}$$

with $\boldsymbol{v}(\boldsymbol{x}, 0) = \boldsymbol{0}$. Due to $\partial_\nu\psi_t|_{\partial\Omega} = 0$, there exists a unique solution pair $(\boldsymbol{v}, q)$ to (3.2) [26].

To effectively control the solution of the heat Eq. (3.1), we assume that (3.3) holds. Under this assumption, we provide a priori estimates for the velocity field $\boldsymbol{v}$ in Lemma 8.

Lemma 8 (Estimate on $\boldsymbol{v}$ [29]). *Assume there exists $\tilde{M} > 0$ such that*

$$\int_0^T \left(\|\psi_{tt}\|^2_{\frac{1}{2},\partial\Omega} + \|\nabla\psi_t\|^2_{\frac{3}{2},\partial\Omega}\right)dt \le \tilde{M}. \tag{3.3}$$

Then if $\boldsymbol{v}$ is the solution of (3.2), there holds

$$\|\nabla\boldsymbol{v}(T)\|^2 + \int_0^T \|\boldsymbol{v}_t\|^2\, dt \le C(\Delta t)^2\int_0^T \left(\|\psi_{tt}\|^2_{\frac{1}{2},\partial\Omega} + \|\nabla\psi_t\|^2_{\frac{3}{2},\partial\Omega}\right)dt \le C(\Delta t)^2.$$

We now denote a new variable $\hat{\boldsymbol{v}}$ as a solution of the heat equation

$$\begin{aligned}
&\hat{\boldsymbol{v}}_t - \Delta\hat{\boldsymbol{v}} = -\nabla\left(q + \Delta t\psi_{tt} - \Delta t\Delta\psi_t\right), && \text{in } \Omega\times J,\\
&\hat{\boldsymbol{v}} = \boldsymbol{0}, && \text{on } \partial\Omega\times J,
\end{aligned} \tag{3.4}$$

where q is the pressure in (3.2) and $\hat{\boldsymbol{v}}(\boldsymbol{x}, 0) = \boldsymbol{0}$. Then the solutions of (3.2) and (3.4) satisfy $\hat{\boldsymbol{v}} = \boldsymbol{v} - \Delta t\nabla\psi_t$. Thanks to Lemma 8, we arrive at Pyo [29]

$$\|\nabla\hat{\boldsymbol{v}}\| + \|\nabla\boldsymbol{v}\| \le C\Delta t, \tag{3.5}$$

which plays a key role in the subsequent analysis.

We now define intermediate variables as

$$\hat{\boldsymbol{U}}(t_{n+1}) = \boldsymbol{u}(t_{n+1}) + \hat{\boldsymbol{v}}(t_{n+1}) \quad \text{and} \quad \boldsymbol{U}(t_{n+1}) = \boldsymbol{u}(t_{n+1}) + \boldsymbol{v}(t_{n+1}). \tag{3.6}$$

Thanks to $\nabla\cdot\boldsymbol{u}(t_{n+1}) = 0$, we have the following lemma.

Lemma 9 (Properties of $\widehat{U}$, $\widehat{v}$, U and v [29]). *The variables $\widehat{U}$, $\widehat{v}$, U defined in (3.6) satisfy the following properties:*

$$\begin{aligned}
&\nabla\cdot\widehat{U}=\nabla\cdot\widehat{v}=-\Delta t\Delta\psi_t, \quad \nabla\cdot U=\nabla\cdot v=0, && \text{in } \Omega,\\
&\widehat{U}=\widehat{v}=0, \quad U\cdot n=v\cdot n=\partial_n\psi_t=0, && \text{on } \partial\Omega.
\end{aligned}$$

Based on the idea of high-order consistency analysis [29,39,45], we present the intermediate momentum equation, which plays a crucial role in the error estimates for the velocity.

Lemma 10 (Intermediate momentum equation). *The variables $\widehat{U}$ and U satisfy*

$$\begin{aligned}
&\frac{\widehat{U}(t_{n+1})-U(t_n)}{\Delta t}+\exp\Big(\frac{t_{n+1}}{T}\Big)Q(t_{n+1})\big(u(t_{n+1})\cdot\nabla\big)u(t_{n+1})\\
&\quad+\nu\Delta\nabla\psi(t_{n+1})-\nu\Delta\widehat{U}(t_{n+1})=f(t_{n+1})+R_u^{n+1}+K^{n+1},
\end{aligned}\tag{3.7}$$

where

$$f(t_{n+1})=\frac{R(t_{n+1})}{\sqrt{E_1(\phi(t_{n+1}))+\delta_0}}\mu(t_{n+1})\nabla\phi(t_{n+1}),\quad R_u^{n+1}=-\frac{1}{\Delta t}\int_{t_n}^{t_{n+1}}\big(t-t_n\big)u_{tt}dt,\quad K^{n+1}=\frac{v(t_{n+1})-v\big(t_n\big)}{\Delta t}-\nu\Delta\widehat{v}(t_{n+1}).$$

In (3.7), the pressure term is appropriately replaced to align with the scheme (2.2) by using the heat Eq. (3.1). We present the error estimates of U and $\widehat{U}$ as follows.

Lemma 11 (Estimates on the auxiliary functions $\widehat{U}$ and U [29]). *Under the assumptions of Theorem 3.1, we have*

$$\|u(t_{n+1})-U(t_{n+1})\|_1+\|u(t_{n+1})-\widehat{U}(t_{n+1})\|_1\le C\Delta t.$$

Let $\big(\widehat{u},u,p,s,\rho\big)^{n+1}$ denote the solution of (2.7d)–(2.7g) and define the corresponding errors as follows:

$$\begin{aligned}
&\widehat{E}^{n+1}=\widehat{U}(t_{n+1})-\widehat{u}^{n+1}, \quad \varepsilon^{n+1}=\Delta\psi(t_{n+1})-s^{n+1},\\
&E^{n+1}=U(t_{n+1})-u^{n+1}, \quad \gamma^{n+1}=\rho^{n+1}-\Delta t\psi_t(t_{n+1}).
\end{aligned}\tag{3.8}$$

From the definition of E^{n+1}, it yields

$$e_u^{n+1}=u(t_{n+1})-U(t_{n+1})+U(t_{n+1})-u^{n+1}=u(t_{n+1})-U(t_{n+1})+E^{n+1}.\tag{3.9}$$

According to Lemma 11, we have established the error estimates for $u(t_{n+1})-U(t_{n+1})$. To derive the error estimate for e_u^{n+1}, it suffices to estimate E^{n+1}.

In conjunction with Lemma 11, we have

$$\widehat{E}^{n+1}=\widehat{U}(t_{n+1})-\widehat{u}^{n+1}=u(t_{n+1})+\widehat{v}(t_{n+1})-\widehat{u}^{n+1}=\widehat{v}(t_{n+1})+\widehat{e}_u^{n+1}.\tag{3.10}$$

If we apply $U(0)=\widehat{U}(0)=u(0)$, then we readily get $E^0=\widehat{E}^0=0$. Before showing the error estimates, we list the following useful lemma which can be proved using (3.6), (3.8) and Lemma 9.

Lemma 12 (Properties of E^{n+1} and $\widehat{E}^{n+1}$ [29]). *The errors E^{n+1} and $\widehat{E}^{n+1}$ defined in (3.8) satisfy the following properties:*

$$\begin{aligned}
&\nabla\cdot E^{n+1}=0, \quad \widehat{E}^{n+1}=E^{n+1}+\nabla\gamma^{n+1}, && \text{in } \Omega,\\
&\|\widehat{E}^{n+1}\|^2=\|E^{n+1}\|^2+\|\nabla\gamma^{n+1}\|^2, && \text{in } \Omega,\\
&E^{n+1}\cdot n=0, \quad \widehat{E}^{n+1}=0, && \text{on } \partial\Omega.
\end{aligned}$$

Based on the above discussions, we are now in position to present the estimates for velocity in the following lemma. The detailed proof is presented in Appendix B.

Lemma 13. *Under the assumptions of Theorem 3.1, we have*

$$\begin{aligned}
&\frac{\big\|E^{n+1}\big\|^2-\|E^n\|^2}{2\Delta t}+\frac{\|\nabla\gamma^{n+1}\|^2}{2\Delta t}+\frac{\nu}{2}\big(\|\varepsilon^{n+1}\|^2-\|\varepsilon^n\|^2\big)+\frac{\nu}{4}\|\nabla\widehat{E}^{n+1}\|^2\\
&\le-\exp\Big(\frac{t_{n+1}}{T}\Big)e_Q^{n+1}\big((u^n\cdot\nabla)u^n,\widehat{e}_u^{n+1}\big)+\frac{e_R^{n+1}}{\sqrt{E_1(\phi^n)+\delta_0}}\big(\mu^n\nabla\phi^n,\widehat{e}_u^{n+1}\big)\\
&\quad+\frac{\nu}{8}\|\nabla\widehat{E}^{n+1}\|^2+\frac{M}{8}\big\|\nabla e_\mu^n\big\|^2+C\|e_\phi^n\|_1^2+C|e_R^{n+1}|^2+C|e_Q^{n+1}|^2+C(\big\|e_u^n\big\|_1^2+1)\|E^n\|^2\\
&\quad+C\big\|e_u^n\big\|_1^2(\Delta t)^2+C\big\|v_t\big\|^2_{L^2(J_n;L^2(\Omega))}+C\|u\|^2_{H^1(J_n;H^2(\Omega))}\Delta t+C\|\psi\|^2_{H^2(J_n;H^2(\Omega))}\Delta t\\
&\quad+C\|\mu\|^2_{H^1(J_n;H^1(\Omega))}\Delta t+C\big(1+\|\varepsilon^n\|^2\big)(\Delta t)^2+C\big(\|u\|^2_{H^2(J_n;H^{-1}(\Omega))}+\|\phi\|^2_{H^1(J_n;H^1(\Omega))}\big)\Delta t.
\end{aligned}\tag{3.11}$$

Remark 3.3. In fact, if we do not introduce the intermediate Eq. (3.7), it seems impossible to obtain the optimal convergence rate. Specifically, if we directly subtract (2.7d) from (1.1c) with $\boldsymbol{g}=0$ at t_{n+1}, we have the following error equation:

$$\begin{aligned}&\frac{\hat{e}_{\boldsymbol{u}}^{n+1}-e_{\boldsymbol{u}}^{n}}{\Delta t}-\nu\Delta\hat{e}_{\boldsymbol{u}}^{n+1}\\&=-\nu\nabla s^{n}+\nabla p(t_{n+1})+\Big(\frac{R(t_{n+1})}{\sqrt{E_1\big(\phi(t_{n+1})\big)+\delta_0}}\mu(t_{n+1})\nabla\phi(t_{n+1})-\frac{R^{n+1}}{\sqrt{E_1(\phi^n)+\delta_0}}\mu^n\nabla\phi^n\Big)\\&\quad+\exp\Big(\frac{t_{n+1}}{T}\Big)\Big(Q^{n+1}(\boldsymbol{u}^n\cdot\nabla)\boldsymbol{u}^n-Q(t_{n+1})\big(\boldsymbol{u}(t_{n+1})\cdot\nabla\big)\boldsymbol{u}(t_{n+1})\Big)+\boldsymbol{R}_u^{n+1}.\end{aligned}\tag{3.12}$$

Notice that s^n is an additional variable introduced by the GU method, which is absent from the original model (2.2f). Taking the inner product of (3.12) with $2\Delta t\hat{e}_{\boldsymbol{u}}^{n+1}$, we have

$$\|\hat{e}_{\boldsymbol{u}}^{n+1}\|^2-\|e_{\boldsymbol{u}}^{n}\|^2+\|\hat{e}_{\boldsymbol{u}}^{n+1}-e_{\boldsymbol{u}}^{n}\|^2+2\Delta t\nu\|\nabla\hat{e}_{\boldsymbol{u}}^{n+1}\|^2=2\nu\Delta t(s^n,\nabla\cdot\hat{e}_{\boldsymbol{u}}^{n+1})-2\Delta t(p(t_{n+1}),\nabla\cdot\hat{e}_{\boldsymbol{u}}^{n+1})+\dots.\tag{3.13}$$

For the first and second terms on the right-hand side of (3.13), we have

$$2\Delta t\big(\nu(s^n,\nabla\cdot\hat{e}_{\boldsymbol{u}}^{n+1})-(p(t_{n+1}),\nabla\cdot\hat{e}_{\boldsymbol{u}}^{n+1})\big)\le\frac{\nu\Delta t}{32}\|\nabla\hat{e}_{\boldsymbol{u}}^{n+1}\|^2+64\nu\|s^n\|^2\Delta t+\frac{64}{\nu}\|p(t_{n+1})\|^2\Delta t.$$

Clearly, $\frac{\nu\Delta t}{32}\|\nabla\hat{e}_{\boldsymbol{u}}^{n+1}\|^2$ can be controlled by $2\Delta t\nu\|\nabla\hat{e}_{\boldsymbol{u}}^{n+1}\|^2$ from the left-hand side of (3.13). As expected, we will apply the telescoping by summing up the estimate from $n=0$ to $n=m$, which results in $\Delta t\sum_{n=0}^{m}(\nu\|s^n\|^2+\frac{1}{\nu}\|p\|^2)$. However, applying Grönwall inequality does not introduce an additional Δt for this term. Since $\sum_{n=0}^{m}\|p\|^2\Delta t$ is only bounded, there is no temporal convergence order for this term. Furthermore, according to the energy stability results in Remark 2.9, only $\nu\Delta t\|s^n\|^2$ is bounded, which implies that $\sum_{n=0}^{m}\nu\|s^n\|^2\Delta t$ might be unbounded. Overall, the term $2\Delta t\big(\nu(s^n,\nabla\cdot\hat{e}_{\boldsymbol{u}}^{n+1})-(p(t_{n+1}),\nabla\cdot\hat{e}_{\boldsymbol{u}}^{n+1})\big)$ introduces an imbalance, leading to a potentially unbounded result. Therefore, it is crucial to introduce the intermediate equations for our estimate.

Note that we did not analyze the first term on the right-hand side of (3.11), which can be canceled out by its counterpart from the equation of the auxiliary variable Q. Following the proof of Lemma 4.4 in Li and Shen [21], we can establish the error estimate for Q.

Lemma 14. *Under the assumptions of Theorem 3.1, for $0\le n\le N_t-1$, we have*

$$\frac{|e_Q^{n+1}|^2-|e_Q^{n}|^2}{2\Delta t}+\frac{1}{2T}|e_Q^{n+1}|^2\le\exp\Big(\frac{t_{n+1}}{T}\Big)e_Q^{n+1}\big(\boldsymbol{u}^n\cdot\nabla\boldsymbol{u}^n,\hat{e}_{\boldsymbol{u}}^{n+1}\big)+C\|\boldsymbol{u}^n\|_1^2\|\boldsymbol{E}^n\|^2+C\big(\Delta t\|\boldsymbol{u}^n\|_1^2+\|\boldsymbol{u}\|^2_{H^1(J_n;L^2(\Omega))}\big)\Delta t.$$

4. Numerical experiments

In this section, we present several numerical examples to validate the convergence rates and reliability of the GU-MSAV schemes for the CHNS system

$$\begin{aligned}&\phi_t+\boldsymbol{u}\cdot\nabla\phi=M\Delta\mu+f_1, && \text{in }\Omega\times J,\\&\mu=-\lambda\chi\Delta\phi+\lambda F'(\phi), && \text{in }\Omega\times J,\\&\boldsymbol{u}_t+\boldsymbol{u}\cdot\nabla\boldsymbol{u}-\nu\Delta\boldsymbol{u}+\nabla p=\mu\nabla\phi+\boldsymbol{f_2}, && \text{in }\Omega\times J,\\&\nabla\cdot\boldsymbol{u}=0, && \text{in }\Omega\times J,\\&\partial_{\boldsymbol{n}}\phi=\partial_{\boldsymbol{n}}\mu=0,\quad\boldsymbol{u}=\boldsymbol{0}, && \text{on }\partial\Omega\times J,\end{aligned}\tag{4.1}$$

with given source terms f_1 and $\boldsymbol{f}_2$, where parameters are specified subsequently.

To enforce the discrete divergence-free condition for velocity, we adopt the second-order marker-and-cell (MAC) method [21,46] on staggered spatial grids. For simplicity, we always take $N_x=N_y\equiv N$ and $h_x=h_y\equiv h$ in our tests, and errors are measured as follows

$$\|e_{\zeta}\|_{\ell^2(L^2)}=\Big(\sum_{n=1}^{N_t}\Delta t\|\zeta^n-\zeta(t_n)\|^2\Big)^{1/2},\quad\|e_{\zeta}\|_{\ell^\infty(L^2)}=\max_{1\le n\le N_t}\|\zeta^n-\zeta(t_n)\|,$$
$$|e_{\theta}|_{\ell^\infty}=\max_{0\le n\le N_t-1}|\theta^{n+1}-\theta(t_{n+1})|,\quad\zeta=\{\phi,\mu,\boldsymbol{u},p\},\ \theta=\{R,Q\}.$$

Moreover, we measure interface resolution by the number of grid cells across the diffuse interface. Unless otherwise specified, the 90 % thickness is resolved by approximately $6\sim11$ cells; finer resolutions are used only in Section 4.1 to verify accuracy.

4.1. Convergence tests

In this subsection, we evaluate the temporal accuracy of the developed schemes. By suitable choices of f_1 and $\boldsymbol{f}_2$ in (4.1), we construct the manufactured exact solutions of the CHNS system as Li and Shen [21]

$$\boldsymbol{u}(x,y,t)=\big[\cos(t)\sin^2(\pi x)\sin(2\pi y),-\cos(t)\sin^2(\pi y)\sin(2\pi x)\big],$$
$$p(x,y,t)=\sin(t)\cos(\pi x)\sin(\pi y),\quad\phi(x,y,t)=\sin(t)\cos(\pi x)\cos(\pi y).$$

Table 1
Errors and convergence rates of the GU-MSAV-BDF1 scheme.

Δt	$\frac{1}{32}$	$\frac{1}{64}$	$\frac{1}{128}$	$\frac{1}{256}$	$\frac{1}{512}$
$\|e_\phi\|_{\ell^\infty(L^2)}$	4.5038 E−2	2.2327 E−2	1.1086 E−2	5.5198 E−3	2.7535 E−3
Rate	—	1.01	1.01	1.01	1.00
$\|\nabla e_\phi\|_{\ell^\infty(L^2)}$	3.0121 E−1	1.4992 E−1	7.4574 E−2	3.7160 E−2	1.8545 E−2
Rate	—	1.01	1.01	1.00	1.00
$\|e_{\boldsymbol{u}}\|_{\ell^\infty(L^2)}$	2.1824 E−2	1.1117 E−2	5.6204 E−3	2.8261 E−3	1.4171 E−3
Rate	—	0.97	0.98	0.99	1.00
$\|\nabla e_{\boldsymbol{u}}\|_{\ell^\infty(L^2)}$	1.9389 E−1	9.4861 E−2	4.6767 E−2	2.3208 E−2	1.1559 E−2
Rate	—	1.03	1.02	1.01	1.01
$\|e_{\boldsymbol{u}}\|_{\ell^\infty(L^\infty)}$	3.3868 E−2	1.7366 E−2	8.8029 E−3	4.4270 E−3	2.2194 E−3
Rate	—	0.96	0.98	0.99	1.00
$\|e_p\|_{\ell^2(L^2)}$	5.2117 E−1	2.6702 E−1	1.3466 E−1	6.7546 E−2	3.3817 E−2
Rate	—	0.96	0.99	1.00	1.00
$\|e_p\|_{\ell^\infty(L^2)}$	6.9098 E−1	3.4828 E−1	1.7414 E−1	8.6949 E−2	4.3426 E−2
Rate	—	0.99	1.00	1.00	1.00
$\|e_R\|_{\ell^\infty}$	3.2820 E−2	1.6373 E−2	8.1719 E−3	4.0840 E−3	2.0418 E−3
Rate	—	1.00	1.00	1.00	1.00
$\|e_Q\|_{\ell^\infty}$	4.4100 E−3	2.6162 E−3	1.3825 E−3	7.0611 E−4	3.5631 E−4
Rate	—	0.75	0.92	0.97	0.99

Table 2
Errors and convergence rates of the GU-MSAV-BDF2 scheme.

Δt	$\frac{1}{32}$	$\frac{1}{64}$	$\frac{1}{128}$	$\frac{1}{256}$	$\frac{1}{512}$
$\|e_\phi\|_{\ell^\infty(L^2)}$	3.8534 E−3	9.4948 E−4	2.3581 E−4	5.8747 E−5	1.4657 E−5
Rate	—	2.02	2.01	2.01	2.00
$\|\nabla e_\phi\|_{\ell^\infty(L^2)}$	2.7593 E−2	6.8782 E−3	1.7211 E−3	4.3065 E−4	1.0774 E−4
Rate	—	2.00	2.00	2.00	2.00
$\|e_{\boldsymbol{u}}\|_{\ell^\infty(L^2)}$	2.2569 E−3	5.4519 E−4	1.3530 E−4	3.3756 E−5	8.4314 E−6
Rate	—	2.05	2.01	2.00	2.00
$\|\nabla e_{\boldsymbol{u}}\|_{\ell^\infty(L^2)}$	2.4095 E−2	5.1375 E−3	1.0449 E−3	2.5657 E−4	6.3598 E−5
Rate	—	2.23	2.30	2.03	2.01
$\|e_{\boldsymbol{u}}\|_{\ell^\infty(L^\infty)}$	3.7398 E−3	9.1627 E−4	2.2450 E−4	5.5740 E−5	1.3901 E−5
Rate	—	2.03	2.03	2.01	2.00
$\|e_p\|_{\ell^2(L^2)}$	3.8130 E−2	9.3208 E−3	2.3076 E−3	5.7386 E−4	1.4315 E−4
Rate	—	2.03	2.01	2.01	2.00
$\|e_p\|_{\ell^\infty(L^2)}$	4.7980 E−2	1.1634 E−2	2.8907 E−3	7.2155 E−4	1.9101 E−4
Rate	—	2.04	2.01	2.00	1.92
$\|e_R\|_{\ell^\infty}$	6.1723 E−3	1.5491 E−3	3.8839 E−4	9.7186 E−5	2.4176 E−5
Rate	—	1.99	2.00	2.00	2.01
$\|e_Q\|_{\ell^\infty}$	6.2454 E−4	1.6748 E−4	4.3550 E−5	1.1121 E−5	2.8139 E−6
Rate	—	1.90	1.94	1.97	1.98

The simulations are performed with $\Omega = [0,1]^2$, $T = 1$, $\nu = \lambda = \chi = 1$, $\epsilon = 0.1$, $M = 10^{-3}$ and $\delta_0 = 0$.

We choose $h = \Delta t$ with $\Delta t = \frac{1}{32}, \frac{1}{64}, \frac{1}{128}, \frac{1}{256}, \frac{1}{512}$ to observe the temporal accuracy. The numerical errors for both the first- and second-order schemes are summarized in Tables 1 and 2. As can be seen, the GU-MSAV-BDF1 scheme (2.7) demonstrates first-order convergence for all variables, in agreement with the theoretical result stated in Theorem 3.1. Furthermore, the GU-MSAV-BDF2 scheme (2.35) achieves second-order convergence rates across all variables. In particular, the velocity error measured in the $\ell^\infty(0,T;L^\infty(\Omega)) \cap \ell^\infty(0,T;H^1(\Omega))$-norm and pressure error in the $\ell^\infty(0,T;L^2(\Omega))$-norm both attain optimal convergence rates. Note that the classical pressure-correction methods [21,23,47] typically suffer from an order reduction for velocity in the $\ell^2(0,T;H^1(\Omega))$-norm and pressure in the $\ell^\infty(0,T;L^2(\Omega))$-norm. However, since our proposed schemes avoid artificial Neumann boundary conditions, they achieve full second-order accuracy.

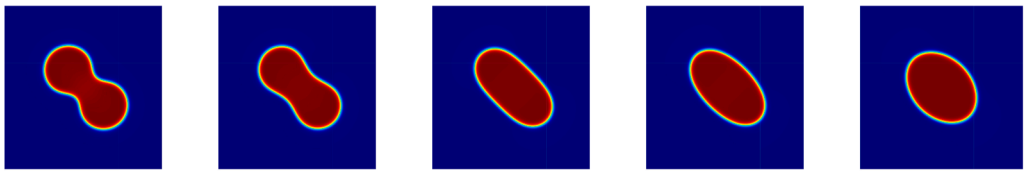
4.2. *Bubbles merging driven by surface tension*

We simulate the merging of two bubbles driven by surface tension, following a similar setup to Zhao and Han [7], Guo et al. [48]. The initial conditions are $\boldsymbol{u}(x,y,0) = \mathbf{0}$ on the domain $\Omega = [0,1]^2$, with the phase field initialized as

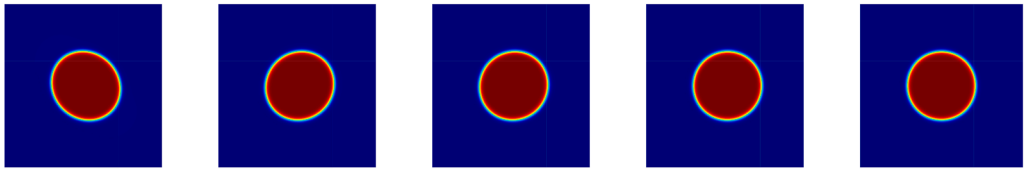
$$\phi(x,y,0) = 1 - \tanh\frac{-r + \sqrt{(x-x_a)^2 + (y-y_a)^2}}{2a\epsilon} - \tanh\frac{-r + \sqrt{(x-x_b)^2 + (y-y_b)^2}}{2a\epsilon},$$

where $x_a = 0.5 - \frac{r}{\sqrt{2}}, y_a = 0.5 + \frac{r}{\sqrt{2}}, x_b = 0.5 + \frac{r}{\sqrt{2}}, y_b = 0.5 - \frac{r}{\sqrt{2}}$, $r = 0.15$ and $a = 10^{-2}$. The parameters are chosen as $M = 10^{-2}$, $\chi = 10^{-4}$, $\delta_0 = 0$, $\epsilon = \lambda = 1$, with $N = 128$ and $\Delta t = 10^{-3}$.

We investigate the dynamics of bubble merging in a hydrodynamic environment using the GU-MSAV-BDF2 scheme (2.35) with viscosities $\nu = 10^{-2}$ and $\nu = 10^{-3}$ at various time instants, as illustrated in Figs. 1 and 2, respectively. Although both bubbles ultimately

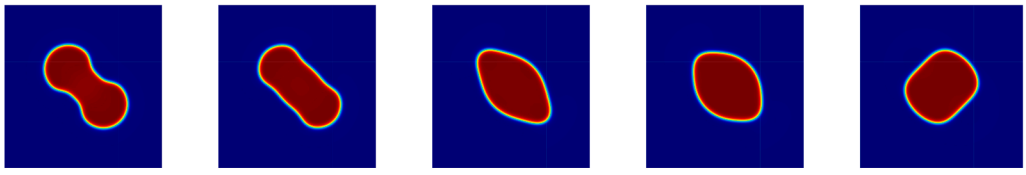


(a) from left to right: $t = 0.2, 0.4, 0.8, 1, 1.5$

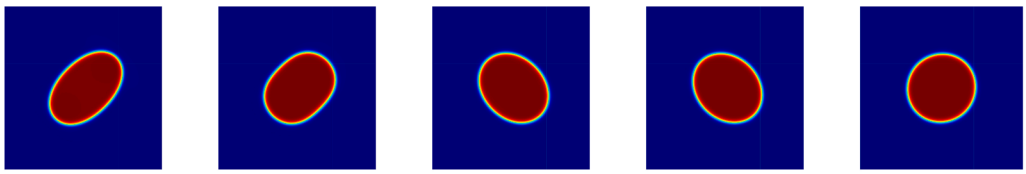


(b) from left to right: $t = 2, 3, 4, 5, 10$

Fig. 1. Simulated dynamics of bubble merging with $\nu = 10^{-2}$ at different times.



(a) from left to right: $t = 0.2, 0.4, 0.8, 1, 1.5$



(b) from left to right: $t = 2, 3, 4, 5, 10$

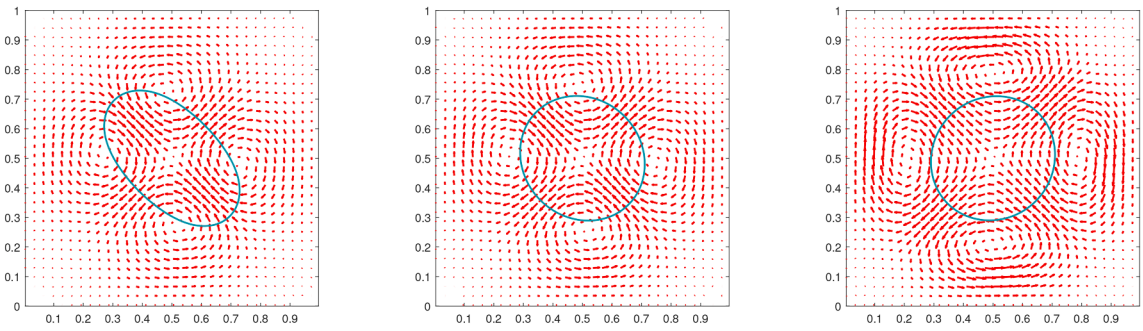
Fig. 2. Simulated dynamics of bubble merging with $\nu = 10^{-3}$ at different times.

merge into a single entity and the resulting droplet reaches a steady state, the merging dynamics differ significantly between the two cases. For $\nu = 10^{-2}$, the bubbles first undergo smoothing along the left diagonal, with the merging interface transitioning from concave curvature to rugby ball geometry. As the merging process progresses, the aspect ratio of the merged elliptical structure monotonically approaches unity, culminating in a stable circular configuration by the equilibrium state. Although the bubbles exhibit a tendency to rotate from the right diagonal between $t = 2$ and $t = 4$, this rotation is subtle. In contrast, the bubbles undergo a more pronounced directional shift for $\nu = 10^{-3}$. They initially move along the left diagonal line, gradually rotate toward the right diagonal line, and eventually stabilize into a circular shape. This distinct trajectory highlights the role of inertia in governing the bubble-merging dynamics. The velocity fields for both cases in Figs. 3 and 4 further verify the above process.

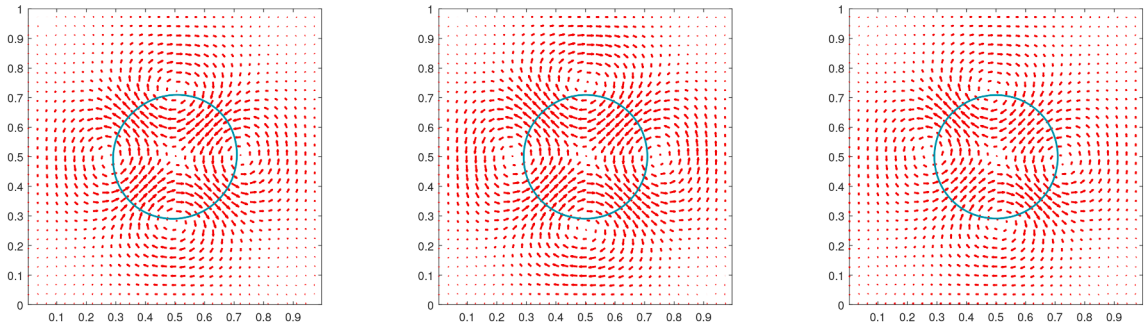
Additionally, we plot the evolution of the original energy $E(\phi, \boldsymbol{u})$ defined in (1.2) with different viscosities over time in Fig. 5, which confirms the energy decay property of the numerical solutions in two cases. It has been observed that a larger viscosity coefficient leads to more gradual changes in the energy evolution. Moreover, as shown in Fig. 6, the evolution of the mass increment in both cases remains at the level of machine precision (around 10^{-15}), which demonstrates the rigorously conservative nature of the proposed method. In this example, the diffuse interface is only resolved by about 5.33 grid points; however, the simulated phase field, velocity field, and energy histories agree closely with [7].

4.3. Spinodal decomposition

In this example, we simulate the spinodal decomposition dynamics in a binary fluid mixture, starting from a perturbed initial concentration field given by $\phi(x, y, 0) = 0.5 + 0.001\mathrm{rand}(x, y)$ in the domain $\Omega = [0, 1]^2$, where $\mathrm{rand}(x, y)$ denotes a random number in $[-1, 1]$. The initial velocity is set to zero. We set the parameters as $\epsilon = \lambda = 10^{-2}$, $M = 10^{-3}$ and $\chi = \nu = 1$. The simulation is performed using the GU-MSAV-BDF2 scheme (2.35) with $\Delta t = 4 \times 10^{-4}$ and $N = 256$.



(a) from left to right: $t = 1, 2, 3$



(b) from left to right: $t = 4, 5, 10$

Fig. 3. Visualization of the simulated velocity field with $\nu = 10^{-2}$ at different times.

Snapshots of the phase variable during the spinodal decomposition are shown in Fig. 7, which clearly illustrates the phase separation process. The corresponding velocity fields are depicted in Fig. 8. As time evolves, the initial chaotic state gradually transforms into multiple small circular domains, which subsequently merge and grow larger. Fig. 9 shows the evolution of original energy, with the energy functional decreasing significantly at early times before gradually stabilizing as the final time is approached. Additionally, it is evident that the scheme holds the mass conservation, in agreement with the theoretical results presented in Theorem 2.13.

4.4. Swirling flow

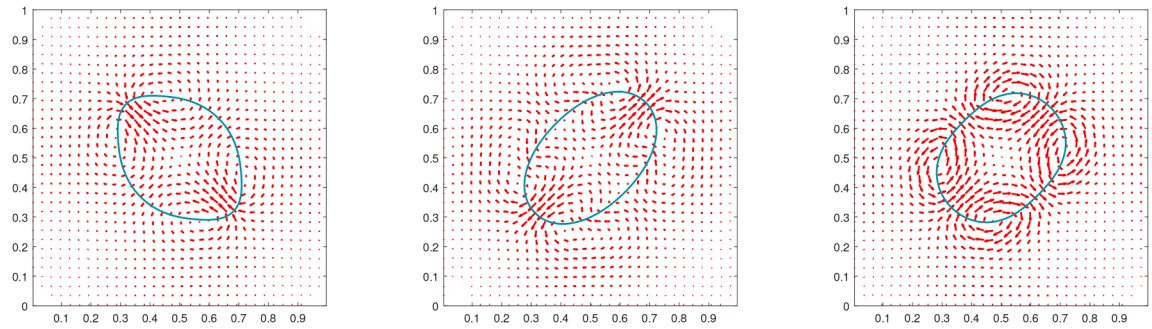
In this example, we consider the canonical single-vortex stirring test [49–51] to further assess the reliability of the developed GU-MSAV-BDF2 scheme. The computational domain is chosen as the unit square $\Omega = [0, 1]^2$, and the initial phase is a circular drop of radius $r = 0.15$ centered at $(0.5, 0.75)$, initialized by a tanh function with interface thickness ϵ. We drive the flow with a solenoidal single-vortex field (persistent stirring) and set the material and numerical parameters as

$$\epsilon^2 = \chi = 5 \times 10^{-3}, \quad \lambda = 10^{-2}, \quad \nu = 10^{-3}, \quad M = 10^{-4}, \quad N = 500, \quad \Delta t = 10^{-4}.$$

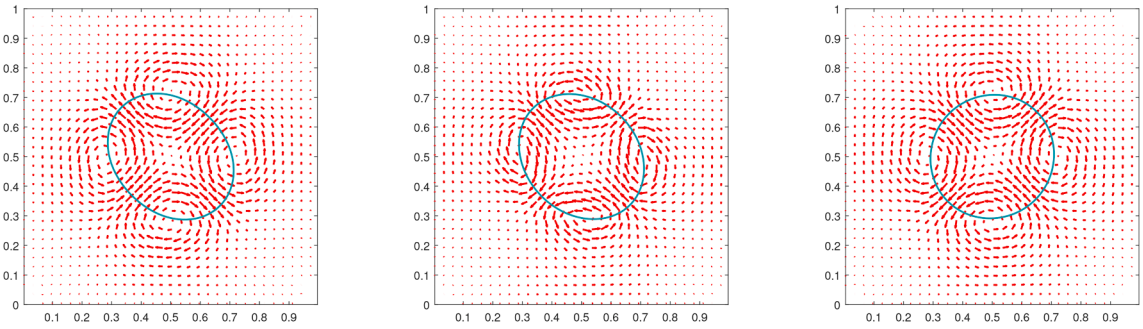
Fig. 10 shows that under persistent stirring the initially circular drop is stretched into a spiral filament that winds toward the vortex center and progressively thins. When the local filament thickness becomes comparable to the grid spacing or to the diffuse-interface width ϵ, the outer-tail region is the first to break up, with satellite droplets forming. This behavior is consistent with the single-vortex results reported in Guo et al. [49]. The faint bluish trails are artifacts of phase-field diffusion in zones of high curvature and shear, an effect exacerbated by local under-resolution. Moreover, Fig. 11 depicts the evolution of the equivalent energy E_{sav} defined in (2.4), the modified discrete energy $E_{\text{num},2}$ defined in (2.40), and the mass increment. It is observed that the two energy functionals are in close agreement with each other and both exhibit monotonic decay, which is in accordance with the energy stability result in Theorem 2.13. Meanwhile, the mass increment stays at the round-off error level, indicating excellent discrete mass conservation. These results again provide further evidence of the accuracy and robustness of the proposed scheme for strongly deforming flows.

4.5. Dripping droplet

Although model (4.1) is derived under matched densities, when the density ratio is small or the two different fluids are comparable, we can adopt the Boussinesq approximation [52–54] of the momentum equation by modeling buoyancy through the specified term



(a) from left to right: $t = 1, 2, 3$



(b) from left to right: $t = 4, 5, 10$

Fig. 4. Visualization of the simulated velocity field with $\nu = 10^{-3}$ at different times.

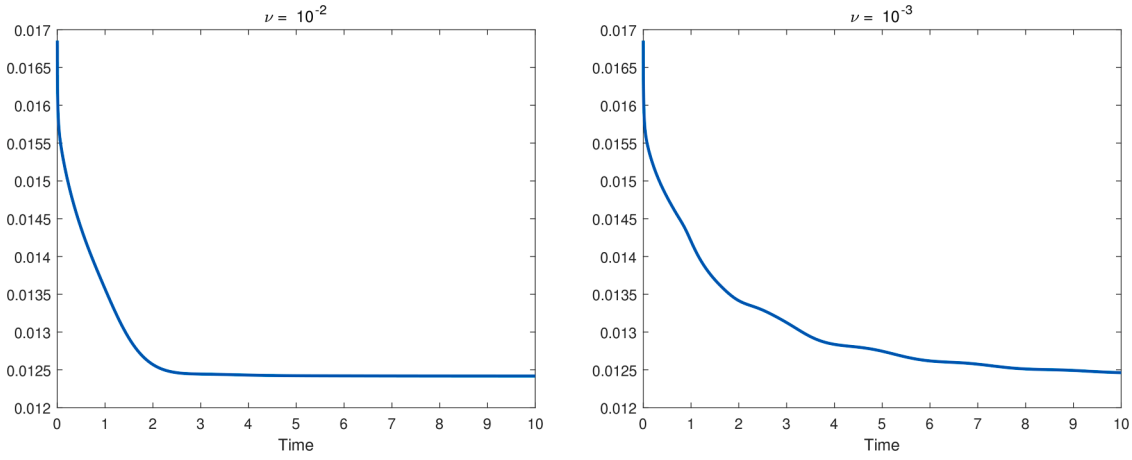


Fig. 5. Evolution of the energy for different ν.

$\boldsymbol{f}_2$. In this section, the gravity-driven detachment of a viscous drop initially attached to the upper wall is simulated using model (4.1) with

$$\boldsymbol{f}_2 = \rho(\phi)\boldsymbol{g}, \quad \rho(\phi) = \phi - \bar{\phi},$$

where $\bar{\phi}$ is a constant (usually the average value of ϕ) and $\boldsymbol{g}$ denotes the gravitational acceleration. In the following simulations, we set $\boldsymbol{g} = [0, -10]^\top$ in the two space dimensions (2D) and $\boldsymbol{g} = [0, 0, -10]^\top$ in the three space dimensions (3D), and all simulations

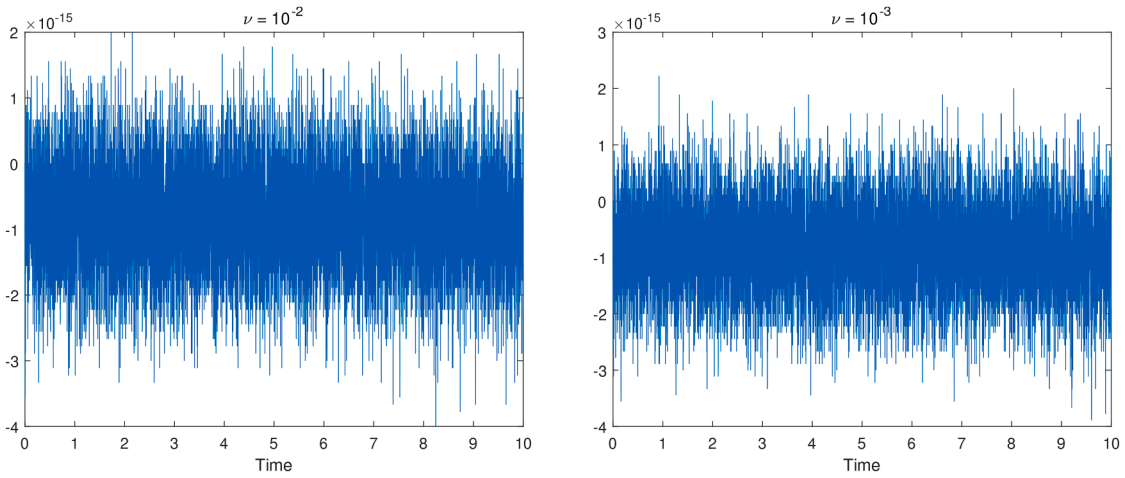
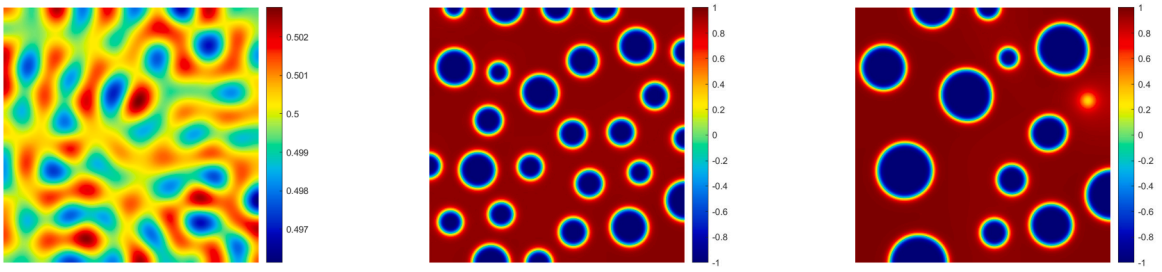
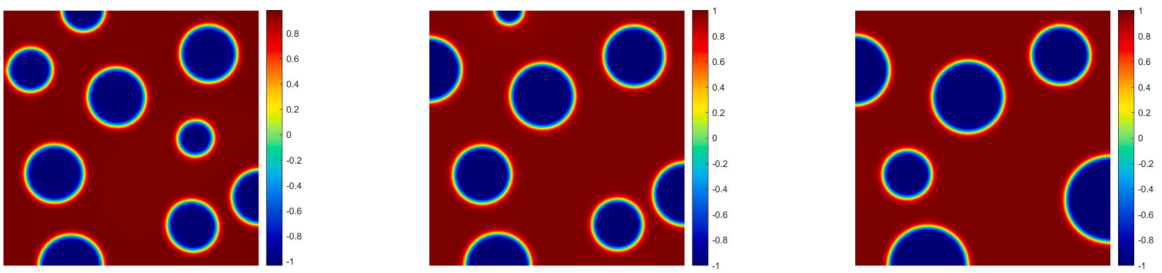


Fig. 6. Evolution of the mass increment for different ν.



(a) from left to right: $t = 0.25,\ 1,\ 2$



(b) from left to right: $t = 3,\ 5,\ 10$

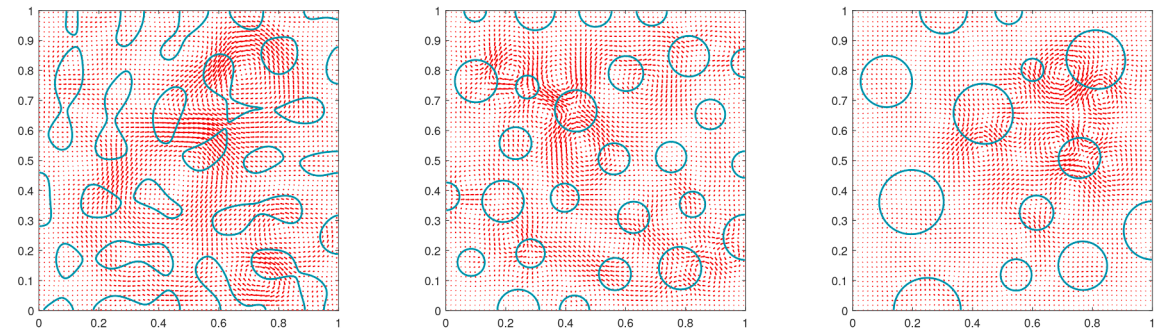
Fig. 7. Snapshots of the simulated phase variable at different times.

are conducted using the proposed GU-MSAV-BDF2 scheme. Droplet detachment under various parameters is investigated to further evaluate the robustness and effectiveness of the proposed method.

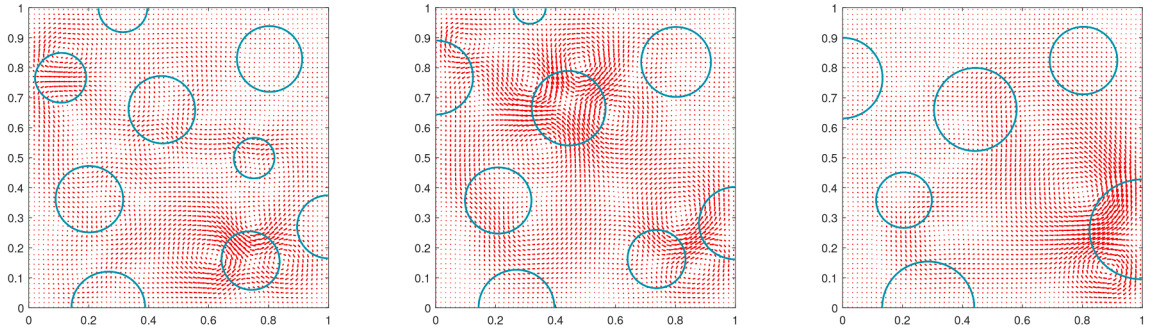
First, we set the computational domain $\Omega = (0, 1) \times (0, 2)$ and choose the initial conditions

$$\phi(x, y, 0) = \tanh\left(\frac{0.32 - \sqrt{(x - 0.5)^2 + (y - 2.1)^2}}{\epsilon}\right), \quad \boldsymbol{u}_0(x, y) = \boldsymbol{0}$$

with parameters $\epsilon = 10^{-2}$, $M = 10^{-2}$, $\lambda = 10^{-3}$, $\bar{\phi} = 0$, $\chi = 1$, $N = 256$ and $\Delta t = 4 \times 10^{-4}$. To assess the Reynolds-number effect, we also vary the kinematic viscosity $\nu = 0.1,\ 0.02$. Snapshots of the phase variable are shown in Figs. 12 and 13 respectively for different ν. In both cases, the droplet elongates, forming a slender neck that eventually leads to pinch-off, after which the detached drop descends. Moreover, as the effective Reynolds number (proportional to $1/\nu$) increases, the onset of pinch-off advances and the falling speed



(a) from left to right: $t = 0.25, 1, 2$



(b) from left to right: $t = 3, 5, 10$

Fig. 8. Visualization of the simulated velocity field at different times.

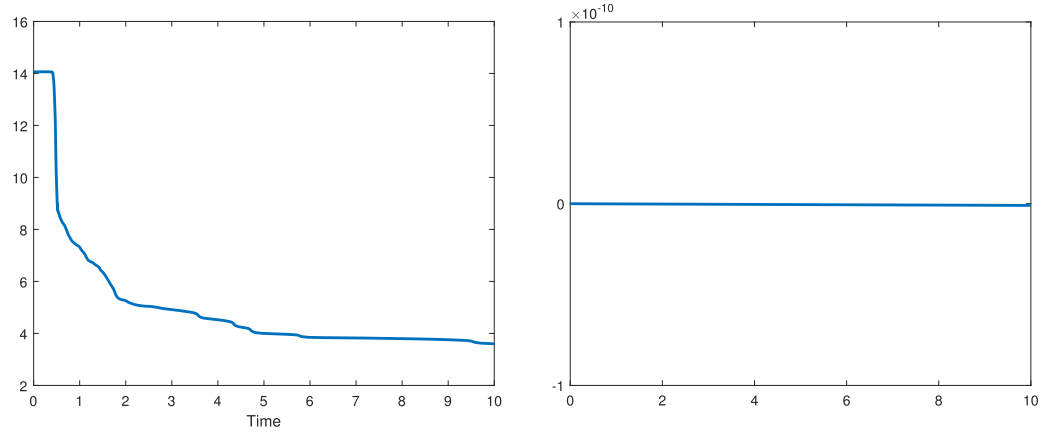
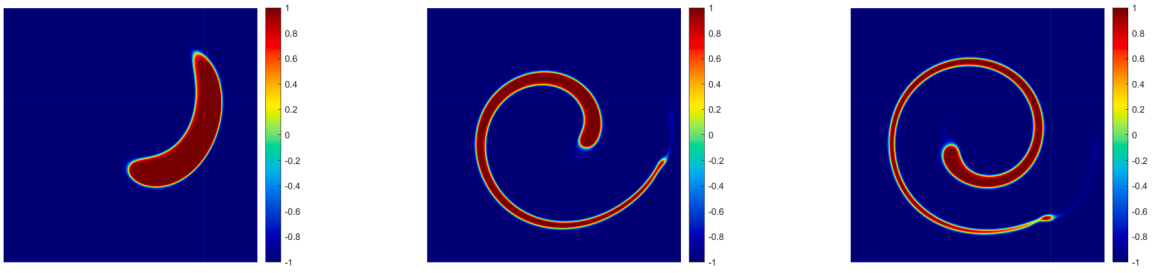


Fig. 9. Evolution of the energy (left) and mass increment (right).

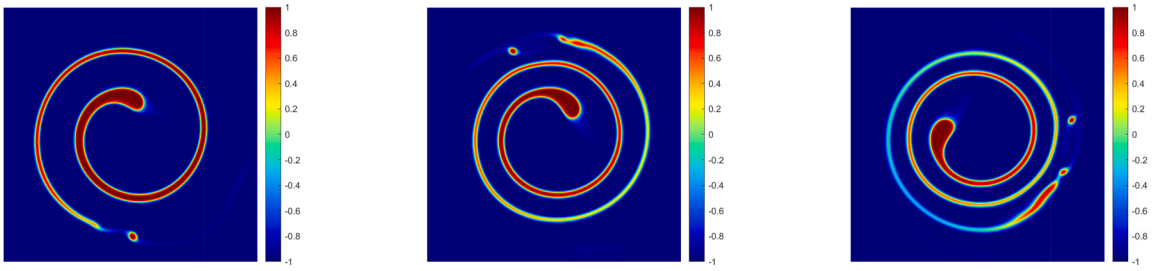
increases, which are consistent with [53,54]. The corresponding velocity fields are plotted in Figs. 14 and 15, which also illustrate these intriguing physical processes.

Next, we set the computational domain $\Omega = (0, 1) \times (0, 1)$ and choose the initial conditions

$$\phi(x, y, 0) = \tanh\left(\frac{0.32 - \sqrt{(x - 0.5)^2 + (y - 1.1)^2}}{2\epsilon}\right), \quad \boldsymbol{u}_0(x, y) = \boldsymbol{0}$$



(a) from left to right: $t = 0.1, 0.3, 0.4$



(b) from left to right: $t = 0.5, 0.8, 1$

Fig. 10. Snapshots of the swirling flow at different times.

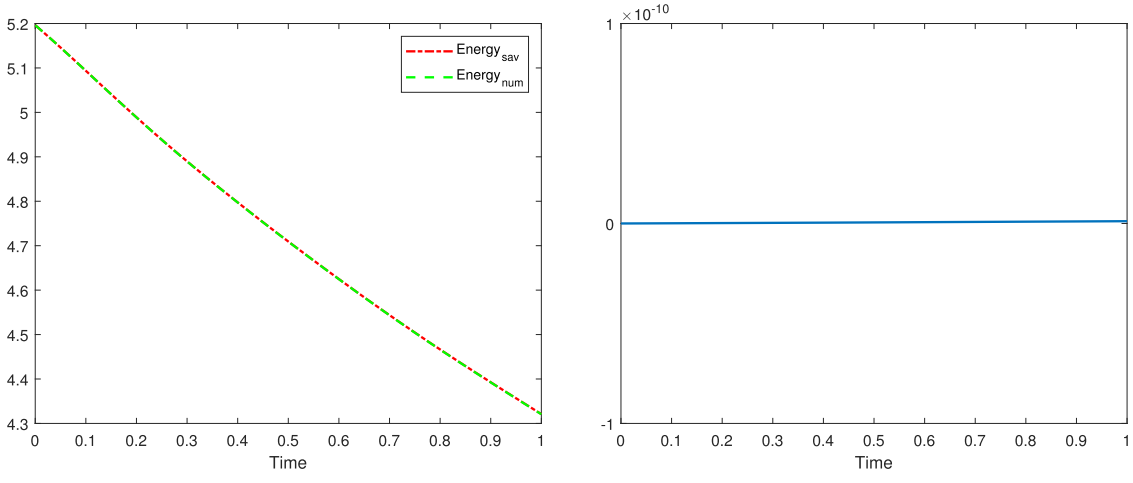


Fig. 11. Evolution of the SAV energy and its numerical approximation (left), and the mass increment (right).

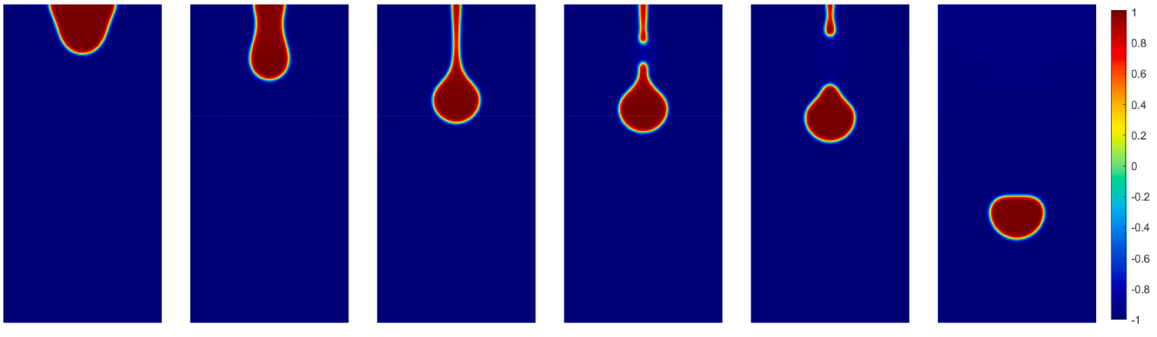


Fig. 12. Evolution of the 2D droplet falling with $\nu = 10^{-1}$ at different times $t = 0.4, 0.8, 1.3, 1.4, 1.5, 2.5$ from left to right.

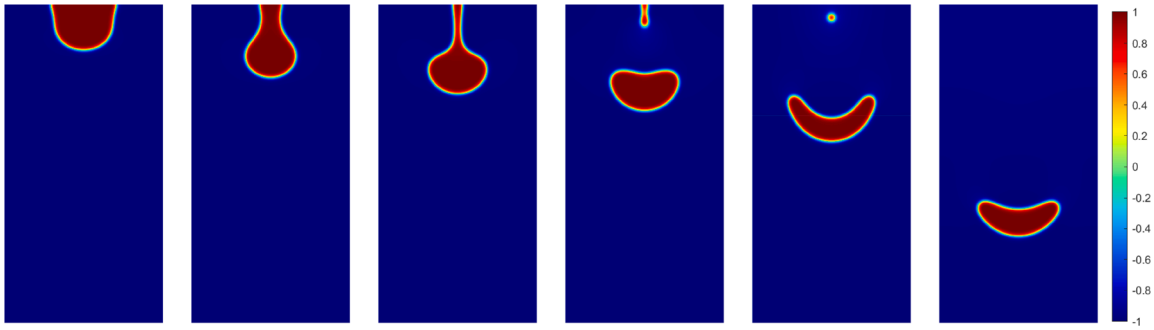


Fig. 13. Evolution of the 2D droplet falling with $\nu = 2 \times 10^{-2}$ at different times $t = 0.2, 0.4, 0.5, 0.6, 0.8, 1.5$ from left to right.

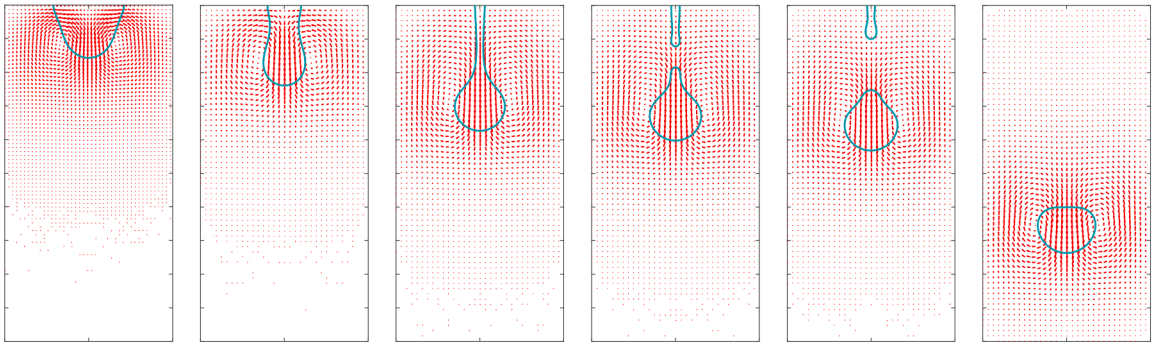


Fig. 14. Visualization of the simulated velocity field with $\nu = 10^{-1}$ at different times $t = 0.4, 0.8, 1.3, 1.4, 1.5, 2.5$ from left to right.

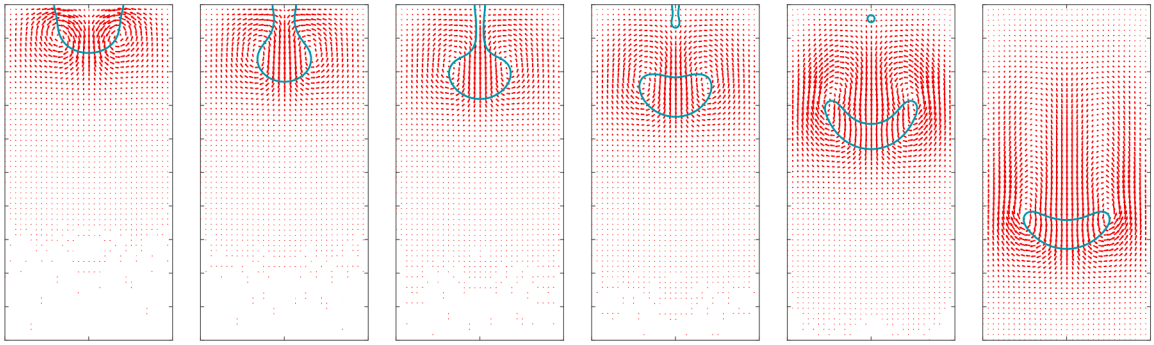


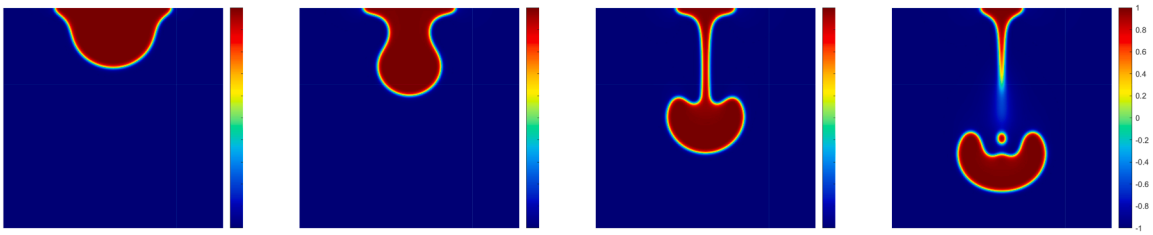
Fig. 15. Visualization of the simulated velocity field with $\nu = 2 \times 10^{-2}$ at different times $t = 0.2, 0.4, 0.5, 0.6, 0.8, 1.5$ from left to right.

with parameters $\epsilon = 7.5 \times 10^{-3}$, $M = 10^{-1}$, $\lambda = 10^{-5}$, $\nu = 5 \times 10^{-2}$, $\chi = 1$, $N = 256$ and $\Delta t = 10^{-3}$. Here $\bar{\phi}$ is the spatial average of ϕ. Snapshots of the phase function and the corresponding velocity are shown in Fig. 16. Pronounced spike-like protrusions emerge when the liquid filament becomes highly elongated prior to detachment. This phenomenon is consistent with that reported in Li et al. [54].

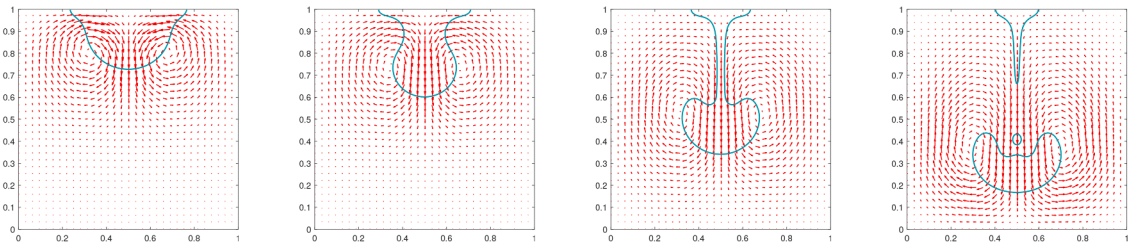
Finally, we extend the simulation to 3D domain $\Omega = (0, 1) \times (0, 1) \times (0, 1.5)$. The initial conditions are chosen as

$$\phi(x, y, z, 0) = \tanh\left(\frac{0.32 - \sqrt{(x-0.5)^2 + (y-0.5)^2 + (z-1.6)^2}}{\epsilon}\right), \quad \boldsymbol{u}_0(x, y, z) = \boldsymbol{0}$$

with parameters $\epsilon = 2.5 \times 10^{-2}$, $M = 10^{-2}$, $\lambda = 10^{-3}$, $\bar{\phi} = 0$, $\nu = 2 \times 10^{-2}$, $\chi = 1$, $N = 64$ and $\Delta t = 5 \times 10^{-4}$. Isosurface snapshots of the interface (i.e., $\phi = 0$) at selected times are shown in Fig. 17. As seen, an elongated liquid filament develops and subsequently undergoes pinch-off at approximately $t \approx 0.7$, yielding a bowl-shaped daughter droplet. This evolution mirrors the planar results and is consistent with the behavior reported in Doan et al. [53].

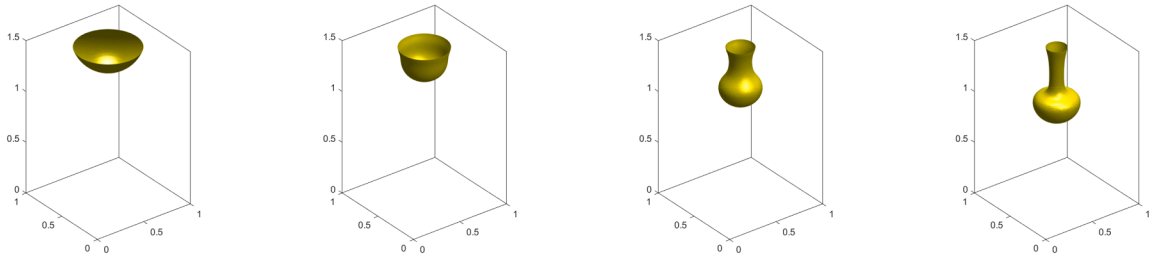


(a) Phase function at $t = 0.2, 0.4, 0.7, 0.9$

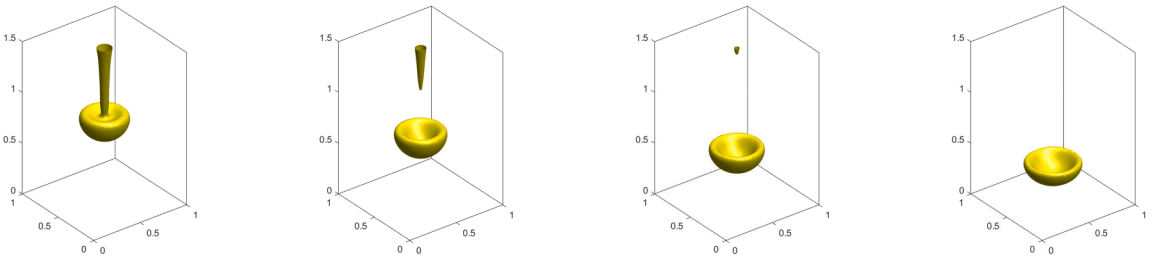


(b) Velocity field at $t = 0.2, 0.4, 0.7, 0.9$

Fig. 16. Snapshots of the phase function and corresponding velocity field at different times.



(a) from left to right: $t = 0, 0.2, 0.4, 0.5$



(b) from left to right: $t = 0.6, 0.7, 0.8, 0.9$

Fig. 17. Evolution of the 3D droplet falling with $\nu = 2 \times 10^{-2}$ at different times.

5. Conclusions

In this paper, we constructed first- and second-order time discretization schemes for the CHNS system based on the MSAV method, the ZEC approach, and the GU method. Both schemes are linear, efficient, mass-conserving and unconditionally energy stable, requiring only the solution of a sequence of Poisson-type equations with constant coefficients at each time step. In contrast to traditional projection-type schemes, our methods avoid imposing artificial boundary conditions on the pressure, thereby preventing the introduction of boundary layers that could reduce accuracy. We rigorously established the error estimates for the first-order scheme, achieving an optimal temporal convergence rate for both the phase variable and the velocity. In fact, the introduced Gauge variable presented significant challenges in error analysis, particularly in formulating the error equation. To overcome these challenges, we

constructed an auxiliary "momentum" equation, which facilitated the attainment of optimal temporal convergence. Finally, ample numerical experiments validated the accuracy and reliability of the developed schemes.

In this work, we focus on the first-order temporal semi-discretization scheme due to its complexity, leaving the fully-discrete error analysis and the estimates of the second-order scheme for our future research. Moreover, robust error estimates with respect to the physical parameters ν and ϵ appear promising [42]. We also plan to explore potential applications and extensions of our numerical methods to realistic physics-based models with high-performance computing, such as models with mismatched densities, variable viscosity, and even variable mobility [19,48,49].

CRediT authorship contribution statement

Baolin Kuang: Writing – review & editing, Writing – original draft, Software, Methodology; **Hongfei Fu:** Writing – review & editing, Methodology, Conceptualization; **Rihui Lan:** Writing – review & editing, Software, Methodology, Conceptualization; **Lili Ju:** Writing – review & editing, Methodology, Conceptualization.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Proof of Lemma 7

Subtracting (2.7a) from (2.2a) at t_{n+1}, we obtain the error equation as follows:

$$\frac{e_\phi^{n+1}-e_\phi^n}{\Delta t}-M\Delta e_\mu^{n+1}=R_\phi^{n+1}+E_N^{n+1}, \tag{A.1}$$

where

$$R_\phi^{n+1}=\frac{\phi(t_{n+1})-\phi(t_n)}{\Delta t}-\phi_t(t_{n+1})=\frac{1}{\Delta t}\int_{t_n}^{t_{n+1}}(t-t_n)\phi_{tt}dt,$$

$$E_N^{n+1}=\frac{R^{n+1}}{\sqrt{E_1(\phi^n)+\delta_0}}(\boldsymbol{u}^n\cdot\nabla)\phi^n-\frac{R(t_{n+1})}{\sqrt{E_1(\phi(t_{n+1}))+\delta_0}}(\boldsymbol{u}(t_{n+1})\cdot\nabla)\phi(t_{n+1}).$$

Taking the inner product of (A.1) with e_μ^{n+1} and e_ϕ^{n+1}, respectively, we have

$$\left(\frac{e_\phi^{n+1}-e_\phi^n}{\Delta t},e_\mu^{n+1}\right)+M\|\nabla e_\mu^{n+1}\|^2=(R_\phi^{n+1},e_\mu^{n+1})+(E_N^{n+1},e_\mu^{n+1}), \tag{A.2}$$

and

$$\frac{\|e_\phi^{n+1}\|^2-\|e_\phi^n\|^2+\|\delta_t e_\phi^{n+1}\|^2}{2\Delta t}=(R_\phi^{n+1},e_\phi^{n+1})+(E_N^{n+1},e_\phi^{n+1})-M(\nabla e_\mu^{n+1},\nabla e_\phi^{n+1}). \tag{A.3}$$

Subtracting (2.7b) from (2.2b) at t_{n+1}, we get

$$e_\mu^{n+1}=-\Delta e_\phi^{n+1}+\frac{e_R^{n+1}}{\sqrt{E_1(\phi^n)+\delta_0}}F'(\phi^n)+E_F^{n+1}, \tag{A.4}$$

where

$$E_F^{n+1}=R(t_{n+1})\left(\frac{F'(\phi(t_{n+1}))}{\sqrt{E_1(\phi(t_{n+1}))+\delta_0}}-\frac{F'(\phi^n)}{\sqrt{E_1(\phi^n)+\delta_0}}\right).$$

Taking the inner product of (A.4) with Me_μ^{n+1} and $\frac{1}{\Delta t}\delta_t e_\phi^{n+1}$, respectively, we obtain

$$M\|e_\mu^{n+1}\|^2 = M(\nabla e_\mu^{n+1}, \nabla e_\phi^{n+1}) + M\frac{e_R^{n+1}}{\sqrt{E_1(\phi^n)+\delta_0}}(F'(\phi^n), e_\mu^{n+1}) + M(E_F^{n+1}, e_\mu^{n+1}),$$

and

$$\left(e_\mu^{n+1}, \frac{1}{\Delta t}\delta_t e_\phi^{n+1}\right) = \frac{1}{2\Delta t}\left(\|\nabla e_\phi^{n+1}\|^2 - \|\nabla e_\phi^n\|^2 + \|\delta_t \nabla e_\phi^{n+1}\|^2\right) + \left(E_F^{n+1}, \frac{1}{\Delta t}\delta_t e_\phi^{n+1}\right) + \frac{e_R^{n+1}}{\sqrt{E_1(\phi^n)+\delta_0}}\left(F'(\phi^n), \frac{1}{\Delta t}\delta_t e_\phi^{n+1}\right). \tag{A.5}$$

Combining (A.2)–(A.5), we can derive

$$\begin{aligned}
&\frac{1}{2\Delta t}\left(\|\nabla e_\phi^{n+1}\|^2 - \|\nabla e_\phi^n\|^2 + \|\delta_t \nabla e_\phi^{n+1}\|^2\right) + M\|\nabla e_\mu^{n+1}\|^2 + \frac{1}{2\Delta t}\left(\|e_\phi^{n+1}\|^2 - \|e_\phi^n\|^2 + \|\delta_t e_\phi^{n+1}\|^2\right) + M\|e_\mu^{n+1}\|^2 \\
&= -\frac{e_R^{n+1}}{\sqrt{E_1(\phi^n)+\delta_0}}\left(F'(\phi^n), \frac{1}{\Delta t}\delta_t e_\phi^{n+1}\right) + M\frac{e_R^{n+1}}{\sqrt{E_1(\phi^n)+\delta_0}}(F'(\phi^n), e_\mu^{n+1}) + \left(E_F^{n+1}, Me_\mu^{n+1} - \frac{1}{\Delta t}\delta_t e_\phi^{n+1}\right) + (E_N^{n+1}, e_\mu^{n+1} + e_\phi^{n+1}) \\
&\quad + (R_\phi^{n+1}, e_\mu^{n+1}) + (R_\phi^{n+1}, e_\phi^{n+1}) =: \sum_{i=1}^{6}\mathcal{A}_i.
\end{aligned} \tag{A.6}$$

The second term on the right-hand side of (A.6) can be estimated as

$$\mathcal{A}_2 \le C|e_R^{n+1}|^2 + \frac{M}{8}\|e_\mu^{n+1}\|^2,$$

where the Cauchy-Schwarz inequality and the boundedness of ϕ^n are utilized. Recalling (A.1), the third term on the right-hand side of (A.6) can be rewritten as

$$\mathcal{A}_3 = \left(E_F^{n+1}, Me_\mu^{n+1} - M\Delta e_\mu^{n+1} - R_\phi^{n+1} - E_N^{n+1}\right) = \left(E_F^{n+1}, Me_\mu^{n+1} - R_\phi^{n+1} - E_N^{n+1}\right) + M\left(\nabla E_F^{n+1}, \nabla e_\mu^{n+1}\right) = \mathcal{A}_{3,1} + \mathcal{A}_{3,2}. \tag{A.7}$$

Using the definition of E_F^{n+1}, we can rewrite it as follows:

$$E_F^{n+1} = \frac{R(t_{n+1})F'(\phi(t_{n+1}))\left(E_1(\phi^n) - E_1(\phi(t_{n+1}))\right)}{\sqrt{E_1(\phi^n)+\delta_0}\sqrt{E_1(\phi(t_{n+1}))+\delta_0}\left(\sqrt{E_1(\phi^n)+\delta_0}+\sqrt{E_1(\phi(t_{n+1}))+\delta_0}\right)} + \frac{R(t_{n+1})}{\sqrt{E_1(\phi^n)+\delta_0}}\left(F'\left(\phi(t_{n+1})\right) - F'(\phi^n)\right).$$

Thus, we obtain

$$\|E_F^{n+1}\| \le C\|e_\phi^n\| + C\|\phi\|_{H^1(J_n;L^2(\Omega))}\sqrt{\Delta t}, \quad \|\nabla E_F^{n+1}\| \le C\|e_\phi^n\|_1 + C\|\phi\|_{H^1(J_n;H^1(\Omega))}\sqrt{\Delta t}.$$

Making use of the product rule and the incompressibility, we have

$$\begin{aligned}
E_N^{n+1} &= \frac{R^{n+1}}{\sqrt{E_1(\phi^n)+\delta_0}}(\boldsymbol{u}^n\cdot\nabla)\phi^n - \frac{R(t_{n+1})}{\sqrt{E_1(\phi(t_{n+1}))+\delta_0}}\left(\boldsymbol{u}(t_{n+1})\cdot\nabla\right)\phi(t_{n+1}) \\
&= \frac{R(t_{n+1})}{\sqrt{E_1(\phi(t_{n+1}))+\delta_0}}\nabla\cdot\left(\boldsymbol{u}(t_n)\phi(t_n) - \boldsymbol{u}(t_{n+1})\phi(t_{n+1})\right) + \left(\frac{R^{n+1}}{\sqrt{E_1(\phi^n)+\delta_0}} - \frac{R(t_{n+1})}{\sqrt{E_1(\phi(t_{n+1}))+\delta_0}}\right)\nabla\cdot\left(\boldsymbol{u}(t_n)\phi(t_n)\right) \\
&\quad - \frac{R^{n+1}}{\sqrt{E_1(\phi^n)+\delta_0}}\nabla\cdot\left(\boldsymbol{u}(t_n)e_\phi^n\right) - \frac{R^{n+1}}{\sqrt{E_1(\phi^n)+\delta_0}}\nabla\cdot\left(e_{\boldsymbol{u}}^n\phi^n\right).
\end{aligned}$$

Thanks to Lemmas 6, 4 and 2.9, $\mathcal{A}_{3,1}$ can be bounded by

$$\begin{aligned}
\mathcal{A}_{3,1} &\le C\|E_F^{n+1}\|\left(\|e_\mu^{n+1}\| + \|R_\phi^{n+1}\|\right) + C\|\nabla E_F^{n+1}\|(\|e_\phi^n\|_{L^4} + \|e_{\boldsymbol{u}}^n\|) \\
&\quad + C\|\nabla E_F^{n+1}\|\left(\|\boldsymbol{u}\|_{H^1(J_n;H^1(\Omega))} + \|\phi\|_{H^1(J_n;H^1(\Omega))}\right)\sqrt{\Delta t} + C\|\nabla E_F^{n+1}\|\left(|e_R^{n+1}| + \|e_\phi^n\|_{L^4} + \|\phi\|_{H^1(J_n;L^2(\Omega))}\sqrt{\Delta t}\right) \\
&\le C|e_R^{n+1}|^2 + \frac{M}{8}\|e_\mu^{n+1}\|^2 + C\|e_{\boldsymbol{u}}^n\|^2 + C\|e_\phi^n\|_1^2 + C\left(\|\boldsymbol{u}\|^2_{H^1(J_n;H^1(\Omega))} + \|\phi\|^2_{H^1(J_n;H^1(\Omega))} + \|\phi\|^2_{H^2(J_n;L^2(\Omega))}\right)\Delta t.
\end{aligned} \tag{A.8}$$

In addition, $\mathcal{A}_{3,2}$ can be estimated by

$$\mathcal{A}_{3,2} \le C\|e_\phi^n\|_1^2 + \frac{M}{8}\|\nabla e_\mu^{n+1}\|^2 + C\|\phi\|^2_{H^1(J_n;H^1(\Omega))}\Delta t.$$

Similar to (A.8), the fourth term on the right-hand side of (A.6) can be bounded by

$$\begin{aligned}
\mathcal{A}_4 &\le C\|\nabla e_\mu^{n+1} + \nabla e_\phi^{n+1}\|\left(\|\boldsymbol{u}\|_{H^1(J_n;H^1(\Omega))} + \|\phi\|_{H^1(J_n;H^1(\Omega))}\right)\sqrt{\Delta t} + C\|\nabla e_\mu^{n+1} + \nabla e_\phi^{n+1}\|\left(|e_R^{n+1}| + \|e_\phi^n\|_{L^4} + \|\phi\|_{H^1(J_n;L^2(\Omega))}\sqrt{\Delta t}\right) \\
&\quad + C\|\nabla e_\mu^{n+1} + \nabla e_\phi^{n+1}\|(\|e_\phi^n\|_{L^4} + \|e_{\boldsymbol{u}}^n\|) \\
&\le C|e_R^{n+1}|^2 + C\|e_\phi^n\|_1^2 + C\|e_{\boldsymbol{u}}^n\|^2 + \frac{M}{8}\|\nabla e_\mu^{n+1}\|^2 + C\|e_\phi^{n+1}\|_1^2 + C\left(\|\boldsymbol{u}\|^2_{H^1(J_n;H^1(\Omega))} + \|\phi\|^2_{H^1(J_n;H^1(\Omega))}\right)\Delta t.
\end{aligned}$$

Applying the Cauchy-Schwarz inequality, we have

$$\mathcal{A}_5 + \mathcal{A}_6 \le C\|\phi\|^2_{H^2(J_n;L^2(\Omega))}\Delta t + C\|e_\phi^{n+1}\|^2 + \frac{M}{8}\|e_\mu^{n+1}\|^2.$$

Combining (A.6) with the estimates of $\mathcal{A}_2$–$\mathcal{A}_6$, we obtain

$$\begin{aligned}
&\frac{1}{2\Delta t}\left(\|e_\phi^{n+1}\|_1^2-\|e_\phi^n\|_1^2\right)+\frac{3M}{4}\|\nabla e_\mu^{n+1}\|^2+\frac{5M}{8}\|e_\mu^{n+1}\|^2\\
&\le-\frac{e_R^{n+1}}{\sqrt{E_1(\phi^n)+\delta_0}}\left(F'(\phi^n),\frac{1}{\Delta t}\delta_t e_\phi^{n+1}\right)+C|e_R^{n+1}|^2+C\|e_\phi^n\|_1^2+C\|e_\phi^{n+1}\|_1^2\\
&\quad+C\left\|e_{\boldsymbol{u}}^n\right\|^2+C\left(\|\boldsymbol{u}\|^2_{H^1(J_n;H^1(\Omega))}+\|\phi\|^2_{H^1(J_n;H^1(\Omega))}+\|\phi\|^2_{H^2(J_n;L^2(\Omega))}\right)\Delta t.
\end{aligned}\tag{A.9}$$

Next, we continue the estimate by establishing an error equation corresponding to the auxiliary variable R^{n+1}. Subtracting (2.7c) from (2.2c) at t_{n+1} and multiplying the equation by $2e_R^{n+1}$ lead to

$$\begin{aligned}
&\frac{1}{\Delta t}\left(|e_R^{n+1}|^2-\left|e_R^n\right|^2+|\delta_t e_R^{n+1}|^2\right)\\
&=\frac{e_R^{n+1}}{\sqrt{E_1(\phi^n)+\delta_0}}\left(F'(\phi^n),\frac{1}{\Delta t}\delta_t e_\phi^{n+1}\right)-\frac{e_R^{n+1}}{\sqrt{E_1(\phi^n)+\delta_0}}\left(F'(\phi^n),R_\phi^{n+1}\right)+\frac{e_R^{n+1}}{\sqrt{E_1(\phi^n)+\delta_0}}\left(F'\left(\phi(t_{n+1})\right)-F'(\phi^n),\phi_t(t_{n+1})\right)\\
&\quad+\left(\frac{e_R^{n+1}}{\sqrt{E_1(\phi(t_{n+1}))+\delta_0}}-\frac{e_R^{n+1}}{\sqrt{E_1(\phi^n)+\delta_0}}\right)\left(F'\left(\phi(t_{n+1})\right),\phi_t(t_{n+1})\right)\\
&\quad+\frac{e_R^{n+1}}{\sqrt{E_1(\phi^n)+\delta_0}}\left(\left(\hat{\boldsymbol{u}}^{n+1},\mu^n\nabla\phi^n\right)-\left(\mu^{n+1},\boldsymbol{u}^n\cdot\nabla\phi^n\right)\right)+2R_r^{n+1}e_R^{n+1}=:\sum_{i=1}^{6}\mathcal{B}_i,
\end{aligned}\tag{A.10}$$

where

$$R_r^{n+1}=\frac{R(t_{n+1})-R\left(t_n\right)}{\Delta t}-R_t(t_{n+1})=\frac{1}{\Delta t}\int_{t_n}^{t_{n+1}}\left(t-t_n\right)R_{tt}dt.$$

By using the Cauchy–Schwarz inequality, the second term on the right-hand side of (A.10) can be recast into

$$\mathcal{B}_2\le C|e_R^{n+1}|^2+C\|\phi\|^2_{H^2(J_n;L^2(\Omega))}\Delta t.$$

Similarly, the third and fourth terms on the right-hand side of (A.10) can be estimated by

$$\mathcal{B}_3+\mathcal{B}_4\le C|e_R^{n+1}|^2+C\|e_\phi^n\|^2+C\|\phi\|^2_{H^1(J_n;L^2(\Omega))}\Delta t.$$

Combining Remark 2.9 and Lemma 6, the fifth term on the right-hand side of (A.10) can be bounded by

$$\begin{aligned}
\mathcal{B}_5&=\frac{e_R^{n+1}}{\sqrt{E_1(\phi^n)+\delta_0}}\left(\left(\mu(t_{n+1}),\boldsymbol{u}(t_{n+1})\cdot\nabla\phi(t_{n+1})\right)-\left(\mu^{n+1},\boldsymbol{u}^n\cdot\nabla\phi^n\right)\right)\\
&\quad+\frac{e_R^{n+1}}{\sqrt{E_1(\phi^n)+\delta_0}}\left(\left(\hat{\boldsymbol{u}}^{n+1},\mu^n\nabla\phi^n\right)-\left(\boldsymbol{u}(t_{n+1}),\mu(t_{n+1})\nabla\phi(t_{n+1})\right)\right)\\
&=\frac{e_R^{n+1}}{\sqrt{E_1(\phi^n)+\delta_0}}\left(e_\mu^{n+1},\boldsymbol{u}^n\cdot\nabla\phi^n\right)+\frac{e_R^{n+1}}{\sqrt{E_1(\phi^n)+\delta_0}}\left(\mu(t_{n+1}),\left(e_{\boldsymbol{u}}^n+\int_{t_n}^{t_{n+1}}\boldsymbol{u}_t dt\right)\cdot\nabla\phi^n\right)\\
&\quad-\frac{e_R^{n+1}}{\sqrt{E_1(\phi^n)+\delta_0}}\left(\hat{e}_{\boldsymbol{u}}^{n+1},\mu^n\nabla\phi^n\right)-\frac{e_R^{n+1}}{\sqrt{E_1(\phi^n)+\delta_0}}\left(\boldsymbol{u}(t_{n+1}),\left(e_\mu^n+\int_{t_n}^{t_{n+1}}\mu_t dt\right)\nabla\phi^n\right)\\
&\le-\frac{e_R^{n+1}}{\sqrt{E_1(\phi^n)+\delta_0}}\left(\hat{e}_{\boldsymbol{u}}^{n+1},\mu^n\nabla\phi^n\right)+C|e_R^{n+1}|^2+\frac{M}{8}\left(\|\nabla e_\mu^{n+1}\|^2+\left\|e_\mu^n\right\|^2\right)\\
&\quad+C\|e_{\boldsymbol{u}}^n\|^2+C\left(\|\boldsymbol{u}\|^2_{H^1(J_n;L^2(\Omega))}+\|\mu\|^2_{H^1(J_n;L^2(\Omega))}\right)\Delta t.
\end{aligned}$$

For the last term on the right-hand side of (A.10), we have

$$\mathcal{B}_6\le C|e_R^{n+1}|^2+C\|R\|^2_{H^2(J_n)}\Delta t.$$

Substituting the estimates of $\mathcal{B}_2$–$\mathcal{B}_6$ into (A.10), we have

$$\begin{aligned}
\frac{1}{\Delta t}\left(|e_R^{n+1}|^2-\left|e_R^n\right|^2\right)&\le\frac{e_R^{n+1}}{\sqrt{E_1(\phi^n)+\delta_0}}\left(F'(\phi^n),\frac{1}{\Delta t}\delta_t e_\phi^{n+1}\right)+C|e_R^{n+1}|^2+\frac{M}{8}(\|\nabla e_\mu^{n+1}\|^2+\|e_\mu^n\|^2)\\
&\quad-\frac{e_R^{n+1}}{\sqrt{E_1(\phi^n)+\delta_0}}\left(\hat{e}_{\boldsymbol{u}}^{n+1},\mu^n\nabla\phi^n\right)+C\left\|e_{\boldsymbol{u}}^n\right\|^2+C\left\|e_\phi^n\right\|^2+C\|\phi\|^2_{H^2(J_n;L^2(\Omega))}\Delta t\\
&\quad+C\left(\|\boldsymbol{u}\|^2_{H^1(J_n;L^2(\Omega))}+\|\mu\|^2_{H^1(J_n;L^2(\Omega))}+\|\phi\|^2_{H^1(J_n;L^2(\Omega))}+\|R\|^2_{H^2(J_n)}\right)\Delta t.
\end{aligned}\tag{A.11}$$

Combining (A.11) with (A.9) gives

$$\begin{aligned}
&\frac{1}{2\Delta t}\left(\|e_\phi^{n+1}\|_1^2-\|e_\phi^n\|_1^2\right)+\frac{5M}{8}\|e_\mu^{n+1}\|_1^2+\frac{1}{\Delta t}\left(|e_R^{n+1}|^2-\left|e_R^n\right|^2\right)\\
&\quad\le C\|e_\phi^n\|_1^2+C\|e_\phi^{n+1}\|_1^2+C|e_R^{n+1}|^2+C\|\boldsymbol{u}\|^2_{H^1(J_n;H^1(\Omega))}\Delta t+\frac{M}{8}\|e_\mu^n\|^2\\
&\quad\quad-\frac{e_R^{n+1}}{\sqrt{E_1(\phi^n)+\delta_0}}\left(\hat{e}_{\boldsymbol{u}}^{n+1},\mu^n\nabla\phi^n\right)+C\left\|e_{\boldsymbol{u}}^n\right\|^2+C\|R\|^2_{H^2(J_n)}\Delta t\\
&\quad\quad+C\left(\|\phi\|^2_{H^1(J_n;H^1(\Omega))}+\|\phi\|^2_{H^2(J_n;L^2(\Omega))}+\|\mu\|^2_{H^1(J_n;L^2(\Omega))}\right)\Delta t,
\end{aligned}$$

which completes the proof of Lemma 7.

Appendix B. Proof of Lemma 13

Subtracting (2.7d) from (3.7), we have

$$\begin{aligned}
\frac{\widehat{\boldsymbol{E}}^{n+1}-\boldsymbol{E}^n}{\Delta t}-\nu\Delta\widehat{\boldsymbol{E}}^{n+1}&=\boldsymbol{R}_u^{n+1}+\boldsymbol{K}^{n+1}-\nu\nabla\left(\Delta\psi\left(t_{n+1}\right)-s^n\right)+\exp\left(\frac{t_{n+1}}{T}\right)\left(Q^{n+1}(\boldsymbol{u}^n\cdot\nabla)\boldsymbol{u}^n-Q(t_{n+1})\left(\boldsymbol{u}\left(t_{n+1}\right)\cdot\nabla\right)\boldsymbol{u}\left(t_{n+1}\right)\right)\\
&\quad+\frac{R(t_{n+1})}{\sqrt{E_1\left(\phi(t_{n+1})\right)+\delta_0}}\mu(t_{n+1})\nabla\phi(t_{n+1})-\frac{R^{n+1}}{\sqrt{E_1(\phi^n)+\delta_0}}\mu^n\nabla\phi^n.
\end{aligned}\tag{B.1}$$

Taking the inner product of (B.1) with $\widehat{\boldsymbol{E}}^{n+1}$ and combining Lemma 12, we obtain

$$\begin{aligned}
&\frac{\left\|\boldsymbol{E}^{n+1}\right\|^2-\|\boldsymbol{E}^n\|^2+\|\widehat{\boldsymbol{E}}^{n+1}-\boldsymbol{E}^n\|^2}{2\Delta t}+\nu\|\nabla\widehat{\boldsymbol{E}}^{n+1}\|^2+\frac{\|\nabla\gamma^{n+1}\|^2}{2\Delta t}\\
&\quad=(\boldsymbol{R}_u^{n+1}+\boldsymbol{K}^{n+1},\widehat{\boldsymbol{E}}^{n+1})+\nu(\Delta\psi\left(t_{n+1}\right)-s^n,\nabla\cdot\widehat{\boldsymbol{E}}^{n+1})\\
&\quad\quad+\exp\left(\frac{t_{n+1}}{T}\right)\left(Q^{n+1}(\boldsymbol{u}^n\cdot\nabla)\boldsymbol{u}^n-Q(t_{n+1})\left(\boldsymbol{u}\left(t_{n+1}\right)\cdot\nabla\right)\boldsymbol{u}\left(t_{n+1}\right),\widehat{\boldsymbol{E}}^{n+1}\right)\\
&\quad\quad+\left(\frac{R(t_{n+1})}{\sqrt{E_1\left(\phi(t_{n+1})\right)+\delta_0}}\mu(t_{n+1})\nabla\phi(t_{n+1})-\frac{R^{n+1}}{\sqrt{E_1(\phi^n)+\delta_0}}\mu^n\nabla\phi^n,\widehat{\boldsymbol{E}}^{n+1}\right)=:\sum_{i=1}^4C_i.
\end{aligned}\tag{B.2}$$

Thanks to (3.5) and Lemma 8, the first term on the right-hand side of (B.2) can be estimated by

$$\begin{aligned}
C_1&\le\frac{\nu}{20}\|\nabla\widehat{\boldsymbol{E}}^{n+1}\|^2+C\left\|\nabla\widehat{\boldsymbol{v}}\right\|^2+C\|\boldsymbol{u}\|^2_{H^2(J_n;H^{-1}(\Omega))}\Delta t+C\left\|\boldsymbol{v}_t\right\|^2_{L^2(J_n;L^2(\Omega))}\\
&\le\frac{\nu}{20}\|\nabla\widehat{\boldsymbol{E}}^{n+1}\|^2+C\|\boldsymbol{u}\|^2_{H^2(J_n;H^{-1}(\Omega))}\Delta t+C\left\|\boldsymbol{v}_t\right\|^2_{L^2(J_n;L^2(\Omega))}+C(\Delta t)^2.
\end{aligned}\tag{B.3}$$

where the Hölder inequality and Lemma 1 (i) are utilized.

Noticing (2.7f) and the definition of ε, Lemma 9 implies

$$\nabla\cdot\widehat{\boldsymbol{E}}^{n+1}=-\Delta t\Delta\psi_t(t_{n+1})+\delta_t s^{n+1}=-\int_{t_n}^{t_{n+1}}\left(t-t_n\right)\Delta\psi_{tt}dt-\delta_t\varepsilon^{n+1},\tag{B.4}$$

then combining Lemma 1 (i), we have

$$\|\delta_t\varepsilon^{n+1}\|^2\le C(\Delta t)^3\int_{t_n}^{t_{n+1}}\left\|\Delta\psi_{tt}\right\|^2dt+\|\nabla\widehat{\boldsymbol{E}}^{n+1}\|^2.\tag{B.5}$$

Using (B.4) and (B.5) and the definition of ε^n given in (3.8), the second term on the right-hand side of (B.2) can be bounded by

$$\begin{aligned}
C_2&=\nu\left(\Delta\left(\psi\left(t_{n+1}\right)-\psi\left(t_n\right)\right)+\varepsilon^n,-\int_{t_n}^{t_{n+1}}\left(t-t_n\right)\Delta\psi_{tt}dt-\delta_t\varepsilon^{n+1}\right)\\
&\le-\nu(\varepsilon^n,\delta_t\varepsilon^{n+1})+\frac{\nu}{40}\|\delta_t\varepsilon^{n+1}\|^2+C\|\varepsilon^n\|^2\Delta t+C\|\psi\|^2_{H^2(J_n;H^2(\Omega))}\Delta t\\
&\le-\frac{\nu}{2}\left(\|\varepsilon^{n+1}\|^2-\|\varepsilon^n\|^2\right)+C\|\varepsilon^n\|^2\Delta t+\frac{21\nu}{40}\|\nabla\widehat{\boldsymbol{E}}^{n+1}\|^2+C\|\psi\|^2_{H^2(J_n;H^2(\Omega))}\Delta t.
\end{aligned}$$

For the third term on the right-hand side of (B.2), we have

$$\begin{aligned}
C_3&=\exp\left(\frac{t_{n+1}}{T}\right)Q(t_{n+1})\left(\left(\boldsymbol{u}^n-\boldsymbol{u}(t_{n+1})\right)\cdot\nabla\boldsymbol{u}(t_{n+1}),\widehat{\boldsymbol{E}}^{n+1}\right)+\exp\left(\frac{t_{n+1}}{T}\right)Q(t_{n+1})\left(\boldsymbol{u}^n\cdot\nabla\left(\boldsymbol{u}^n-\boldsymbol{u}(t_{n+1})\right),\widehat{\boldsymbol{E}}^{n+1}\right)\\
&\quad-\exp\left(\frac{t_{n+1}}{T}\right)e_Q^{n+1}\left((\boldsymbol{u}^n\cdot\nabla)\boldsymbol{u}^n,\widehat{\boldsymbol{E}}^{n+1}\right)=:\sum_{j=1}^3C_{3,j}.
\end{aligned}\tag{B.6}$$

Thanks to Lemma 2, the first term on the right-hand side of (B.6) can be estimated by

$$\begin{aligned}C_{3,1} &\le c_2(1+c_1)\|\boldsymbol{u}(t_{n+1})-\boldsymbol{u}^n\|\|\boldsymbol{u}(t_{n+1})\|_2\|\nabla\hat{\boldsymbol{E}}^{n+1}\|\\ &\le \frac{\nu}{40}\|\nabla\hat{\boldsymbol{E}}^{n+1}\|^2+C\|\boldsymbol{u}\|^2_{L^\infty(0,T;H^2(\Omega))}\|e^n_{\boldsymbol{u}}\|^2+C\|\boldsymbol{u}\|^2_{L^\infty(0,T;H^2(\Omega))}\|\boldsymbol{u}\|^2_{H^1(J_n;L^2(\Omega))}\Delta t\\ &\le \frac{\nu}{40}\|\nabla\hat{\boldsymbol{E}}^{n+1}\|^2+C\|\boldsymbol{E}^n\|^2+C(\Delta t)^2+C\|\boldsymbol{u}\|^2_{H^1(J_n;L^2(\Omega))\cap L^\infty(0,T;H^2(\Omega))}\Delta t,\end{aligned}$$

where Lemma 11 and (3.9) have been applied.

Employing the stability results in Remark 2.9 and (3.9), the second term on the right-hand side of (B.6) can be estimated by

$$\begin{aligned}C_{3,2} &= \exp\Big(\frac{t_{n+1}}{T}\Big)Q(t_{n+1})\big(\boldsymbol{u}^n\cdot\nabla\big(\boldsymbol{u}(t_n)-\boldsymbol{u}(t_{n+1})\big),\hat{\boldsymbol{E}}^{n+1}\big)\\ &\quad+\exp\Big(\frac{t_{n+1}}{T}\Big)Q(t_{n+1})\big(e^n_{\boldsymbol{u}}\cdot\nabla e^n_{\boldsymbol{u}},\hat{\boldsymbol{E}}^{n+1}\big)-\exp\big(\tfrac{t_{n+1}}{T}\big)Q(t_{n+1})\big(\boldsymbol{u}(t_n)\cdot\nabla e^n_{\boldsymbol{u}},\hat{\boldsymbol{E}}^{n+1}\big)\\ &\le c_2(1+c_1)\|\nabla\hat{\boldsymbol{E}}^{n+1}\|\|\boldsymbol{u}^n\|\Big\|\int_{t_n}^{t_{n+1}}\boldsymbol{u}_t dt\Big\|_2+c_2(1+c_1)\|e^n_{\boldsymbol{u}}\|^{1/2}\|e^n_{\boldsymbol{u}}\|_1^{1/2}\|e^n_{\boldsymbol{u}}\|^{1/2}\|e^n_{\boldsymbol{u}}\|_1^{1/2}\|\nabla\hat{\boldsymbol{E}}^{n+1}\|\\ &\quad+c_2(1+c_1)\|\nabla\hat{\boldsymbol{E}}^{n+1}\|\|e^n_{\boldsymbol{u}}\|\|\boldsymbol{u}(t_n)\|_2\\ &\le \frac{\nu}{20}\|\nabla\hat{\boldsymbol{E}}^{n+1}\|^2+C\big(\|\boldsymbol{u}\|^2_{L^\infty(0,T;H^2(\Omega))}+\|e^n_{\boldsymbol{u}}\|_1^2\big)\|e^n_{\boldsymbol{u}}\|^2+C\|\boldsymbol{u}\|^2_{H^1(J_n;H^2(\Omega))\cap L^\infty(0,T;H^2(\Omega))}\Delta t\\ &\le \frac{\nu}{20}\|\nabla\hat{\boldsymbol{E}}^{n+1}\|^2+C\big(1+\|e^n_{\boldsymbol{u}}\|_1^2\big)\|\boldsymbol{E}^n\|^2+C\big(1+\|e^n_{\boldsymbol{u}}\|_1^2\big)(\Delta t)^2+C\|\boldsymbol{u}\|^2_{H^1(J_n;H^2(\Omega))\cap L^\infty(0,T;H^2(\Omega))}\Delta t.\end{aligned}$$

Noticing (3.10), the third term on the right-hand side of (B.6) can be estimated by

$$C_{3,3} = -\exp\Big(\frac{t_{n+1}}{T}\Big)e_Q^{n+1}\big((\boldsymbol{u}^n\cdot\nabla)\boldsymbol{u}^n,\hat{\boldsymbol{v}}(t_{n+1})\big)-\exp\Big(\frac{t_{n+1}}{T}\Big)e_Q^{n+1}\big((\boldsymbol{u}^n\cdot\nabla)\boldsymbol{u}^n,\hat{e}^{n+1}_{\boldsymbol{u}}\big).\tag{B.7}$$

Making use of $\boldsymbol{u}^n=\boldsymbol{u}(t_n)+\boldsymbol{v}(t_n)-\boldsymbol{E}^n$ and (3.5), the first term on the right-hand side of (B.7) can be estimated by

$$\begin{aligned}&-\exp\Big(\frac{t_{n+1}}{T}\Big)e_Q^{n+1}\big((\boldsymbol{u}^n\cdot\nabla)\boldsymbol{u}^n,\hat{\boldsymbol{v}}(t_{n+1})\big)\\ &=-\exp\Big(\frac{t_{n+1}}{T}\Big)e_Q^{n+1}\Big[\big((\boldsymbol{u}(t_n)+\boldsymbol{v}(t_n))\cdot\nabla(\boldsymbol{u}(t_n)+\boldsymbol{v}(t_n)),\hat{\boldsymbol{v}}(t_{n+1})\big)-\big((\boldsymbol{u}(t_n)+\boldsymbol{v}(t_n))\cdot\nabla\boldsymbol{E}^n,\hat{\boldsymbol{v}}(t_{n+1})\big)\\ &\quad-\big(\boldsymbol{E}^n\cdot\nabla\big(\boldsymbol{u}(t_n)+\boldsymbol{v}(t_n)\big),\hat{\boldsymbol{v}}(t_{n+1})\big)+\big(\boldsymbol{E}^n\cdot\nabla\boldsymbol{E}^n,\hat{\boldsymbol{v}}(t_{n+1})\big)\Big]\\ &\le C|e_Q^{n+1}|\|\boldsymbol{u}(t_n)+\boldsymbol{v}(t_n)\|_1^2\|\nabla\hat{\boldsymbol{v}}(t_{n+1})\|+C|e_Q^{n+1}|\|\boldsymbol{u}(t_n)+\boldsymbol{v}(t_n)\|_1\|\nabla\boldsymbol{E}^n\|\|\nabla\hat{\boldsymbol{v}}(t_{n+1})\|\\ &\quad+C|e_Q^{n+1}|\|\nabla\boldsymbol{E}^n\|\|\boldsymbol{u}(t_n)+\boldsymbol{v}(t_n)\|_1\|\nabla\hat{\boldsymbol{v}}(t_{n+1})\|+C|e_Q^{n+1}|\|\nabla\boldsymbol{E}^n\|^{\frac12}\|\boldsymbol{E}^n\|^{\frac12}\|\nabla\boldsymbol{E}^n\|^{\frac12}\|\boldsymbol{E}^n\|^{\frac12}\|\nabla\hat{\boldsymbol{v}}(t_{n+1})\|.\end{aligned}\tag{B.8}$$

Combining Lemmas 12 and 1 (ii), we have

$$\|\Delta\gamma^n\|^2=\|\nabla\cdot\hat{\boldsymbol{E}}^n\|^2\le\|\nabla\hat{\boldsymbol{E}}^n\|^2,\quad\|\nabla\boldsymbol{E}^n\|^2\le C\|\nabla\hat{\boldsymbol{E}}^n\|^2+C\|\Delta\gamma^n\|^2\le 2C\|\nabla\hat{\boldsymbol{E}}^n\|^2.\tag{B.9}$$

In conjunction with Remark 2.9 and $Q(t)=\exp(-\frac{t}{T})$, it yields $|e_Q^{n+1}|\le 1+L$ and

$$\|\boldsymbol{E}^n\|=\|U(t_n)-\boldsymbol{u}^n\|\le C\Delta t+\|e^n_{\boldsymbol{u}}\|\le C.\tag{B.10}$$

Noticing $\|\nabla\hat{\boldsymbol{v}}(t_{n+1})\|\le C\Delta t$ and combining (B.8)–(B.10), we obtain

$$C_{3,3}\le-\exp\Big(\frac{t_{n+1}}{T}\Big)e_Q^{n+1}\big((\boldsymbol{u}^n\cdot\nabla)\boldsymbol{u}^n,\hat{e}^{n+1}_{\boldsymbol{u}}\big)+C|e_Q^{n+1}|^2+\frac{\nu}{8}\|\nabla\hat{\boldsymbol{E}}^n\|^2+C(\Delta t)^2.$$

By using Remark 2.9, Lemmas 6 and 4, the fourth term on the right-hand side of (B.2) can be bounded by

$$\begin{aligned}C_4 &= \Big(\tfrac{R(t_{n+1})}{\sqrt{E_1(\phi^n)+\delta_0}}-\tfrac{R^{n+1}}{\sqrt{E_1(\phi^n)+\delta_0}}\Big)(\mu^n\nabla\phi^n,\hat{\boldsymbol{E}}^{n+1})+R(t_{n+1})\Big(\tfrac{1}{\sqrt{E_1(\phi(t_{n+1}))+\delta_0}}-\tfrac{1}{\sqrt{E_1(\phi^n)+\delta_0}}\Big)(\mu^n\nabla\phi^n,\hat{\boldsymbol{E}}^{n+1})\\ &\quad+\tfrac{R(t_{n+1})}{\sqrt{E_1(\phi(t_{n+1}))+\delta_0}}(\mu(t_{n+1})\nabla\phi(t_{n+1})-\mu^n\nabla\phi^n,\hat{\boldsymbol{E}}^{n+1})\\ &\le \tfrac{e_R^{n+1}}{\sqrt{E_1(\phi^n)+\delta_0}}(\mu^n\nabla\phi^n,\hat{\boldsymbol{E}}^{n+1})+\|\mu(t_{n+1})-\mu^n\|_{L^4}\|\nabla\phi^n\|_{L^4}\|\hat{\boldsymbol{E}}^{n+1}\|\\ &\quad+C\Big|\tfrac{1}{\sqrt{E_1(\phi^n)+\delta_0}}-\tfrac{1}{\sqrt{E_1(\phi(t_{n+1}))+\delta_0}}\Big|\|\mu^n\|\|\nabla\phi^n\|_{L^4}\|\nabla\hat{\boldsymbol{E}}^{n+1}\|\\ &\quad+\|\mu(t_{n+1})\|_{L^4}\|\nabla\phi(t_{n+1})-\nabla\phi^n\|\|\hat{\boldsymbol{E}}^{n+1}\|_{L^4}\\ &\le \tfrac{e_R^{n+1}}{\sqrt{E_1(\phi^n)+\delta_0}}\big(\mu^n\nabla\phi^n,\hat{e}^{n+1}_{\boldsymbol{u}}\big)+C|e_R^{n+1}|^2+C\|e^n_\phi\|_1^2+\tfrac{M}{8}\Big\|\nabla e^n_\mu\Big\|^2+\tfrac{\nu}{10}\|\nabla\hat{\boldsymbol{E}}^{n+1}\|^2\\ &\quad+K_2(\|\boldsymbol{E}^n\|^2+\|\hat{\boldsymbol{E}}^{n+1}-\boldsymbol{E}^n\|^2)+C\Big(\|\mu\|^2_{H^1(J_n;H^1(\Omega))}+\|\phi\|^2_{H^1(J_n;H^1(\Omega))}\Big)\Delta t+C(\Delta t)^2,\end{aligned}\tag{B.11}$$

where the estimate $\|\hat{\boldsymbol{E}}^{n+1}\|^2\le\|\boldsymbol{E}^n\|^2+\|\hat{\boldsymbol{E}}^{n+1}-\boldsymbol{E}^n\|^2$ is also utilized and K_2 is a positive constant which is independent of Δt.

Combining (B.2) with the estimates of C_2–C_4, if Δt is small enough, s.t. $\Delta t \le \frac{2}{K_2}$, we have

$$\begin{aligned}
&\frac{\left\|\boldsymbol{E}^{n+1}\right\|^2-\|\boldsymbol{E}^n\|^2}{2\Delta t}+\frac{\|\nabla\gamma^{n+1}\|^2}{2\Delta t}+\frac{\nu}{2}\left(\|\varepsilon^{n+1}\|^2-\|\varepsilon^n\|^2\right)+\frac{\nu}{4}\|\nabla\widehat{\boldsymbol{E}}^{n+1}\|^2\\
&\le-\exp\left(\tfrac{t_{n+1}}{T}\right)e_Q^{n+1}\left((\boldsymbol{u}^n\cdot\nabla)\boldsymbol{u}^n,\widehat{e}_{\boldsymbol{u}}^{n+1}\right)+\frac{e_R^{n+1}}{\sqrt{E_1(\phi^n)+\delta_0}}\left(\mu^n\nabla\phi^n,\widehat{e}_{\boldsymbol{u}}^{n+1}\right)+\frac{\nu}{8}\|\nabla\widehat{\boldsymbol{E}}^n\|^2\\
&\quad+\frac{M}{8}\left\|\nabla e_\mu^n\right\|^2+C\|e_\phi^n\|_1^2+C|e_R^{n+1}|^2+C|e_Q^{n+1}|^2+C(\left\|e_{\boldsymbol{u}}^n\right\|_1^2+1)\|\boldsymbol{E}^n\|^2+C\left\|e_{\boldsymbol{u}}^n\right\|_1^2(\Delta t)^2\\
&\quad+C\|\boldsymbol{u}\|^2_{H^1(J_n;H^2(\Omega))\cap L^\infty(0,T;H^2(\Omega))}\Delta t+C\|\psi\|^2_{H^2(J_n;H^2(\Omega))}\Delta t+C\|\mu\|^2_{H^1(J_n;H^1(\Omega))}\Delta t\\
&\quad+C\left\|\boldsymbol{v}_t\right\|^2_{L^2(J_n;L^2(\Omega))}+C(\|\boldsymbol{u}\|^2_{H^2(J_n;H^{-1}(\Omega))}+\|\phi\|_{H^1(J_n;H^1(\Omega))}+\|\varepsilon^n\|^2)\Delta t+C(\Delta t)^2.
\end{aligned}$$

Then we complete the proof of Lemma 13.

Appendix C. Proof of Theorem 3.1

Note that Lemma 5 implies that $\|\boldsymbol{u}^{n+1}\|_1=\|P_H\widehat{\boldsymbol{u}}^{n+1}\|_1\le c_5\|\widehat{\boldsymbol{u}}^{n+1}\|_1$. Define

$$\begin{aligned}
C_0&=C\|R\|^2_{H^2(J_n)}+C\|\psi\|^2_{H^2(J_n;H^2(\Omega))}+C\left(1+\|\boldsymbol{u}^n\|_1^2+\left\|e_{\boldsymbol{u}}^n\right\|_1^2\right)\Delta t+C\left(\|\phi\|^2_{H^1(J_n;H^1(\Omega))}+\|\phi\|^2_{H^2(J_n;L^2(\Omega))}+\|\mu\|^2_{H^1(J_n;H^1(\Omega))}\right)\\
&\quad+C\left(\|\boldsymbol{u}\|^2_{H^1(J_n;H^2(\Omega))\cap L^\infty(0,T;H^2(\Omega))}+\|\boldsymbol{u}\|^2_{H^2(J_n;H^{-1}(\Omega))}\right),
\end{aligned}$$

and Remark 2.9 implies that $\sum_{n=0}^{N_t}C_0\le C(1+L)$, where L is independent of Δt. Thus, combining Lemmas 7, 13 and 14, and multiplying it by $2\Delta t$, we have

$$\begin{aligned}
&\|e_\phi^{n+1}\|_1^2-\|e_\phi^n\|_1^2+\frac{5M}{4}\Delta t\|e_\mu^{n+1}\|_1^2+2\left(|e_R^{n+1}|^2-\left|e_R^n\right|^2\right)+\left\|\boldsymbol{E}^{n+1}\right\|^2-\|\boldsymbol{E}^n\|^2\\
&\quad+\frac{\nu}{2}\Delta t\|\nabla\widehat{\boldsymbol{E}}^{n+1}\|^2-\frac{\nu}{4}\Delta t\|\nabla\widehat{\boldsymbol{E}}^{n+1}\|^2+\|\nabla\gamma^{n+1}\|^2+\nu\Delta t\left(\|\varepsilon^{n+1}\|^2-\|\varepsilon^n\|^2\right)+|e_Q^{n+1}|^2-\left|e_Q^n\right|^2\\
&\le C\Delta t(\|e_\phi^n\|_1^2+\nu\Delta t\|\varepsilon^n\|^2)+C_1\Delta t\left(\|e_\phi^{n+1}\|_1^2+|e_R^{n+1}|^2+|e_Q^{n+1}|^2\right)+\frac{M}{4}\Delta t\left\|e_\mu^n\right\|_1^2\\
&\quad+C\Delta t(1+\|\boldsymbol{u}^n\|_1^2+\left\|e_{\boldsymbol{u}}^n\right\|_1^2)\|\boldsymbol{E}^n\|^2+C_0(\Delta t)^2+C\Delta t\left\|\boldsymbol{v}_t\right\|^2_{L^2(J_n;L^2(\Omega))},
\end{aligned}\tag{C.1}$$

where Lemmas 5 and 8 are used.

Then, summing (C.1) over n from 0 to $m-1$, and applying Lemma 3 and Remark 2.9, if $\Delta t\le\frac{1}{2C_1}$ we obtain

$$\|e_\phi^m\|_1^2+\Delta t\sum_{n=0}^{m-1}\|e_\mu^{n+1}\|_1^2+\|\boldsymbol{E}^m\|^2+\Delta t\sum_{n=0}^{m-1}\|\nabla\widehat{\boldsymbol{E}}^{n+1}\|^2+\Delta t\|\varepsilon^m\|^2+|e_R^m|^2+|e_Q^m|^2\le C(\Delta t)^2,\tag{C.2}$$

where we have used the fact from Remark 2.9 that

$$C\Delta t\sum_{n=0}^{m-1}(\|\boldsymbol{u}^n\|_1^2+\left\|e_{\boldsymbol{u}}^n\right\|_1^2)\|\boldsymbol{E}^n\|^2\le C\Delta t\sum_{n=0}^{m-1}(2\|\widehat{\boldsymbol{u}}^n\|_1^2+\|\boldsymbol{u}\|_1^2)\|\boldsymbol{E}^n\|^2\le C\|\boldsymbol{E}^n\|^2.$$

On the other hand, by combining Lemma 11, (3.9) and (C.2), we obtain the estimate

$$\|e_{\boldsymbol{u}}^{m+1}\|^2\le C(\Delta t)^2.\tag{C.3}$$

Thanks to (3.5), we have

$$\Delta t\sum_{n=0}^{m}\|\nabla\widehat{e}_{\boldsymbol{u}}^{n+1}\|^2=\Delta t\sum_{n=0}^{m}\left(\|\nabla\widehat{\boldsymbol{E}}^{n+1}-\nabla\widehat{\boldsymbol{v}}(t_{n+1})\|^2\right)\le 2\Delta t\sum_{n=0}^{m}\left(\|\nabla\widehat{\boldsymbol{E}}^{n+1}\|^2+\|\nabla\widehat{\boldsymbol{v}}(t_{n+1})\|^2\right)\le C(\Delta t)^2.\tag{C.4}$$

Then we complete the proof of Theorem 3.1 by combining (C.2)–(C.4).

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